

Generalized Bayesian Composite Quantile Regression with an Application to Equity Premium Forecasting

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Abstract

Composite quantile regression (CQR) is a robust and efficient estimator under heavy-tailed and contaminated errors. Existing Bayesian extensions rely on working likelihoods that require latent-variable augmentation and can deliver poorly calibrated credible intervals. We develop generalized Bayesian CQR, which exponentiates the composite quantile loss directly, targeting the same objective as frequentist CQR. Because generalized Bayes replaces point optimization with posterior averaging over the loss surface, it is especially relevant under heavy-tailed errors where the composite quantile loss flattens near its minimum. In generalized Bayes posterior dispersion depends on a learning rate that we calibrate by matching marginal variances to their frequentist sandwich counterparts. The resulting credible intervals achieve near-nominal coverage in cross-sectional settings and substantially reduce the undercoverage of i.i.d. intervals under serial dependence, with a residual shortfall under high persistence that mirrors the finite-sample bias of frequentist HAC inference. The calibration has a closed-form solution under flat priors and extends to normal and spike-and-slab LASSO priors for shrinkage and variable selection. Sampling uses standard Metropolis-Hastings with no latent variables, achieving roughly 100-fold computational gains over likelihood-based Bayesian CQR at a common quantile grid. Monte Carlo experiments show competitive or improved point estimation relative to frequentist CQR, reliable coverage, and robust variable selection across Gaussian, heavy-tailed, and contaminated error distributions. An equity premium forecasting application demonstrates that the efficiency and robustness gains translate into economically meaningful improvements in out-of-sample portfolio performance.

Keywords: Composite quantile regression, Gibbs posterior, Generalized Bayes, Learning rate calibration, Equity premium forecasting, Spike-and-slab priors.

JEL Classification: C11, C14, C21, C52, C53, E37, G17.

1 Introduction

Economic outcomes are frequently characterized by complex conditional distributions with heavy tails, heteroskedasticity, and extreme events. The global financial crisis and the COVID-19 pandemic showed that tail dynamics, poorly captured by mean regression, can dominate risk assessment and policy design. Composite quantile regression (CQR), introduced by [Zou and Yuan \(2008\)](#), provides a robust, semi-parametric alternative that estimates a common slope vector by simultaneously fitting a grid of conditional quantiles under a location-shift model. CQR minimizes a composite loss that sums check functions over observations and quantile levels, inheriting the robustness properties of quantile regression (bounded influence function, resistance to outliers) while achieving efficiency gains through information pooling across quantile levels¹.

Despite its appealing frequentist properties, Bayesian implementation of CQR has remained challenging. The Bayesian quantile regression literature relies primarily on the asymmetric Laplace distribution (ALD) working likelihood ([Yu and Moyeed, 2001](#); [Kozumi and Kobayashi, 2011](#)). For CQR, [Huang and Chen \(2015\)](#) construct a composite working likelihood by combining ALD densities across quantile levels and introducing latent mixture allocations. A related Bayesian approach is that of [Arnroth \(2025\)](#), who generalizes the working-likelihood framework using a skewed exponential power distribution, thereby extending the implied loss from the quantile case (L^1) to L^p -quantiles while retaining a composite mixture structure. In these approaches, the resulting Markov chain Monte Carlo (MCMC) schemes become burdensome in moderate dimensions, and incorporating shrinkage priors adds further computational complexity. More fundamentally, the induced posterior mode from a likelihood-based approximation does not coincide with the CQR estimator in finite samples, and inference for extreme quantiles (where accurate tail assessment is most critical) can be unreliable due to misspecification of the residual distribution ([Sriram et al., 2013](#); [Yang and He, 2012](#); [Yang et al., 2016](#)).

This article develops Generalized Bayesian Composite Quantile Regression (GBCQR), which exponentiates the composite quantile loss directly within the decision-theoretic framework of [Bissiri et al. \(2016\)](#). The approach relies on standard random-walk Metropolis-Hastings sampling, avoiding both the distributional assumptions and latent variable augmentation of ALD-based methods. We calibrate the learning rate by matching posterior marginal variances to their frequentist sandwich counterparts. Under flat priors this approach gives a closed-form solution, while under penalized priors it requires only a one-dimensional optimization. The framework naturally accommodates spike-and-slab priors for variable selection with full posterior inference, a feature computationally impractical under ALD. The proposed generalized Bayes posterior sampler is roughly two orders of magnitude faster than ALD-based Bayesian CQR. At the same time, GBCQR can offer a computational advantage over frequentist penalized CQR in moderate-to-high-dimensional settings. While frequentist CQR with variable selection requires solving a non-smooth optimization problem jointly over K quantile levels, the MCMC

¹Under standard regularity conditions, [Zou and Yuan \(2008\)](#) show that CQR achieves high asymptotic efficiency relative to OLS across a broad class of error distributions, with larger gains under heavy tails where OLS performance degrades.

sampler of GBCQR only evaluates the composite loss at each proposed parameter value, with the prior encoding the regularization directly.

Generalized Bayesian inference is an alternative paradigm that replaces the likelihood with a loss function encoding the inferential objective (Bissiri et al., 2016)². The object of interest, known as a Gibbs posterior, takes the form

$$\pi(\beta \mid y) \propto \exp(-\eta \cdot L_n(\beta)) \cdot \pi(\beta), \quad (1)$$

where $L_n(\beta)$ is an empirical loss, $\pi(\beta)$ is a prior, and $\eta > 0$ is a learning rate controlling posterior concentration. Bissiri et al. (2016) show that the Gibbs posterior formula is the unique coherent belief update arising from a decision-theoretic criterion that minimizes expected loss relative to the prior while penalizing departures from prior beliefs through the Kullback–Leibler divergence. This approach is useful when the likelihood is unknown, intractable, or potentially misspecified³, which is precisely the situation in composite quantile regression, where the check loss characterizes the inferential target but need not correspond to any correctly specified probability model.

Since the loss $L_n(\beta)$ is not derived from a fully specified sampling distribution, the Gibbs posterior is identified only up to a scale parameter and η plays this role by determining the mapping from empirical loss differences into posterior dispersion. As discussed in Frazier et al. (2023), among others, the learning rate η has no default value and inappropriate choices yield over-concentrated or over-dispersed posteriors. Existing calibration methods, including the loss-likelihood bootstrap (Lyddon et al., 2019) and generalized posterior calibration (Syring and Martin, 2019), develop asymptotic theory under flat priors and often target joint credible regions⁴. In econometrics, however, prior influence may be non-negligible, and coefficient-by-coefficient inference is standard. This paper calibrates the learning rate so that each posterior marginal variance matches, as closely as possible, its frequentist sandwich counterpart, yielding approximately nominal frequentist coverage for individual coefficients (the standard object of inference in applied econometrics)⁵. The framework extends to normal and spike-and-slab LASSO priors (Ročková and George, 2018) by adjusting targets to reflect each prior’s frequentist analogue. In practice, the difficulty of calibrating the learning rate has been the main obstacle to the adoption of generalized Bayesian methods in applied work, where the existing literature is predominantly theoretical. A secondary contribution of this paper is to demonstrate that a computationally simple, closed-form calibration can deliver reliable inference across a wide range of settings, including under prior regularization and serial dependence.

Monte Carlo simulations indicate that GBCQR performs comparably to frequentist CQR across diverse error distributions, achieving competitive RMSE, near-nominal coverage in cross-sectional designs, and fast computation. Under Cauchy errors, GBCQR delivers meaningful RMSE reductions relative to frequentist CQR again arising from posterior averaging over flat

²Related constructions also arise in the econometric literature on quasi-Bayesian inference based on extremum estimators (Chernozhukov and Hong, 2003).

³See, for example, Jiang and Tanner (2008); Alquier et al. (2016); Syring and Martin (2022).

⁴See Wu and Martin (2023) for a comprehensive comparison.

⁵This parallels the dispersion-matching approach in Agnoletto et al. (2025).

regions of the loss surface⁶. For variable selection, GBCQR with spike-and-slab priors achieves selection rates comparable to frequentist penalized CQR while maintaining performance under heavy tails where standard Bayesian regression with spike-and-slab prior fails.

The empirical application contributes to the equity premium forecasting literature (Goyal and Welch, 2008; Goyal et al., 2024) by bringing composite quantile methods to bear on return prediction. Financial returns exhibit heavy tails and occasional extreme observations that can destabilize OLS-based forecasts. CQR provides an alternative. Under the common-slope location-shift model, it estimates the conditional location more efficiently than OLS whenever the error distribution departs from Gaussianity, with substantial gains under heavy tails (Zou and Yuan, 2008). The Bayesian framework further stabilizes coefficient estimates through posterior averaging, which is especially valuable when the composite loss surface is relatively flat near its minimum, a pattern consistent with the Monte Carlo evidence in Section 3 under heavy-tailed errors. Quantile-based methods have also attracted interest in finance for their ability to capture predictability that may vary across the conditional distribution (Linton and Whang, 2007; Han et al., 2016). Extending GBCQR to quantile-specific slope models is a natural direction for future work, but falls outside the scope of the present paper.

Using the 31 predictors from the updated Goyal et al. (2024) dataset, we find that GBCQR improves upon OLS in out-of-sample R^2 for the majority of predictors, although statistically significant gains over the historical mean remain elusive. The economic evaluation, however, reveals more pronounced differences. GBCQR delivers average CER improvements of 0.7 percentage points and Sharpe ratio gains of approximately 20% relative to both OLS and the historical mean, with lower portfolio turnover reflecting the stability of quantile-based estimates. Across the 31 predictors, GBCQR produces the highest Sharpe ratio in 87% of cases and improves upon the historical mean in CER terms for 74% of variables. A variable selection exercise using the spike-and-slab prior further demonstrates that GBCQR can identify predictive signals in a multivariate setting where standard Bayesian regression fails under heavy tails.

The remainder of the paper proceeds as follows. Section 2 presents the GBCQR framework, including the composite quantile loss, the Gibbs posterior construction, and the parameter-by-parameter variance matching calibration for flat, normal, and spike-and-slab LASSO priors. Section 3 reports Monte Carlo evidence on estimation accuracy, coverage, variable selection, and computational performance. Section 4 illustrates the methodology with an empirical application. Section 5 concludes.

2 Methodology

2.1 Composite quantile regression as an econometric model

Economic time series exhibit heavy tails, outlier contamination, and distributional asymmetry that render conditional mean models inadequate for risk assessment. Quantile regression (Koenker and Bassett, 1978) addresses this by estimating conditional quantiles directly,

⁶Both Bayesian and frequentist CQR are resistant to outliers. The main advantage of the Bayesian approach is not extra robustness, but improved stability when the optimum is weakly identified.

without distributional assumptions on the error term. In financial economics, for instance, characterizing the lower tail of return distributions is central to Growth-at-Risk analysis (Adrian et al., 2019) and portfolio risk management, where the relevant objects of inference are conditional quantiles rather than conditional means. More broadly, quantile regression has become a standard tool in applied econometrics whenever the effect of covariates varies across the conditional distribution of outcomes; see Koenker (2005) for a comprehensive treatment.

Composite quantile regression (CQR), introduced by Zou and Yuan (2008), extends this framework by estimating multiple quantiles simultaneously under a common slope restriction. The motivation is both statistical and economic. From a statistical perspective, pooling information across quantile levels produces efficiency gains. Under the linear location-shift model commonly assumed in applied work, the slope coefficients are identified by every quantile, and CQR exploits this by aggregating information rather than discarding it. Zou and Yuan (2008) show that the resulting estimator achieves high asymptotic relative efficiency (ARE) compared to OLS under Gaussian errors ($\text{ARE} \approx 0.96$), while offering dramatic improvements under heavy tails. Under Student- t_3 errors, for example, the ARE of CQR relative to OLS rises to approximately 1.9, and under Cauchy errors (where the OLS estimator is not even consistent) CQR remains root- n consistent and asymptotically normal. From an economic perspective, CQR is well suited to applications where the researcher believes that a common set of slope coefficients governs the conditional distribution, but the error distribution is unknown and potentially non-Gaussian. This setting is pervasive in macroeconomic forecasting and financial risk assessment, where heavy tails and outlier contamination are the norm rather than the exception.

To fix notation, let $\{(y_i, x_i)\}_{i=1}^n$ denote the observed data, where $y_i \in \mathbb{R}$ is a scalar response and $x_i \in \mathbb{R}^p$ is a vector of covariates. Consider the linear location-shift model

$$y_i = \alpha^* + x_i^\top \beta^* + \varepsilon_i, \quad i = 1, \dots, n, \quad (2)$$

where $\alpha^* \in \mathbb{R}$ is an intercept, $\beta^* \in \mathbb{R}^p$ is the slope vector of interest, and the errors satisfy $\varepsilon_i = \sigma z_i$ with $\sigma > 0$ and $\{z_i\}$ i.i.d. from a continuous, symmetric distribution F with density f . For clarity, we work throughout with centered regressors satisfying $\mathbb{E}[x_i] = 0$ and $\mathbb{E}[x_i x_i^\top] = C \succ 0$, with $x_i \perp \varepsilon_i$. Centering is a normalization that simplifies exposition by orthogonalizing slope and intercept score components; the general case with $\mathbb{E}[x_i] \neq 0$ requires only a Schur complement correction in the asymptotic covariance, as detailed in Appendix A.2.

Under (2), the conditional τ_k -quantile of y_i given x_i takes the form $Q_{y|x}(\tau_k | x_i) = \theta_k^* + x_i^\top \beta^*$, where $\theta_k^* = \alpha^* + \sigma F^{-1}(\tau_k)$ is a quantile-specific intercept. The slope β^* is common across all quantile levels, and CQR estimates it by minimizing the composite loss that aggregates the check function $\rho_\tau(u) = u(\tau - \mathbb{1}\{u < 0\})$ across a grid of quantile levels $0 < \tau_1 < \dots < \tau_K < 1$:

$$L_n(\beta, \theta) = \sum_{k=1}^K \sum_{i=1}^n \rho_{\tau_k}(y_i - \theta_k - x_i^\top \beta), \quad (3)$$

where $\theta = (\theta_1, \dots, \theta_K)^\top$. Since the slope vector β is the inferential target, the quantile-specific intercepts are profiled out. For fixed β , the minimizer over θ_k is the sample τ_k -quantile of

the partial residuals $\{y_i - x_i^\top \beta\}_{i=1}^n$, so the profiled loss is $L_n(\beta) := L_n(\beta, \hat{\theta}(\beta))$ and the CQR estimator is $\hat{\beta}_n = \arg \min_{\beta \in \mathbb{R}^p} L_n(\beta)$. Under the centering condition $\mathbb{E}[x_i] = 0$, the slope and intercept score components are orthogonal in expectation, so profiling out the intercepts does not affect the first-order asymptotic distribution of $\hat{\beta}_n$; see Appendix A.2 for the formal argument.

Under standard regularity conditions (stated in Appendix A.1), the CQR estimator satisfies

$$\sqrt{n}(\hat{\beta}_n - \beta^*) \xrightarrow{d} \mathcal{N}(0, \Sigma_{\text{CQR}}), \quad \Sigma_{\text{CQR}} = J^{-1} I J^{-1}, \quad (4)$$

where the curvature and score covariance matrices are

$$J = \frac{A_1}{\sigma} C, \quad I = A_2 C, \quad (5)$$

with scalars

$$A_1 = \sum_{k=1}^K f(F^{-1}(\tau_k)), \quad A_2 = \sum_{k=1}^K \sum_{\ell=1}^K (\min(\tau_k, \tau_\ell) - \tau_k \tau_\ell). \quad (6)$$

The sandwich form (4) is the standard M-estimator covariance (Huber, 1967; White, 1982), and substituting (5) recovers the expression in Zou and Yuan (2008). We retain the sandwich decomposition because the curvature J and score covariance I enter separately into the learning rate calibration developed below.

Two features of J and I deserve attention. The curvature J aggregates the error density evaluated at each quantile. Intuitively, $f(F^{-1}(\tau_k))$ measures the sharpness of the check loss at the τ_k -quantile, so that quantile levels where the density is high contribute more information. When the error distribution has heavy tails, $f(F^{-1}(\tau_k))$ is smaller at more extreme quantiles, reducing the overall curvature and flattening the composite quantile loss surface near the optimum. In such settings, a Gibbs posterior can improve finite-sample stability by averaging over a set of near-minimizers rather than selecting a single optimizer. The Monte Carlo results in Section 3 are consistent with this mechanism, particularly under Cauchy errors where GBCQR improves upon frequentist CQR in RMSE. The score covariance I depends on the cross-quantile terms $\min(\tau_k, \tau_\ell) - \tau_k \tau_\ell$, which equal $\text{Cov}(\mathbb{1}\{\varepsilon_i < \sigma F^{-1}(\tau_k)\}, \mathbb{1}\{\varepsilon_i < \sigma F^{-1}(\tau_\ell)\})$. These terms capture the statistical dependence between quantile score contributions at different levels, and are always non-negative.

2.2 Why Bayesian CQR, and why existing approaches fall short

There are several reasons to pursue Bayesian inference for CQR beyond what frequentist methods offer. First, in applied macroeconomics and finance, the number of candidate predictors often exceeds the effective sample size once the data are split by forecast horizon and quantile level. Shrinkage and variable selection are then essential, and Bayesian methods provide a principled framework for both through the prior distribution. Second, full posterior distributions enable coherent uncertainty quantification for non-linear functions of parameters (conditional quantile forecasts, tail risk measures, and portfolio allocations) without recourse to the delta method or bootstrap resampling. Third, in sequential forecasting applications, the posterior from one vintage of data serves as a natural prior for the next, facilitating real-time updating.

The dominant approach to Bayesian quantile regression, due to Yu and Moyeed (2001), casts

the check loss as the kernel of an asymmetric Laplace distribution (ALD), treating the ALD as a working likelihood. This construction has been extended to composite quantile regression by [Huang and Chen \(2015\)](#), who combine ALD densities across quantile levels with latent mixture allocations, and more recently by [Arnroth \(2025\)](#), who generalize the working likelihood to a skewed exponential power family. The ALD approach has been enormously influential and remains the most widely used Bayesian quantile regression method.

However, the ALD approach has well-documented limitations that motivate the present work. The first is inferential. Because the ALD is a working likelihood rather than the true data-generating process, the resulting posterior can be poorly calibrated. [Sriram et al. \(2013\)](#) and [Yang and He \(2012\)](#); [Yang et al. \(2016\)](#) establish that ALD-based credible intervals can exhibit systematic under-coverage, particularly for tail quantiles where accurate uncertainty quantification matters most. The posterior’s asymptotic covariance is dictated by the ALD Fisher information rather than the sandwich covariance that governs the sampling distribution of the quantile regression estimator, and the two coincide only when the true error distribution happens to be asymmetric Laplace. The second limitation is computational. The ALD working likelihood requires data augmentation with latent scale variables ([Kozumi and Kobayashi, 2011](#)), and the extension to CQR further introduces latent mixture allocations across quantile levels ([Huang and Chen, 2015](#)). The resulting MCMC schemes involve sampling $2n+K$ latent variables and evaluating nK mixture log-probabilities per iteration, where n is the sample size and K is the number of quantile levels, making the approach computationally burdensome in the moderate-dimensional settings typical of macroeconomic applications. As our Monte Carlo experiments in Section 3 demonstrate, GBCQR achieves computation roughly two orders of magnitude faster than the ALD-based approach of [Huang and Chen \(2015\)](#). The third limitation concerns extensibility. Incorporating structured priors for variable selection (such as the spike-and-slab LASSO of [Ročková and George \(2018\)](#)) within the ALD framework requires non-trivial modifications to the data augmentation scheme, whereas the loss-based approach developed below accommodates such priors naturally.

2.3 Generalized Bayesian inference: from a loss function to a Bayesian posterior

The generalized Bayesian framework of [Bissiri et al. \(2016\)](#) resolves the tension between loss-based estimation and Bayesian inference by treating the loss function as the primitive for belief updating. Instead of constructing a working likelihood that mimics the loss, one exponentiates the loss directly to obtain a posterior distribution. [Bissiri et al. \(2016\)](#) establish that this construction (known as a Gibbs posterior) is the unique coherent belief update satisfying a set of decision-theoretic axioms: the posterior minimizes expected loss under the prior, subject to a Kullback-Leibler penalty for deviation from prior beliefs. Related constructions appear in the econometrics literature on quasi-Bayesian inference for extremum estimators ([Chernozhukov and Hong, 2003](#)) and in the PAC-Bayes framework for aggregation ([Catoni, 2004](#); [Alquier et al., 2016](#)). The recent survey by [Frazier et al. \(2023\)](#) provides a comprehensive treatment of the statistical foundations.

What makes this framework particularly attractive in the case of CQR is its fidelity to the

frequentist estimand. The CQR estimator $\hat{\beta}_n$ is defined as the minimizer of the composite quantile loss. Constructing a Gibbs posterior from the same loss ensures that the posterior is centered on the same target as the frequentist procedure, inheriting its robustness properties by construction rather than through a distributional assumption. This approach is in contrast to the ALD approach, where the posterior targets the maximum of a working likelihood that need not coincide with the loss minimizer in finite samples.

Given a prior $\pi(\beta)$ on $\mathbb{R}^p$ and the profiled composite loss $L_n(\beta)$ from (3), the Gibbs posterior is

$$\pi_n^{(\eta)}(\beta | y) \propto \exp\{-\eta L_n(\beta)\} \pi(\beta), \quad \eta > 0. \quad (7)$$

When $L_n(\beta)$ coincides with a negative log-likelihood and $\eta = 1$, the Gibbs posterior reduces to the standard Bayes posterior. In the present setting, the composite quantile loss does not derive from a probability model, so no default value of η exists and the learning rate must be calibrated. The scalar $\eta > 0$ controls the rate at which the posterior concentrates around the loss minimizer. In the log-posterior kernel,

$$\ell_n(\beta; \eta) := -\eta L_n(\beta) + \log \pi(\beta), \quad (8)$$

so that larger values of η give more weight to the data relative to prior beliefs.

To understand the role of η and to develop the calibration strategy, it is useful to examine the behavior of the Gibbs posterior in a neighbourhood of the loss minimizer. A standard Laplace approximation of (7) around the posterior mode $\tilde{\beta}_n^{(\eta)} := \arg \max_{\beta} \ell_n(\beta; \eta)$ yields

$$\pi_n^{(\eta)}(\beta | y) \approx \mathcal{N}\left(\tilde{\beta}_n^{(\eta)}, V_{\text{post}}(\eta)\right), \quad V_{\text{post}}(\eta) := [\eta n \hat{J}_n + P]^{-1}, \quad (9)$$

where $P := -\nabla_{\beta}^2 \log \pi(\beta)|_{\beta=\tilde{\beta}_n^{(\eta)}}$ is the prior precision matrix and $\hat{J}_n = n^{-1} \nabla_{\beta}^2 L_n(\hat{\beta}_n)$ is the normalized curvature of the loss at the CQR estimator. A technical subtlety arises because the check loss is piecewise linear: the Hessian of $L_n(\beta)$ does not exist in the classical sense. Following the standard approach in the quantile regression literature (Powell, 1991; Koenker, 2005), we replace the indicator function in the quantile score by a kernel-smoothed approximation $\Phi(-u/h_n)$, where Φ is the standard normal distribution function and $h_n \rightarrow 0$, $nh_n \rightarrow \infty$. The resulting smoothed curvature estimator $\hat{J}_n$ is consistent for the population curvature J and delivers a well-defined Hessian for the Laplace approximation. Details of the kernel smoothing construction and bandwidth selection are given in Appendix A.3.

The Laplace covariance (9) reveals precisely why η must be calibrated. Under a flat prior ($P = 0$), the posterior covariance reduces to $V_{\text{post}}(\eta) = (\eta n \hat{J}_n)^{-1}$, which is proportional to $\hat{J}_n^{-1}$. But the frequentist sampling distribution of $\hat{\beta}_n$ is governed by the sandwich covariance $\hat{J}_n^{-1} \hat{I}_n \hat{J}_n^{-1}$, not by $\hat{J}_n^{-1}$ alone. In a correctly specified likelihood model, $J = I$ and the two coincide, but for the composite quantile loss this equality fails generically: J depends on the error density through A_1 while I depends on cross-quantile indicator covariances through A_2 . Setting $\eta = 1$ naively would produce posterior credible intervals with incorrect frequentist coverage, a pathology first identified in the generalized Bayes literature by Grünwald and van Ommen (2017) and studied extensively by Syring and Martin (2022) and Frazier et al. (2023).

Throughout the paper, we adopt the convention that $L_n(\beta)$ denotes the total profiled composite loss, so $L_n(\beta) = O_p(n)$, and the normalized quantities $\hat{J}_n$ and $\hat{I}_n := n^{-1} \sum_{i=1}^n \hat{s}_i \hat{s}_i^\top$ (where $\hat{s}_i$ is the score contribution of observation i) are both $O_p(1)$. This ensures that the calibrated learning rate is of order $O_p(1)$.

2.4 Learning rate calibration

We have established that inappropriate choices of the learning rate parameter η yield posterior credible intervals that are either too narrow (over-concentration) or too wide (under-concentration). Several calibration methods have been proposed in the literature to address this important problem, including the loss-likelihood bootstrap of [Lyddon et al. \(2019\)](#), the generalized posterior calibration of [Syring and Martin \(2019\)](#), the Safe Bayes approach of [Grünwald and van Ommen \(2017\)](#), and the dispersion matching of [Agnolotto et al. \(2025\)](#). These methods typically target joint posterior properties (such as the trace or determinant of the posterior covariance) and develop theory under flat priors.

In econometrics, however, inference typically focuses on individual regression coefficients, and prior information is often substantive. We therefore adopt a calibration strategy based on diagonal variance matching: the learning rate is chosen so that each posterior marginal variance matches, as closely as possible, the corresponding frequentist marginal variance. This ensures that posterior credible intervals for individual coefficients achieve approximately nominal frequentist coverage, the standard object of inference in applied econometrics.

Each prior specification has a natural frequentist analogue: the unpenalized CQR estimator $\hat{\beta}_n$ under the flat prior, a ridge-CQR estimator $\hat{\beta}_n^R$ under the normal prior, and an adaptive CQR-LASSO estimator $\hat{\beta}_n^A$ under the spike-and-slab LASSO prior. Let $\hat{\beta}$ denote the relevant analogue generically. Define the scaled posterior marginal variances and frequentist targets

$$d_j(\eta) := n [V_{\text{post}}(\eta)]_{jj}, \quad t_j := [\hat{\Sigma}]_{jj}, \quad (10)$$

where $V_{\text{post}}(\eta) = [\eta n \hat{J}_n + P]^{-1}$ is the Laplace covariance and $\hat{\Sigma}$ is the penalized sandwich covariance appropriate to the prior. Both $d_j(\eta)$ and t_j are $O_p(1)$ under our normalization conventions. The learning rate is selected by minimizing

$$\eta^* = \arg \min_{\eta > 0} \sum_{j \in \mathcal{J}} \left(\frac{d_j(\eta)}{t_j} - 1 \right)^2, \quad (11)$$

where $\mathcal{J}$ is an index set that depends on the prior (all coordinates for flat and normal priors; the active set for the spike-and-slab LASSO). The relative form $d_j(\eta)/t_j - 1$ renders each coordinate's contribution scale-free, so that coordinates with different variance magnitudes enter the objective on equal footing; a levels objective $\sum_j (d_j(\eta) - t_j)^2$ would instead be dominated by the highest-variance coordinates and would not admit the closed-form minimizer derived below for the flat prior.

As shown in [Appendix A.4](#), each $d_j(\eta)$ is continuous and strictly decreasing in η whenever $\hat{J}_n \succ 0$. Under a flat prior, $d_j(\eta)$ diverges as $\eta \rightarrow 0^+$, so the objective diverges at the left boundary. As $\eta \rightarrow \infty$, $d_j(\eta)$ converges to zero and the objective converges to $|\mathcal{J}|$. Since each

d_j decreases continuously from infinity to zero, it equals t_j at some interior value of η , where the j th term in (11) is zero. The objective therefore falls strictly below $|\mathcal{J}|$ for interior values of η , which guarantees an interior minimizer. Under proper priors, an interior minimizer exists as long as $\lim_{\eta \rightarrow 0^+} d_j(\eta) > t_j$ for at least one $j \in \mathcal{J}$.

The curvature and score covariance are evaluated at the frequentist analogue $\hat{\beta}$ rather than at the posterior mode $\tilde{\beta}_n^{(\eta)}$. This simplifies the calibration from a fixed-point problem to a one-dimensional optimization, and the approximation is asymptotically innocuous: under posterior concentration, $\tilde{\beta}_n^{(\eta)} - \hat{\beta} = O_p(n^{-1/2})$, and the kernel-smoothed curvature estimator is locally Lipschitz, so the induced change in $d_j(\eta)$ is $o_p(1)$.

Two features of the calibration merit explicit discussion. First, the Laplace approximation in (9) serves as a local device for calibrating η and constructing the RWMH proposal, not as a substitute for exact posterior inference. The MCMC sampler of Section 2.6 draws from the exact Gibbs posterior (7), and all reported credible intervals and posterior inclusion probabilities are computed from these draws. The Laplace covariance predicts the posterior dispersion at the calibrated η^* only to the extent that the posterior is approximately Gaussian near its mode. In regimes where this approximation is less accurate (for instance, under Cauchy errors where the composite loss surface is flatter), the coverage results in Section 3 provide the relevant finite-sample validation. Coverage is near-nominal across all error distributions and across the full range of sample sizes and design correlations considered in the Supplementary Appendix ($n \in \{100, 200\}$, $\rho \in \{0.25, 0.50, 0.75\}$), suggesting that the Laplace proxy is reliable for calibration purposes even when it may not accurately describe the full posterior geometry.

Second, the diagonal variance matching criterion is coordinate-specific: it calibrates marginal inference on the regression coefficients as parametrized, and is not invariant to reparametrizations that rotate or rescale the regressors. This is a feature shared with any coordinate-wise inference procedure, including standard heteroskedasticity-robust standard errors. In applied work, standardizing regressors is standard practice in penalized regression and is recommended here as well. That said, the Monte Carlo experiments provide indirect evidence of robustness to the design correlation structure: the Toeplitz designs with $\rho \in \{0.25, 0.50, 0.75\}$ span a wide range of eigenvalue ratios in the design matrix, and coverage remains within 1–2 percentage points of nominal throughout (Supplementary Appendix). This stability suggests that the calibrated η^* adapts to the design without sensitivity to the specific correlation level.

Appendix A.6 establishes a Bernstein–von Mises result and an exact-marginal-calibration property for the flat-prior case: under the maintained location-shift model, the calibrated Gibbs posterior is asymptotically Gaussian and its marginal credible intervals coincide with the frequentist sandwich coordinate by coordinate. A general theory covering normal and spike-and-slab LASSO priors is an important open problem that falls outside the scope of the present paper. The development of such theory is a natural companion project, and recent results on related Gibbs posteriors (Syring and Martin, 2022; Agnoletto et al., 2025; Winter et al., 2026) suggest that the theoretical foundations can be established under conditions similar to those maintained here. The present paper establishes the methodology computationally and provides extensive finite-sample evidence of its properties across a broad range of settings.

We now specialize the calibration to each of the three prior specifications considered in this paper.

2.4.1 Flat prior

The flat prior $\pi^F(\beta) \propto 1$ imposes no regularization ($P = 0$), so the posterior mode coincides with the CQR estimator $\hat{\beta}_n$. The Laplace covariance simplifies to

$$V_{\text{post}}^F(\eta) = \frac{1}{\eta n} \hat{J}_n^{-1}, \quad d_j^F(\eta) = \frac{[\hat{J}_n^{-1}]_{jj}}{\eta}, \quad (12)$$

and the frequentist target is the unpenalized CQR sandwich:

$$\hat{\Sigma}^F = \hat{J}_n^{-1} \hat{I}_n \hat{J}_n^{-1}, \quad t_j^F = [\hat{J}_n^{-1} \hat{I}_n \hat{J}_n^{-1}]_{jj}. \quad (13)$$

Defining the variance ratios $r_j := [\hat{J}_n^{-1}]_{jj} / t_j^F$ and substituting into (11) yields a closed-form solution (derivation in Appendix A.4):

$$\eta^* = \frac{\sum_{j=1}^p r_j^2}{\sum_{j=1}^p r_j}. \quad (14)$$

This is a weighted average of the variance ratios with weights proportional to r_j itself, so that coordinates with larger posterior-to-frequentist variance ratios contribute more to the calibrated learning rate. This specification serves as the baseline: it isolates the effect of η from any prior-induced shrinkage.

It is instructive to compare (14) with the learning rate implied by the loss-likelihood bootstrap (LLB) of Lyddon et al. (2019). Under a flat prior, the LLB yields

$$\eta_{\text{LLB}}^* = \frac{\text{tr}(\hat{J}_n \hat{I}_n^{-1} \hat{J}_n)}{\text{tr}(\hat{J}_n)}, \quad (15)$$

which matches the total Fisher information by calibrating the trace of the inverse covariance. The LLB approach ensures correct aggregate posterior spread but may leave individual coordinates miscalibrated whenever the eigenstructure of $\hat{J}_n$ differs from that of $\hat{I}_n$. The diagonal objective (11) targets coordinate-wise calibration instead, which is the relevant criterion when inference focuses on individual regression coefficients.

2.4.2 Normal prior

The Gaussian prior $\pi^N(\beta) \propto \exp(-\|\beta\|_2^2 / (2\tau^2))$ induces ridge-type shrinkage, with prior precision $P = \tau^{-2} I_p$. Its frequentist analogue is the ridge-CQR estimator

$$\hat{\beta}_n^R = \arg \min_{\beta \in \mathbb{R}^p} \left\{ L_n(\beta) + \frac{1}{2\tau^2} \|\beta\|_2^2 \right\}. \quad (16)$$

The Laplace covariance becomes

$$V_{\text{post}}^N(\eta) = (\eta n \hat{J}_n + \tau^{-2} I_p)^{-1}, \quad d_j^N(\eta) = n [(\eta n \hat{J}_n + \tau^{-2} I_p)^{-1}]_{jj} = [(\eta \hat{J}_n + n^{-1} \tau^{-2} I_p)^{-1}]_{jj}, \quad (17)$$

and the frequentist target is the $\sqrt{n}$ -scaled sandwich covariance of the ridge-CQR estimator:

$$\hat{\Sigma}^N = (\hat{J}_n + n^{-1} \tau^{-2} I_p)^{-1} \hat{I}_n (\hat{J}_n + n^{-1} \tau^{-2} I_p)^{-1}, \quad t_j^N = [\hat{\Sigma}^N]_{jj}. \quad (18)$$

The factor n^{-1} multiplying the prior precision arises because the ridge-CQR objective (16) sums a total loss of order $O(n)$ with a penalty of order $O(1)$; expressing the sandwich in terms of the normalized curvature $\hat{J}_n = n^{-1} \nabla^2 L_n$ accordingly rescales the penalty contribution. Note that η does not appear in the target: the learning rate calibrates the loss contribution in the generalized posterior, not the frequentist ridge estimator. Unlike the flat prior, no closed-form minimizer exists because $d_j^N(\eta)$ depends on η through a ridge-shifted matrix inverse. Computation is accelerated by the eigendecomposition $\hat{J}_n = Q \text{diag}(\mu_1, \dots, \mu_p) Q^\top$, which gives

$$d_j^N(\eta) = \sum_{m=1}^p \frac{Q_{jm}^2}{\eta \mu_m + n^{-1} \tau^{-2}}, \quad (19)$$

so evaluating $d_j^N(\eta)$ for each candidate η requires no repeated matrix inversions. The minimizer is obtained by golden-section search.

The hyperparameter τ^2 controls the strength of regularization and is selected independently of η . Under a flat prior, the learning rate affects only posterior dispersion, leaving the mode unchanged. Under the normal prior, η plays a dual role: the posterior mode solves $\min_{\beta} \{L_n(\beta) + \|\beta\|^2 / (2\eta\tau^2)\}$, so the effective penalty $\lambda_{\text{eff}} = 1/(2\eta\tau^2)$ depends on η , and the mode shifts as η varies. The calibration remains well-defined because the frequentist target (18) is evaluated at the fixed ridge-CQR estimator $\hat{\beta}_n^R$ with penalty $1/(2\tau^2)$, not at the η -dependent posterior mode. This evaluation convention, discussed after equation (11), simplifies the calibration from a fixed-point problem to a one-dimensional optimization. The approximation is asymptotically innocuous: the prior contribution to the mode is $O(n^{-1})$ relative to the posterior scale $O(n^{-1/2})$, so the η -induced mode shift vanishes in the Bernstein–von Mises regime.

2.4.3 Spike-and-slab LASSO prior

Variable selection is a central concern in applied macroeconomic forecasting, where the number of candidate predictors may exceed the effective sample size. The spike-and-slab LASSO (SSL) prior of Ročková and George (2018) provides a continuous relaxation of the discrete spike-and-slab formulation, assigning each coordinate an independent Laplace mixture:

$$\pi_{\text{SSL}}(\beta_j \mid \gamma_j) = (1 - \gamma_j) \frac{\lambda_0}{2} e^{-\lambda_0 |\beta_j|} + \gamma_j \frac{\lambda_1}{2} e^{-\lambda_1 |\beta_j|}, \quad \gamma_j \mid \theta \sim \text{Bernoulli}(\theta), \quad (20)$$

where $\lambda_0 \gg \lambda_1 > 0$ and $\theta \in (0, 1)$ is the prior inclusion probability. When $\gamma_j = 0$, the spike component (λ_0) concentrates β_j near zero; when $\gamma_j = 1$, the slab component (λ_1) provides mild regularization, preserving signals. In all experiments, we set $\lambda_0 = \sqrt{n}$, $\lambda_1 = 1$, and $\theta = 1/p$,

following the default recommendations in [Ročková and George \(2018\)](#).

Because $|\beta_j|$ is not twice differentiable at zero, the Laplace approximation requires a smooth surrogate. We adopt $a_\epsilon(\beta_j) := (\beta_j^2 + \epsilon)^{1/2}$ with small $\epsilon > 0$, yielding a diagonal prior precision $D_{\text{SSL}}(\beta) := -\nabla_\beta^2 \log \pi_{\text{SSL}}(\beta) = \text{diag}([D_{\text{SSL}}]_{11}, \dots, [D_{\text{SSL}}]_{pp})$. Because the SSL prior is a mixture, the second derivative of $\log \pi_{\text{SSL}}(\beta_j)$ does not reduce to a weighted average of the spike and slab Laplace Hessians: it carries an additional term from the dependence of the mixture weights on β_j . Differentiating $\log \pi_{\text{SSL}}(\beta_j)$ twice yields

$$[D_{\text{SSL}}]_{jj} = \bar{\lambda}(\beta_j) a_\epsilon''(\beta_j) - (\lambda_0 - \lambda_1)^2 p_0(\beta_j) p_1(\beta_j) [a_\epsilon'(\beta_j)]^2,$$

where $p_1(\beta_j) \in (0, 1)$ is the relative weight of the slab component at β_j ,

$$p_1(\beta_j) = \frac{\theta (\lambda_1/2) e^{-\lambda_1 a_\epsilon(\beta_j)}}{\theta (\lambda_1/2) e^{-\lambda_1 a_\epsilon(\beta_j)} + (1 - \theta) (\lambda_0/2) e^{-\lambda_0 a_\epsilon(\beta_j)}},$$

$p_0(\beta_j) = 1 - p_1(\beta_j)$, and $\bar{\lambda}(\beta_j) = p_0(\beta_j)\lambda_0 + p_1(\beta_j)\lambda_1$ is the weighted penalty slope. The first term is the standard smoothed- ℓ_1 curvature and is non-negative. The second term can push $[D_{\text{SSL}}]_{jj}$ slightly negative in narrow transition regions where the spike and slab both carry appreciable weight; in these cases the entry is floored at zero to keep the Laplace covariance positive definite, which has no effect away from the transition zone. $[D_{\text{SSL}}]_{jj}$ is thus the diagonal second derivative of the marginal log-mixture prior, not a per-component quantity or one evaluated conditional on γ_j . The Laplace covariance is

$$V_{\text{post}}^{\text{SSL}}(\eta) = (\eta n \hat{J}_n + D_{\text{SSL}})^{-1}, \quad d_j^{\text{SSL}}(\eta) = n [(\eta n \hat{J}_n + D_{\text{SSL}})^{-1}]_{jj} = [(\eta \hat{J}_n + n^{-1} D_{\text{SSL}})^{-1}]_{jj}. \quad (21)$$

A natural frequentist counterpart is the adaptive LASSO of [Zou \(2006\)](#), but in CQR form. The adaptive CQR-LASSO estimator solves

$$\hat{\beta}_n^A = \arg \min_{\beta \in \mathbb{R}^p} \left\{ L_n(\beta) + \lambda \sum_{j=1}^p w_j |\beta_j| \right\}, \quad w_j = |\hat{\beta}_{n,j}^A|^{-\zeta}, \quad (22)$$

where $\hat{\beta}_n$ is the unpenalized CQR estimate from Section 2.4.1, λ is selected by BIC, and $\zeta > 0$ controls the penalty contrast between active and inactive coefficients. All experiments use $\zeta = 2$, following [Zou and Yuan \(2008\)](#). The weights w_j assign larger penalties to coordinates with small unpenalized estimates, inducing selection behaviour that mirrors the SSL structure. The sandwich covariance of the adaptive CQR-LASSO is

$$\hat{\Sigma}^{\text{SSL}} = (\hat{J}_n + n^{-1} D_A)^{-1} \hat{I}_n (\hat{J}_n + n^{-1} D_A)^{-1}, \quad t_j^{\text{SSL}} = [\hat{\Sigma}^{\text{SSL}}]_{jj}, \quad (23)$$

where $D_A = \text{diag}(\lambda w_j h_j)$ is the adaptive penalty Hessian and $h_j = \epsilon (\hat{\beta}_{n,j}^A)^2 + \epsilon)^{-3/2}$ is the smoothed ℓ_1 curvature evaluated at the adaptive solution, obtained by applying the same local quadratic approximation $a_\epsilon(\beta_j) = (\beta_j^2 + \epsilon)^{1/2}$ used for D_{SSL} above. Because the ℓ_1 penalty is not twice differentiable at zero, equation (23) is the sandwich covariance of the smoothed penalized objective, which serves as an approximate calibration target for the non-smooth estimator (22).

The approximation is exact on the active set $\{j : \hat{\beta}_{n,j}^A \neq 0\}$ as $\epsilon \rightarrow 0$, where $h_j \rightarrow 0$ and $\hat{\Sigma}^{\text{SSL}}$ reduces to the unpenalized sandwich restricted to active coordinates. As in (18), the factor n^{-1} reflects the $O(1)/O(n)$ scaling of the penalty relative to the total loss.

The choice of the adaptive CQR-LASSO as the calibration target reflects a structural correspondence between the Bayesian and frequentist penalties. D_{SSL} evaluated at the adaptive solution exhibits two regimes: large diagonal entries $[D_{\text{SSL}}]_{jj}$ for inactive coefficients (dominated by the spike) and small entries for active coefficients (dominated by the slab). Throughout the asymptotic statements below, $\epsilon > 0$ is treated as a fixed numerical smoothing constant and orders are taken in $n \rightarrow \infty$ at fixed ϵ . The adaptive penalty Hessian D_A exhibits an analogous two-regime pattern: $[D_A]_{jj} = \lambda w_j h_j$ is large for inactive coefficients, where $w_j = O_p(n^{\zeta/2})$ diverges because the unpenalized CQR estimator $\hat{\beta}_{n,j}$ contracts at rate $n^{-1/2}$ and $h_j = O(\epsilon^{-1/2})$ is large because $\hat{\beta}_{n,j}^A = 0$ at the sparse adaptive solution; and small for active coefficients, where $w_j \rightarrow |\beta_j^*|^{-\zeta} = O_p(1)$ is bounded away from zero but $h_j = O(\epsilon)$ vanishes as $\epsilon \rightarrow 0$ because $\hat{\beta}_{n,j}^A$ is bounded away from zero. This alignment ensures that the diagonal calibration $d_j^{\text{SSL}}(\eta)/t_j^{\text{SSL}} \approx 1$ compares like with like across both regimes.

That the calibration is anchored to the adaptive LASSO active set is a deliberate design choice, consistent with the principle maintained throughout the paper that each Bayesian prior has a natural frequentist analogue and calibration matches the two. On the active set, the calibration ensures that posterior credible intervals for non-zero coefficients have approximately correct frequentist coverage. On the inactive set, where $\hat{\beta}_{n,j}^A \approx 0$, the surrogate precision $[D_{\text{SSL}}]_{jj}$ used in the Laplace calibration is dominated by the spike contribution, limiting the leverage of η -matching on the calibration target at these coordinates. In the exact Gibbs posterior sampled by the MCMC, the spike penalty and the loss curvature both enter at the same $O_p(1)$ order on the local scale $u = \sqrt{n} \beta_j$, so η does influence the dispersion of inactive coordinates; however, posterior inclusion probabilities are determined by the spike-versus-slab density ratio at the sampled β_j via the Gibbs update (30), and are therefore governed primarily by the prior hyperparameters $(\lambda_0, \lambda_1, \theta)$ rather than by the precise posterior dispersion of β_j . The key empirical evidence that GBCQR-SSL does not mechanically inherit the frequentist selection pattern is provided by the Cauchy DGP in Section 3: GBCQR-SSL maintains TPR = 100% in a regime where BR-SSL collapses to TPR = 3% and the frequentist CQR-LASSO drops to 93%. The posterior-based selection is therefore not simply a repackaging of the frequentist active set.

2.5 Learning rate calibration in time series

In macroeconomic and financial applications, $\{(y_t, x_t^\top)\}_{t=1}^n$ typically exhibit serial dependence through both autocorrelation and conditional heteroskedasticity in the forecast errors. The calibration strategy of Section 2.4 extends naturally to this setting with a single modification: the score covariance $\hat{I}_n$ is replaced by a heteroskedasticity and autocorrelation consistent (HAC) estimator. The curvature J involves the expected conditional density of the errors at the quantile-specific intercepts, and its sample analogue $\hat{J}_n$ converges under an ergodic law of large numbers. The score covariance, by contrast, governs the variance of the $\sqrt{n}$ -scaled CQR estimator and must account for the full auto-covariance structure of the composite score process.

Replacing Assumptions 2–3 with their time series counterparts—strict stationarity, α -mixing

with polynomial decay, mutual independence of $\{x_t\}$ and $\{\varepsilon_t\}$, and existence of moments of order $2 + \delta$ (Assumptions 4–5 in the Appendix)—the sandwich CLT (4) generalizes to

$$\sqrt{n}(\hat{\beta}_n - \beta^*) \xrightarrow{d} \mathcal{N}(0, J^{-1} I_{\text{LR}} J^{-1}), \quad (24)$$

where

$$I_{\text{LR}} = \sum_{h=-\infty}^{\infty} \Gamma_h, \quad \Gamma_h = \text{Cov}(s_t, s_{t-h}), \quad (25)$$

is the long-run covariance of the composite score $s_t = \sum_{k=1}^K x_t(\tau_k - \mathbb{1}\{y_t - \theta_k^* - x_t^\top \beta^* < 0\})$, as defined in Appendix A.3. Because $\{x_t\}$ and $\{\varepsilon_t\}$ are mutually independent, the lag- h autocovariance factors as

$$\Gamma_h = \mathbb{E}[\psi(\varepsilon_t) \psi(\varepsilon_{t-h})] \cdot \mathbb{E}[x_t x_{t-h}^\top], \quad h \geq 1, \quad (26)$$

where $\psi(\varepsilon_t) := \sum_{k=1}^K (\tau_k - \mathbb{1}\{\varepsilon_t < \sigma F^{-1}(\tau_k)\})$ is the scalar error score. The first factor captures how the quantile indicators at time t co-move with those at time $t - h$; the second captures the temporal persistence of the regressors. For I_{LR} to differ from $I = \Gamma_0$, both factors must be non-zero at some $h \geq 1$.

The calibrated learning rate is then obtained by replacing $\hat{I}_n$ with $\hat{I}_n^{\text{HAC}}$ in the frequentist targets—equations (13), (18), and (23)—leaving the Laplace covariances $d_j(\eta)$ and the calibration objective (11) unchanged. Under the flat prior, the closed-form solution (14) carries over with the HAC variance ratios $r_j^{\text{HAC}} := [\hat{J}_n^{-1}]_{jj} / [\hat{J}_n^{-1} \hat{I}_n^{\text{HAC}} \hat{J}_n^{-1}]_{jj}$; under the normal and SSL priors, only the targets in the golden-section search change. For the SSL target (23), the substitution $\hat{I}_n \rightarrow \hat{I}_n^{\text{HAC}}$ is the sole time-series correction: the pilot unpenalized CQR estimate, the adaptive weights w_j , the BIC-selected penalty λ , and the adaptive CQR-LASSO point estimate $\hat{\beta}_n^{\text{A}}$ (and hence the penalty Hessian D_A) are computed exactly as in the i.i.d. case and held fixed. Serial dependence enters the calibration only through the long-run score covariance; the frequentist penalty structure is unaffected.

When serial dependence inflates the long-run score variance so that $I_{\text{LR}} \succeq I$, the HAC targets exceed their i.i.d. counterparts. A larger target requires a smaller learning rate to match the increased sampling variability, giving the ordering $\eta_{\text{HAC}}^* \leq \eta_{\text{i.i.d.}}^*$. Intuitively, the true sampling variance of $\hat{\beta}_n$ is larger than the i.i.d. approximation suggests, and the calibrated posterior widens its credible intervals accordingly.

2.6 Posterior computation

Posterior inference proceeds via adaptive random-walk Metropolis-Hastings (RWMH)⁷. At iteration m , a candidate is drawn from

$$\beta' = \beta^{(m-1)} + \delta \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, V_{\text{post}}(\eta^*)), \quad (27)$$

⁷The adequacy of the random-walk Metropolis-Hastings sampler is assessed in Supplementary Appendix, which reports effective sample size per second across the Monte Carlo designs. The results indicate that GBCQR attains reasonable mixing and delivers one to three orders of magnitude higher ESS per second than BCQR-ALD across all error distributions considered.

where $V_{\text{post}}(\eta^*)$ is the Laplace covariance evaluated at the calibrated learning rate. Using the Laplace covariance as the proposal covariance aligns the proposal shape with the local posterior geometry, which is particularly important when the posterior is anisotropic under shrinkage priors. The candidate is accepted with probability

$$\bar{\alpha}(\beta^{(m-1)}, \beta') = \min \left\{ 1, \frac{\exp\{\ell_n(\beta'; \eta^*)\}}{\exp\{\ell_n(\beta^{(m-1)}; \eta^*)\}} \right\}, \quad (28)$$

where $\ell_n(\beta; \eta) = -\eta L_n(\beta) + \log \pi(\beta)$. The step size δ is adapted during burn-in via the Robbins-Monro recursion

$$\log \delta^{(m)} = \log \delta^{(m-1)} + c_0 m^{-c_1} (\bar{\alpha}^{(m)} - \bar{\alpha}^*), \quad (29)$$

targeting the asymptotically optimal acceptance rate $\bar{\alpha}^* = 0.234$ (Gelman et al., 1997). After burn-in, δ is fixed for the sampling phase.

Under the spike-and-slab LASSO prior, each MCMC iteration augments the β -update with a Gibbs draw for the inclusion indicators. Conditional on $\beta^{(m)}$ and the prior inclusion probability θ , each γ_j is sampled from

$$\mathbb{P}(\gamma_j = 1 \mid \beta_j^{(m)}, \theta) = \frac{\theta \frac{\lambda_1}{2} e^{-\lambda_1 |\beta_j^{(m)}|}}{\theta \frac{\lambda_1}{2} e^{-\lambda_1 |\beta_j^{(m)}|} + (1 - \theta) \frac{\lambda_0}{2} e^{-\lambda_0 |\beta_j^{(m)}|}}, \quad j = 1, \dots, p. \quad (30)$$

Posterior inclusion probabilities are computed as $\text{PIP}_j = M^{-1} \sum_{m=1}^M \gamma_j^{(m)}$ from the M post-burn-in draws, and variable selection is based on the median probability model ($\text{PIP}_j > 0.5$).

A key computational advantage of GBCQR over ALD-based Bayesian CQR deserves emphasis. The ALD mixture of Huang and Chen (2015) introduces $2n + K$ latent variables per iteration (one allocation indicator z_i and one scale variable t_i per observation, plus K quantile-specific intercepts), and the per-iteration cost is $O(nKp)$ because sampling each z_i requires evaluating K mixture log-probabilities involving the full parameter vector. In contrast, the GBCQR sampler evaluates only the composite loss $L_n(\beta)$, a sum of nK check functions, with no latent variable augmentation, yielding a per-iteration cost of $O(nKp)$ for the loss evaluation plus $O(p^2)$ for the proposal. The computational gap arises not from the asymptotic order but from the constant: the ALD sampler must draw n inverse-Gaussian variates, sample n discrete indicators, and update K intercepts at each sweep, overhead that is absent from the single-block random-walk Metropolis–Hastings step of GBCQR. Within each loss evaluation, profiling θ_k at the proposed β reduces to computing K sample quantiles of n partial residuals, an $O(nK \log n)$ operation that accounts for less than 5% of per-iteration runtime in our experiments.

A further computational advantage emerges relative to frequentist CQR with variable selection. Penalized CQR requires solving a non-smooth optimization problem in which K check functions must be jointly minimized subject to an ℓ_1 -type penalty, and the adaptive weights and regularization parameter typically require cross-validation or information-criterion search over a solution path. As the number of regressors grows, this optimization becomes increasingly demanding: interior-point or ADMM solvers must handle nK residuals per

iteration, and warm-starting across the regularization path is complicated by the non-smoothness of the objective. By contrast, the GBCQR sampler at each MCMC iteration merely evaluates the composite loss at the proposed parameter value, a computation that is straightforward regardless of the penalty structure because no optimization over β is required within the iteration. The prior encodes the regularization directly, and the posterior automatically integrates over the penalty landscape. This evaluation-versus-optimization distinction makes the Bayesian approach especially attractive in moderate-to-high-dimensional settings where penalized CQR optimization is burdensome.

The RWMH sampler developed in the Section is deliberately simple. It requires no gradient tuning, no mass-matrix adaptation beyond the Laplace proposal covariance, and no software beyond a standard check-loss evaluation, making it immediately accessible to applied economists working with the moderate-dimensional predictor sets typical of macroeconomic forecasting. In high and ultra-high-dimensional settings where single-block random-walk proposals mix slowly, gradient-based samplers such as Hamiltonian Monte Carlo applied to a smoothed composite loss, or variational Bayes approximations, offer well-established alternatives within the generalized Bayesian framework and represent a natural next step. In the next two Sections we validate empirically our algorithm by establishing its excellent performance in a variety of settings, using synthetic and real datasets. Alternative calibration targets and formal asymptotic theory for the calibrated Gibbs posterior remain promising companion projects.

3 Monte Carlo Simulations

3.1 Design

The simulation adopts the regression design of [Zou and Yuan \(2008\)](#), with extensions to additional error distributions, sample sizes, estimation methods, and time series dependence structures. The data-generating process is

$$y_i = \alpha^* + x_i^\top \beta^* + \varepsilon_i, \quad \alpha^* = 0, \quad i = 1, \dots, n, \quad (31)$$

where $x_i = (x_{i1}, \dots, x_{i8})^\top \in \mathbb{R}^8$ denotes the covariate vector, $\beta^* \in \mathbb{R}^8$ is the true coefficient vector, and ε_i represents the error term. Following [Zou and Yuan \(2008\)](#), the true coefficient vector is

$$\beta^* = (3, 1.5, 0, 0, 2, 0, 0, 0)^\top, \quad (32)$$

which defines an active set $\mathcal{A} = \{1, 2, 5\}$ containing three non-zero coefficients and an inactive set $\mathcal{A}^c = \{3, 4, 6, 7, 8\}$ containing five zero coefficients. This canonical sparse regression setting provides coefficients of varying magnitudes for assessing both estimation and variable selection performance. An intercept is included in all estimated models; since $\alpha^* = 0$, it is excluded from reported evaluation metrics. The cross-sectional covariance matrix Σ_x follows a Toeplitz structure with $(\Sigma_x)_{jk} = \rho^{|j-k|}$ for $1 \leq j, k \leq 8$, with $\rho = 0.50$ in the baseline configuration.

The Monte Carlo experiments are organised into two groups. The first group is a cross-sectional design that considers i.i.d. covariates and i.i.d. errors, isolating the effect of error distribution shape on estimation accuracy, coverage, and variable selection. The second group

is a time series design that introduces serial dependence in both covariates and errors, validating the HAC-calibrated learning rate developed in Section 2.5 under dependence structures. Both groups share the coefficient vector (32), the Toeplitz correlation structure, and the evaluation metrics defined below. For each group, $R = 500$ Monte Carlo replications are conducted. All composite quantile regression methods aggregate information across $K = 19$ quantile levels⁸:

$$\tau = (0.05, 0.10, \dots, 0.90, 0.95)^\top. \quad (33)$$

In the first group, the covariate vector is drawn independently from a multivariate normal distribution

$$x_i \stackrel{\text{iid}}{\sim} \mathcal{N}_8(0, \Sigma_x), \quad (34)$$

and the errors ε_i are drawn independently across observations. The experiments consider a grid over sample sizes $n \in \{100, 200\}$ and correlation parameters $\rho \in \{0.25, 0.50, 0.75\}$. Five error distributions are considered, progressing from the most favourable setting for squared-error methods (Gaussian) through increasingly heavy tails (Student- t_3 , Cauchy), outlier contamination, and asymmetry.

DGP 1 (Normal) $\varepsilon_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$ with $\sigma^2 = 3$. This specification corresponds to Example 1 in Zou and Yuan (2008). Under Gaussian errors, OLS achieves the Cramér–Rao lower bound, so this DGP represents the most favourable setting for squared-error methods. Zou and Yuan (2008) show that the limiting asymptotic relative efficiency (ARE) of CQR with respect to OLS is approximately 0.96, implying a theoretical RMSE ratio $\text{RMSE}_{\text{OLS}}/\text{RMSE}_{\text{CQR}} \approx 0.98$, so that OLS retains a modest 2% RMSE advantage. This DGP therefore serves as the baseline where the efficiency cost of using CQR is at its theoretical maximum.

DGP 2 (Student- t_3) $\varepsilon_i \stackrel{\text{iid}}{\sim} t_3$. This specification corresponds to Example 4 (Corollary 3.1) in Zou and Yuan (2008). The t_3 distribution possesses finite variance $\text{Var}(\varepsilon_i) = 3$ but infinite fourth moment, so the asymptotic variance of OLS exists but standard variance estimators may exhibit substantial finite-sample instability. Zou and Yuan (2008) establish that the limiting ARE of CQR with respect to OLS is approximately 1.9, implying a theoretical RMSE ratio $\text{RMSE}_{\text{OLS}}/\text{RMSE}_{\text{CQR}} \approx 1.38$. This gain arises because the composite loss aggregates density evaluations $f(F^{-1}(\tau_k))$ that remain well-behaved in the tails of the t_3 distribution, whereas the squared-error loss assigns disproportionate weight to extreme observations.

DGP 3 (Cauchy) $\varepsilon_i \stackrel{\text{iid}}{\sim} \text{Cauchy}(0, 1)$. The Cauchy distribution possesses no finite moments: neither mean nor variance exist. Consequently, the population squared-error risk is undefined, the OLS estimator is not root- n consistent, and no finite ARE exists. As Zou and Yuan (2008) emphasize (Section 1.2 and the simulations in Section 5), this is precisely the setting that motivates CQR: the CQR estimator remains root- n consistent and asymptotically normal under the Cauchy distribution, because its asymptotic theory

⁸The Bayesian CQR model of Huang and Chen (2015) uses a smaller grid of $K = 5$, as its latent-variable augmentation scheme introduces $O(nK)$ mixture evaluations per Gibbs iteration, making $K = 19$ computationally prohibitive. The Supplementary Appendix verifies that setting $K = 5$ for all methods yields the same qualitative conclusions.

requires only continuity and positivity of the error density, not finite variance. This DGP provides the sharpest contrast between CQR-based methods (which retain valid inference) and LS-based methods, for which the RMSE diverges with n .

DGP 4 (Contaminated Normal) $\varepsilon_i \stackrel{\text{iid}}{\sim} 0.9 \cdot \mathcal{N}(0, 1) + 0.1 \cdot \mathcal{N}(0, 25)$. With probability 0.9, observations arise from a standard normal component; with probability 0.1, from a high-variance component ($\sigma^2 = 25$). The unconditional variance equals 3.4 and excess kurtosis is approximately 13.45. Scale-contaminated normal mixtures of this form are a classical test case for robustness in regression analysis. In the context of local CQR, [Kai et al. \(2010\)](#) consider a closely related contaminated normal specification and report substantial efficiency gains of local CQR over local LS (Table 1 therein). This DGP examines robustness to outlier contamination commonly encountered in economic and financial data, where the error distribution can be well-behaved for the majority of observations but occasionally generates extreme values.

DGP 5 (Skewed) $\varepsilon_i \stackrel{\text{iid}}{\sim} 0.7 \cdot \mathcal{N}(0, 1) + 0.3 \cdot \mathcal{N}(3, 4)$. This mixture generates right-skewed errors with unconditional mean 0.9 and variance 3.78. The nonzero error mean is absorbed by the intercept in OLS-based methods and by the K quantile-specific intercepts in CQR-based methods, leaving the slope coefficients β^* identified in both cases. Asymmetric error distributions are a standard setup in the quantile regression literature; for instance, [Huang and Chen \(2015\)](#) employ gamma and beta distributions to evaluate Bayesian CQR under skewness. This DGP complements DGPs 1–4 by introducing distributional asymmetry while maintaining finite moments, thereby testing whether the composite quantile loss—which, for the symmetric grid used here, places equal weight on quantile levels above and below the median—remains competitive when the error distribution is itself asymmetric.

The second group of DGPs introduces serial dependence in both covariates and errors, creating the autocorrelated composite score process that motivates the HAC-calibrated learning rate. The covariates follow a vector autoregressive process

$$x_t = \varphi_x x_{t-1} + v_t, \quad v_t \stackrel{\text{iid}}{\sim} \mathcal{N}_8(0, (1 - \varphi_x^2) \Sigma_x), \quad (35)$$

where $\varphi_x \in (0, 1)$ controls the persistence of the covariates. The innovation covariance $(1 - \varphi_x^2) \Sigma_x$ ensures that the marginal distribution of x_t coincides with $\mathcal{N}_8(0, \Sigma_x)$, so the cross-sectional covariance structure is identical to DGPs 1–5 and only the temporal dependence is new. In order to stress the effects of persistence, the baseline sets $\varphi_x = 0.85$.

Five error processes are considered, progressing from first-moment dependence (AR) through symmetric and asymmetric volatility clustering (GARCH, GJR-GARCH) and their combination with heavy-tailed innovations.

DGP 6 (AR) $\varepsilon_t = \varphi_\varepsilon \varepsilon_{t-1} + u_t$, $u_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1 - \varphi_\varepsilon^2)$, with $\varphi_\varepsilon = 0.85$. First-moment persistence in the errors induces nonzero lag autocovariances $\Gamma_h = E[\psi(\varepsilon_t)\psi(\varepsilon_{t-h})] \cdot E[x_t x_{t-h}^\top]$ because the quantile indicators $\mathbb{1}\{\varepsilon_t < \sigma F^{-1}(\tau_k)\}$ at time t depend on ε_t which itself depends on its own lags. As a result, the long-run score covariance $I_{\text{LR}} = \sum_{h=-\infty}^{\infty} \Gamma_h$ substantially

exceeds the contemporaneous covariance $I = \Gamma_0$, and the i.i.d. calibration is expected to produce severely over-concentrated posteriors.

DGP 7 (GARCH) $\varepsilon_t = \sigma_t z_t$, $z_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$, with $\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$, where $\omega = 0.05$, $\alpha_1 = 0.10$, and $\beta_1 = 0.85$. This GARCH(1,1) specification, with persistence $\alpha_1 + \beta_1 = 0.95$, generates substantial volatility clustering while preserving conditional symmetry. Unlike DGP 6, pure volatility clustering has negligible impact on the composite score autocovariance. While individual indicators $\mathbb{1}\{\varepsilon_t < c\}$ are serially dependent through σ_t —since $P(\varepsilon_t < c \mid \mathcal{F}_{t-1}) = \Phi(c/\sigma_t)$ varies with the conditional volatility—the aggregated composite score $\psi(\varepsilon_t) = \sum_k (\tau_k - \mathbb{1}\{\varepsilon_t < c_k\})$ is approximately a martingale difference sequence under the symmetric quantile grid used here: pairing τ_k with $1 - \tau_k$ yields $E[\psi(\varepsilon_t) \mid \sigma_t] \approx 0$ regardless of σ_t , so that $I_{\text{LR}} \approx I$ and the HAC correction is expected to be immaterial for coverage.

DGP 8 (AR-GARCH) $\varepsilon_t = \varphi_\varepsilon \varepsilon_{t-1} + \sigma_t z_t$, $z_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$, with $\sigma_t^2 = \omega + \alpha_1 u_{t-1}^2 + \beta_1 \sigma_{t-1}^2$ and $u_t = \varepsilon_t - \varphi_\varepsilon \varepsilon_{t-1}$, using $\varphi_\varepsilon = 0.85$ and GARCH parameters as in DGP 7. This combines the first-moment persistence of DGP 6 with the volatility clustering of DGP 7, following the AR-GARCH specification.

DGP 9 (AR-GARCH- t) Same specification as DGP 8, but with $z_t \stackrel{\text{iid}}{\sim} t_\nu / \sqrt{\nu/(\nu-2)}$, $\nu = 5$, so that z_t has unit variance. The Student- t_5 innovation introduces excess kurtosis into the conditional distribution. Under heavy-tailed innovations, the unconditional distribution of ε_t inherits both the kurtosis of z_t and the variance amplification from GARCH persistence.

DGP 10 (AR-GJR- t) $\varepsilon_t = \varphi_\varepsilon \varepsilon_{t-1} + \sigma_t z_t$, $z_t \stackrel{\text{iid}}{\sim} t_5 / \sqrt{5/3}$, with

$$\sigma_t^2 = \omega + \alpha_1 u_{t-1}^2 + \gamma u_{t-1}^2 \mathbb{1}\{u_{t-1} < 0\} + \beta_1 \sigma_{t-1}^2, \quad (36)$$

where $\alpha_1 = 0.05$, $\gamma = 0.10$, $\beta_1 = 0.85$, and $u_t = \varepsilon_t - \varphi_\varepsilon \varepsilon_{t-1}$. This is the GJR-GARCH specification of [Glosten et al. \(1993\)](#), which captures the asymmetric volatility response (leverage effect) widely documented in equity returns: negative shocks ($u_{t-1} < 0$) contribute $\alpha_1 + \gamma = 0.15$ to σ_t^2 , while positive shocks contribute only $\alpha_1 = 0.05$.

The baseline time series configuration sets $n = 200$, $\rho = 0.50$, $\varphi_x = 0.85$, and $\varphi_\varepsilon = 0.85$ (for DGPs with AR dynamics). Results for additional configurations ($n \in 100, 200$, and $\varphi_\varepsilon \in \{0.25, 0.50, 0.85\}$) are reported in the Supplementary Appendix.

For each slope coefficient β_j^* , $j = 1, \dots, 8$, the following metrics are computed across the $R = 500$ replications. Let $\hat{\beta}_j^{(r)}$ denote the point estimate from replication r , and let $[\hat{\beta}_{j,L}^{(r)}, \hat{\beta}_{j,U}^{(r)}]$ denote the corresponding $(1 - \alpha)$ interval with $\alpha = 0.05$. The *root mean squared error* is

$$\text{RMSE}_j = \sqrt{\frac{1}{R} \sum_{r=1}^R (\hat{\beta}_j^{(r)} - \beta_j^*)^2}. \quad (37)$$

The *empirical coverage rate* is

$$\text{Coverage}_j = \frac{1}{R} \sum_{r=1}^R \mathbb{1}\{\hat{\beta}_{j,L}^{(r)} \leq \beta_j^* \leq \hat{\beta}_{j,U}^{(r)}\}. \quad (38)$$

Under correct calibration, the nominal 95% coverage should be attained. Summary statistics are reported as averages across the eight slope coefficients⁹.

For variable selection methods, performance is assessed using two metrics computed relative to the true active set $\mathcal{A}$ and inactive set $\mathcal{A}^c$. Let $\hat{\mathcal{A}}^{(r)}$ denote the set of variables selected in replication r . The *true positive rate* (TPR) measures the proportion of active coefficients correctly identified:

$$\text{TPR} = \frac{1}{R} \sum_{r=1}^R \frac{|\hat{\mathcal{A}}^{(r)} \cap \mathcal{A}|}{|\mathcal{A}|}. \quad (39)$$

The *true negative rate* (TNR) measures the proportion of inactive coefficients correctly excluded:

$$\text{TNR} = \frac{1}{R} \sum_{r=1}^R \frac{|(\hat{\mathcal{A}}^{(r)})^c \cap \mathcal{A}^c|}{|\mathcal{A}^c|}. \quad (40)$$

The Monte Carlo simulation compares methods grouped into three categories: Generalized Bayesian methods based on the composite quantile loss (GBCQR-Flat, GBCQR-Normal, GBCQR-SSL), frequentist composite quantile regression estimators (CQR, CQR-LASSO), and likelihood-based Bayesian methods (BR, BRT, BR-SSL, BCQR-ALD). All Bayesian methods use 55,000 total MCMC iterations with a burn-in of 5,000 and thinning by 5, yielding $M = 10,000$ posterior draws. The likelihood-based methods (BR, BRT, BR-SSL, BCQR-ALD) use Gibbs sampling with conjugate full conditionals.

Generalized Bayesian Methods. In all three GBCQR specifications, scale is estimated as $\hat{\sigma} = \text{MAD}(\hat{\varepsilon}_i)/0.675$ from the median CQR residuals, and the loss Hessian $\hat{J}_n$ and score covariance $\hat{I}_n$ are computed at the relevant frequentist point estimate, with $\hat{J}_n$ evaluated using a Gaussian kernel with Silverman’s rule-of-thumb bandwidth $h = 0.9 \hat{\sigma} n^{-1/5}$. The constant 0.9 is calibrated for Gaussian shape, but the coverage results in Tables 1 and 2 are stable across five error distributions with very different tail profiles, providing indirect evidence of robustness to the bandwidth choice. The three methods employ the adaptive random-walk Metropolis–Hastings sampler of Section 2.6.

1. GBCQR-Flat. The flat-prior specification sets $\pi(\beta) \propto 1$. The learning rate admits the closed-form solution (14), with the frequentist target given by the unpenalised CQR sandwich (13), both evaluated at the CQR point estimate $\hat{\beta}_n$.

2. GBCQR-Normal. The normal (ridge) prior sets $\beta \sim \mathcal{N}(\mathbf{0}, \tau^2 I_p)$ with diffuse variance $\tau^2 = 100$. The learning rate is obtained by golden-section search on objective (11), with the Laplace covariance (17) and the ridge-CQR sandwich target (18), both evaluated at $\hat{\beta}_n^R$.

⁹The Supplementary Appendix reports results separately for active and inactive coefficients.

3. GBCQR-SSL. The spike-and-slab LASSO prior (20) of Ročková and George (2018) is placed on the slope coefficients with $\lambda_0 = \sqrt{n}$ (spike), $\lambda_1 = 1$ (slab), and $\theta = 1/p$. The learning rate is obtained by golden-section search on objective (11), with the Laplace covariance (21) and the adaptive CQR-LASSO sandwich target (23), both evaluated at the adaptive CQR-LASSO estimate $\hat{\beta}_n^A$. At each MCMC iteration, the inclusion indicators $\{\gamma_j\}_{j=1}^p$ are updated via the Gibbs step (30), and variable selection is based on $\text{PIP}_j > 0.5$, with PIP_j computed as in Section 2.6.

The simulations also consider GBCQR specifications fixing $\eta = 1$ (no calibration). With the diffuse prior variance $\tau^2 = 100$ used here, the dominant effect of calibration is the correction of posterior dispersion rather than posterior location (see Section 2.4), so these ablations isolate the coverage cost of ignoring the learning-rate problem. For the time series designs, each GBCQR specification is additionally estimated with the HAC-calibrated learning rate. The HAC estimator uses the quadratic spectral kernel with the data-driven bandwidth of Andrews (1991).

Frequentist Methods

4. CQR. The frequentist CQR estimator of Zou and Yuan (2008) is computed via the MM algorithm with profiled quantile-specific intercepts. Two variants are reported: CQR with i.i.d. sandwich standard errors (CQR) and CQR with HAC sandwich standard errors (CQR-HAC), using the same quadratic spectral kernel as the GBCQR-HAC variants. The HAC-based frequentist intervals provide the natural benchmark against which the HAC-calibrated Gibbs posterior is assessed.

5. CQR-LASSO. The adaptive CQR-LASSO estimator solves

$$\hat{\beta}_{\text{ACQR}} = \arg \min_{\beta} \sum_{k=1}^K \sum_{i=1}^n \rho_{\tau_k}(y_i - \hat{\alpha}_{\tau_k}(\beta) - \mathbf{x}_i' \beta) + \lambda_n \sum_{j=1}^p w_j |\beta_j|, \quad (41)$$

where $w_j = |\hat{\beta}_j^{\text{init}}|^{-\zeta}$ are adaptive weights constructed from an unpenalised CQR initial estimate (Zou and Yuan, 2008), with $\zeta = 2$. The regularisation parameter λ_n is selected by BIC. Coefficient j is classified as active if $|\hat{\beta}_j| > 10^{-4}$, and this active-set indicator is the basis for the TPR and TNR reported in Tables 1 and 2.

Likelihood-Based Bayesian Methods

6. BR. Bayesian linear regression with conjugate priors: $(\alpha, \beta')' \sim \mathcal{N}(\mathbf{0}, \tau^2 I_{p+1})$ with $\tau^2 = 100$ and $\sigma^2 \sim \text{IG}(a_0, b_0)$ with $a_0 = b_0 = 0.01$. The Gibbs sampler proceeds with conjugate normal and inverse-gamma updates.

7. BRT. Bayesian linear regression with Student- t_ν errors, using a scale-mixture-of-normals representation:

$$y_i \mid \lambda_i \sim \mathcal{N}(\alpha + \mathbf{x}_i' \beta, \sigma^2 / \lambda_i), \quad \lambda_i \sim \text{Gamma}(\nu/2, \nu/2), \quad (42)$$

where λ_i is an observation-specific precision weight. Marginalising over λ_i recovers a Student- $t_\nu(\alpha + \mathbf{x}'_i\beta, \sigma^2)$ likelihood for y_i . The priors are: $(\alpha, \beta')' \sim \mathcal{N}(\mathbf{0}, 100 I_{p+1})$ and $\sigma^2 \sim \text{IG}(0.01, 0.01)$. The Gibbs sampler augments the standard conjugate updates with a draw of $\lambda_i \mid \cdot \sim \text{Gamma}((\nu + 1)/2, (\nu + (y_i - \alpha - \mathbf{x}'_i\beta)^2/\sigma^2)/2)$ for each observation. The degrees of freedom are set to $\nu = 5$ throughout.

8. BR-SSL. Bayesian linear regression with the canonical spike-and-slab LASSO prior of Ročková and George (2018). The prior hierarchy couples the Laplace penalty to the error variance: $\beta_j \mid \gamma_j, \sigma^2 \sim \text{Laplace}(0, \sigma/\lambda_{\gamma_j})$, where $\lambda_0 = \sqrt{n}$ (spike) and $\lambda_1 = 1$ (slab), $\gamma_j \mid \theta \sim \text{Bernoulli}(\theta)$ with $\theta = 1/p$, and $\sigma^2 \sim \text{IG}(0.01, 0.01)$. This σ^2 -coupled parametrisation follows the canonical specification; it differs from the GBCQR-SSL prior (20), which is independent of any scale parameter. Selection is based on $\text{PIP}_j > 0.5$, computed as in Section 2.6.

9. BCQR-ALD. The Bayesian composite quantile regression method of Huang and Chen (2015) models the conditional density as a finite mixture of asymmetric Laplace distributions (ALDs):

$$f(y_i \mid \beta, \mathbf{b}, \sigma, \boldsymbol{\pi}) = \sum_{k=1}^K \pi_k \frac{\tau_k(1 - \tau_k)}{\sigma} \exp\left(-\frac{\rho_{\tau_k}(y_i - b_k - \mathbf{x}'_i\beta)}{\sigma}\right), \quad (43)$$

where b_k is the τ_k -specific intercept, $\sigma > 0$ is a common scale, and $\boldsymbol{\pi} = (\pi_1, \dots, \pi_K)'$ are mixture weights. Each ALD component admits a normal-exponential scale mixture representation (see Kozumi and Kobayashi, 2011): conditional on a latent indicator $z_i \in \{1, \dots, K\}$ and an exponential mixing variable $t_i \sim \text{Exp}(\sigma)$,

$$y_i \mid z_i=k, t_i \sim \mathcal{N}(b_k + \mathbf{x}'_i\beta + \xi_{1,k} t_i, \sigma \xi_{2,k}^2 t_i), \quad (44)$$

with $\xi_{1,k} = (1 - 2\tau_k)/[\tau_k(1 - \tau_k)]$ and $\xi_{2,k} = \sqrt{2/[\tau_k(1 - \tau_k)]}$. This hierarchical form yields conjugate full conditionals for all parameters, enabling a fully Gibbs sampler. The priors are: $\beta \sim \mathcal{N}(\mathbf{0}, 100 I_p)$, $\sigma \sim \text{IG}(0.01, 0.01)$, and $\boldsymbol{\pi} \sim \text{Dirichlet}(\mathbf{1}_K)$. The mixture weights receive conjugate Dirichlet-multinomial updates: $\boldsymbol{\pi} \mid \mathbf{z} \sim \text{Dirichlet}(1 + n_1, \dots, 1 + n_K)$ with $n_k = \#\{i : z_i = k\}$.

3.2 Monte Carlo Results

The Monte Carlo results support three main conclusions. First, learning-rate calibration is essential for valid uncertainty quantification: across the cross-sectional DGPs, calibrated GBCQR delivers empirical coverage close to the nominal 95% level, whereas the uncalibrated variant ($\eta = 1$) leads to severe undercoverage around 56%. Second, under Gaussian errors GBCQR incurs at most a small efficiency cost relative to Gaussian Bayesian regression, but under heavy tails it matches or improves upon frequentist CQR and sharply outperforms Gaussian likelihood-based methods, with the largest gains under Cauchy errors. Third, the SSL version of GBCQR achieves reliable variable selection across all DGPs, including cases where Gaussian Bayesian spike-and-slab regression fails because extreme residuals distort the

variance-driven shrinkage mechanism.

Table 1 presents Monte Carlo results for the cross-sectional designs with $n = 200$ and $\rho = 0.50$. Because we are using a diffuse prior for GBCQR-Normal, results are very similar to GBCQR-Flat. To save space, we report results for GBCQR-Flat in the Supplementary Appendix.

Panel A reports RMSE relative to frequentist CQR, so that values below unity indicate superior accuracy¹⁰. Under the Gaussian DGP, BR achieves the lowest RMSE ratio (0.983), consistent with the ARE of approximately 0.96 established in Zou and Yuan (2008), which implies a theoretical RMSE ratio $\text{RMSE}_{\text{OLS}}/\text{RMSE}_{\text{CQR}} \approx 0.98$, a modest 2% efficiency advantage for OLS.

This ordering reverses sharply under departures from normality. With the sole exception of the Gaussian DGP, GBCQR-Normal exhibits lower RMSE than both BR and frequentist CQR. For the Student- t_3 distribution, BR deteriorates to a RMSE ratio of 1.31. This is consistent with Zou and Yuan (2008), who establish a limiting ARE of approximately 1.9 for CQR over OLS under t_3 errors, implying a theoretical RMSE ratio $\text{RMSE}_{\text{OLS}}/\text{RMSE}_{\text{CQR}} \approx 1.38$. GBCQR also improves on frequentist CQR, although these gains are small (approximately 3%). Moreover, these gains become substantially more pronounced under the Cauchy DGP. As a consequence of the squared-error loss, BR exhibits catastrophic deterioration, with RMSE exceeding ten times that of frequentist CQR. GBCQR methods achieve relative RMSEs of approximately 0.88, improving upon frequentist CQR itself by a substantial margin. The BCQR-ALD achieves the lowest RMSE ratio under the Cauchy DGP, outperforming all other methods. This advantage may stem from the ALD mixture’s adaptive allocation of observations to quantile components via latent indicators, which implicitly concentrates posterior mass on quantile levels most informative for the data at hand.

More broadly, both GBCQR and BCQR-ALD outperform frequentist CQR in most heavy-tailed DGPs. A contributing mechanism is the geometry of the composite quantile loss surface under heavy tails. When only a small fraction of observations lie near any given conditional quantile, the loss surface becomes relatively flat around its minimum. Frequentist optimization selects a single point on this plateau, whereas posterior sampling averages over a set of nearly equivalent minimizers which can improve finite-sample stability. This posterior averaging provides finite-sample stabilization beyond the robustness already conferred by the check loss itself. That said, the BCQR-ALD’s point estimation advantage under Cauchy errors comes at a cost: it requires roughly 100× the computation of GBCQR (Panel C) and its coverage is not uniformly better (Panel B shows BCQR-ALD undercovers at 90.7% under Gaussian errors, while GBCQR tracks the nominal level closely throughout). Two additional comparisons in Panel A deserve comment. First, the uncalibrated variant GBCQR-Normal($\eta=1$) achieves RMSE ratios comparable to its calibrated counterpart, because with the diffuse prior variance $\tau^2 = 100$ used here, η primarily affects posterior dispersion rather than the posterior mean. Combined with the coverage results in Panel B, this confirms that calibration corrects posterior dispersion, not posterior location. Second, BRT achieves

¹⁰The Supplementary Appendix reports RMSE results separately for active and inactive coefficients; the differences are negligible.

competitive RMSE (average 0.962), outperforming BR under all non-Gaussian DGPs and confirming the value of accommodating heavy tails through the Student- t likelihood. However, BRT deteriorates under the Skewed DGP (1.018) because the Student- t likelihood assumes symmetric errors.

Panel B reports empirical coverage of nominal 95% credible and confidence intervals. GBCQR-Normal achieves empirical coverage between 94.5% and 95.7% across all five DGPs, closely tracking the nominal level. Given 500 replications, the Monte Carlo standard error for a coverage rate near 0.95 is approximately one percentage point, so differences of this magnitude should not be over-interpreted; what matters is that GBCQR coverage is consistently within the range expected under correct calibration. As expected, GBCQR-Normal_($\eta=1$) exhibits severe undercoverage (around 56%), confirming that the learning rate is essential for valid uncertainty quantification even though it has only a minor effect on point estimation. This comparison highlights a practical message: if the researcher needs only point forecasts (posterior means), fixing $\eta = 1$ is a reasonable and computationally convenient strategy that avoids calibration entirely; if credible intervals, posterior inclusion probabilities, or any other measure of posterior uncertainty is needed, calibration is indispensable.

Panel C reports computation time relative to BR. To put the composite-quantile methods on a common footing with BCQR-ALD—whose latent-variable augmentation makes $K = 19$ computationally prohibitive—all CQR methods in Panel C are re-run at $K = 5$. The timings therefore reflect a standardized comparison grid, not the operational cost of each method as deployed in Panels A, B, and D. At $K = 5$, GBCQR-Normal requires roughly 1.2 times the computation of BR, BRT requires approximately 2.9 times (reflecting the additional Gibbs step for the n latent scale variables λ_i), and BCQR-ALD requires approximately 124 $\times$ more computation, driven by the cost of evaluating mixture probabilities and sampling $2n + K$ latent quantities per MCMC iteration. The Supplementary Appendix reports effective sample size (ESS) per second for all methods. Under the baseline design ($n = 200$, $\rho = 0.50$), GBCQR achieves a minimum ESS/sec of approximately 830–905 across all DGPs, compared to 1.9–7.7 for BCQR-ALD, confirming that the timing advantage translates into a comparable gain in effective posterior samples.

Panel D reports variable selection performance via true positive and true negative rates. GBCQR-SSL achieves near-perfect selection across all DGPs, correctly identifying all active coefficients (TPR = 100%) while excluding inactive predictors with high accuracy (TNR = 95–100%). CQR-LASSO performs comparably, with perfect TNR and TPR ranging from 93% to 100%. Under the Cauchy DGP, the TPR of BR-SSL collapses to 3%, a near-complete failure to detect true signals. This pathology arises from the σ^2 -coupled prior parametrisation: extreme residuals inflate the posterior estimate of σ^2 , which propagates through the hierarchical prior to inflate the slab variance, systematically shrinking all coefficients toward zero and preventing any variable from exceeding the inclusion threshold.

Table 1: Monte Carlo Results

Method	Normal	t_3	Contam.	Cauchy	Skewed	Avg.
<i>Panel A: RMSE (all coefficients, relative to CQR)</i>						
GBCQR-Normal	1.011	0.968	0.949	0.881	0.956	0.953
GBCQR-Normal _($\eta=1$)	1.013	0.966	0.948	0.853	0.944	0.945
BCQR-ALD	1.105	1.006	0.961	0.708	0.851	0.926
BRT	1.027	0.957	0.918	0.890	1.018	0.962
BR	0.983	1.309	1.471	11.290	1.128	3.236
<i>Panel B: Coverage of 95% Intervals</i>						
GBCQR-Normal	0.945	0.952	0.953	0.957	0.955	0.952
GBCQR-Normal _($\eta=1$)	0.563	0.566	0.561	0.558	0.561	0.562
BCQR-ALD	0.907	0.934	0.945	0.976	0.926	0.938
CQR	0.942	0.951	0.952	0.953	0.946	0.949
BRT	0.936	0.960	0.965	0.990	0.955	0.961
BR	0.944	0.951	0.954	0.959	0.951	0.952
<i>Panel C: Computation Time (relative to BR, $K = 5$)</i>						
GBCQR-Normal	1.22	1.22	1.22	1.23	1.22	1.22
BCQR-ALD	124.45	124.38	124.56	123.62	124.34	124.27
BRT	2.91	2.91	2.91	2.91	2.91	2.91
<i>Panel D: Variable Selection (TPR, TNR)</i>						
GBCQR-SSL	(100, 96)	(100, 99)	(100, 100)	(100, 95)	(100, 98)	(100, 98)
BR-SSL	(100, 100)	(100, 100)	(100, 100)	(3, 100)	(100, 100)	(81, 100)
CQR-LASSO	(100, 100)	(100, 100)	(100, 100)	(93, 100)	(100, 100)	(99, 100)

Notes: Results based on 500 Monte Carlo replications and a sample size of $n = 200$. Design follows [Zou and Yuan \(2008\)](#): $p = 8$ covariates with Toeplitz correlation $\Sigma_{jk} = \rho^{|j-k|}$, where $\rho = 0.50$, and true coefficients $\beta^* = (3, 1.5, 0, 0, 2, 0, 0, 0)^\top$; active set $\mathcal{A} = \{1, 2, 5\}$, inactive set $\mathcal{A}^c = \{3, 4, 6, 7, 8\}$. Panels A, B, and D use $K = 19$ equally spaced quantile levels for all CQR methods, except BCQR-ALD, which uses $K = 5$ throughout because $K = 19$ is computationally infeasible for its latent-variable augmentation. Panel C separately re-runs all CQR methods at $K = 5$ to report computation times on a common grid. RMSE panels report values relative to frequentist CQR; bold entries indicate values below unity (superior accuracy). Coverage panels report empirical coverage of nominal 95% credible/confidence intervals. “ $\eta = 1$ ” denotes GBCQR variants with the learning rate fixed at unity (no calibration). TPR: fraction of active coefficients correctly identified; TNR: fraction of inactive coefficients correctly excluded. GBCQR-SSL and BR-SSL select via $\text{PIP}_j > 0.5$. GBCQR: Generalized Bayesian Composite Quantile Regression; BCQR-ALD: mixture of Asymmetric Laplace likelihoods ([Huang and Chen, 2015](#)); BR: Gaussian likelihood with diffuse normal prior; BRT: Gaussian likelihood with Student- t errors; SSL: spike-and-slab LASSO prior ([Ročková and George, 2018](#)).

Table 2 presents Monte Carlo results for the time series designs with $n = 200$, $\rho = 0.50$, and strong persistence of $\varphi_x = \varphi_\varepsilon = 0.85$. Again, results for GBCQR-Flat are quantitatively very similar to GBCQR-Normal, and they can be found in the Supplementary Appendix. Panel A indicates that GBCQR and BRT achieve RMSE ratios close to unity, with differences across methods that are generally negligible. BR exhibits again the worst average performance (1.026).

Coverage is the main result from the time series designs. Under the four DGPs with AR dynamics (DGPs 6, 8, 9, 10), the i.i.d.-calibrated GBCQR-Normal achieves an average coverage of approximately 58%. This under-coverage arises because the score covariance $\hat{I}_n$ underestimates the true long-run variance $I_{\text{LR}} = \sum_{h=-\infty}^{\infty} \Gamma_h$ when both covariates and errors exhibit positive autocorrelation. With $\varphi_x = \varphi_\varepsilon = 0.85$, both the error score factor $E[\psi(\varepsilon_t)\psi(\varepsilon_{t-h})]$ and the regressor auto-covariance $E[x_t x_{t-h}^\top]$ are positive for all $h \geq 1$, so that $I_{\text{LR}} \succ I$ and the i.i.d. calibration produces an η that is too large, yielding an over-concentrated posterior. The HAC estimator restores substantial coverage. GBCQR-Normal (HAC) achieves coverage of 82–93% under the five DGPs, representing an average improvement of roughly 20 percentage points. While this coverage remains below the

nominal 95%, the shortfall is not exclusive to the GBCQR calibration strategy: frequentist CQR with HAC standard errors (CQR HAC) achieves comparable coverage of 83–93% under the same DGPs. This parallel suggests that the residual under-coverage is attributable to the finite-sample bias of HAC estimators, rather than to a deficiency in the calibration itself.

To isolate the source of the remaining under-coverage, DGP 7 introduces conditional heteroskedasticity without first-moment dependence. While individual indicators $\mathbb{1}\{\varepsilon_t < c\}$ are serially dependent through σ_t under GARCH dynamics, the aggregated composite score $\psi(\varepsilon_t) = \sum_k (\tau_k - \mathbb{1}\{\varepsilon_t < c_k\})$ is approximately a martingale difference sequence under the symmetric quantile grid, so that $I_{LR} \approx I$ and the i.i.d. calibration should suffice. The results confirm this: GBCQR-Normal achieves 94.0% coverage under the GARCH DGP, closely tracking the nominal level, while the HAC-calibrated variant achieves a marginally lower 92.9%. The near-nominal coverage under DGP 7 confirms that the diagonal variance matching is not the binding constraint on coverage. The parallel under-coverage of both GBCQR (HAC) and CQR (HAC) under AR dynamics identifies the finite-sample bias of HAC estimators as the common source. Improved HAC implementations (for example, pre-whitening or small-sample bandwidth corrections) could narrow this gap and represent a useful avenue for future work; however, such refinements would benefit frequentist and generalized Bayesian inference equally and are not specific to the calibration strategy proposed here. The likelihood-based Bayesian benchmarks are not equipped with a time-series correction in this simulation, so in the dependent designs they serve primarily as RMSE benchmarks rather than credible-interval benchmarks.

The median calibrated learning rates corroborate the coverage findings. Under the i.i.d. calibration, the median $\hat{\eta}$ ranges from 0.160 to 0.170 across DGPs. Under the HAC calibration, $\hat{\eta}$ drops to approximately one-third of these values for the four AR-containing DGPs: 0.059–0.063. This reduction is a direct consequence of the inflated long-run score variance $I_{LR} \succ I$: a larger frequentist target requires a smaller learning rate to match the increased sampling variability, producing wider credible intervals. For the pure GARCH DGP, the HAC-calibrated $\hat{\eta}$ (0.178) is comparable to its i.i.d. counterpart (0.166), confirming that the HAC estimator correctly detects the absence of score-level serial dependence.

Panel D reports variable-selection performance in the time-series designs. While the ideal benchmark is that both TPR and TNR be close to 100%, in forecasting applications preserving a high TPR is often the more important criterion, since omitting genuinely predictive variables may be more costly than retaining a small number of irrelevant ones. The results show that GBCQR-SSL maintains essentially perfect TPR across all DGPs, with HAC calibration having negligible effect on selection. TNR is lower, particularly in the AR-containing designs, indicating some over-selection among inactive but persistent regressors. This pattern suggests that under serial dependence the method is deliberately conservative about discarding potential signals, a feature that is arguably desirable in forecasting applications where false exclusions may be more costly than modest over-inclusion.

Table 2: Monte Carlo results: Time Series

Method	AR(1)	GARCH	AR-GARCH	AR-GARCH- t	AR-GJR- t	Average
<i>Panel A: RMSE relative to CQR</i>						
GBCQR-Normal	1.004	1.012	0.995	0.983	0.986	0.996
GBCQR-Normal (HAC)	1.002	1.013	0.995	0.988	0.991	0.998
GBCQR-Normal ($\eta = 1$)	1.005	1.016	0.996	0.981	0.985	0.997
BR	0.996	0.997	1.015	1.071	1.052	1.026
BRT	1.011	1.027	0.997	0.978	0.984	0.999
BCQR-ALD	1.012	1.132	1.003	0.946	0.952	1.009
<i>Panel B: Empirical coverage (95%)</i>						
GBCQR-Normal	0.587	0.940	0.577	0.587	0.583	0.655
GBCQR-Normal (HAC)	0.822	0.929	0.820	0.835	0.827	0.847
GBCQR-Normal ($\eta = 1$)	0.258	0.567	0.254	0.259	0.256	0.319
CQR	0.579	0.935	0.582	0.592	0.584	0.654
CQR (HAC)	0.829	0.925	0.839	0.851	0.835	0.856
BR	0.592	0.952	0.577	0.590	0.596	0.661
BRT	0.581	0.935	0.572	0.590	0.590	0.654
BCQR-ALD	0.571	0.903	0.562	0.589	0.587	0.643
<i>Panel C: Median learning rate $\hat{\eta}$</i>						
GBCQR-Normal	0.170	0.166	0.163	0.160	0.161	0.164
GBCQR-Normal (HAC)	0.063	0.178	0.060	0.060	0.059	0.084
<i>Panel D: Variable selection (TPR%, TNR%)</i>						
GBCQR-SSL	(100.0,88.1)	(100.0,99.6)	(99.9,62.7)	(99.9,68.5)	(99.8,67.5)	(99.9,77.3)
GBCQR-SSL (HAC)	(100.0,88.4)	(100.0,99.6)	(99.9,63.0)	(99.9,68.8)	(99.8,67.5)	(99.9,77.4)
BR-SSL	(100.0,84.4)	(100.0,99.8)	(99.6,88.8)	(98.5,89.4)	(98.7,87.4)	(99.4,90.0)
CQR-LASSO	(100.0,93.4)	(99.9,99.1)	(96.5,85.8)	(96.2,88.1)	(96.7,86.2)	(97.9,90.5)

Notes: Results are based on 500 Monte Carlo replications and a sample size of $n = 200$. Covariates follow $x_t = \phi_x x_{t-1} + v_t$ with $\phi_x = 0.85$ and a Toeplitz cross-sectional correlation $\rho = 0.5$. True coefficients are $\beta^* = [3, 1.5, 0, 0, 2, 0, 0, 0]'$ with active set $\{1, 2, 5\}$. All GBCQR and CQR methods use $K = 19$ quantile levels $\tau \in \{0.05, 0.10, \dots, 0.95\}$; BCQR-ALD uses a coarser grid of $K = 5$. MCMC samplers run 50,000 iterations after a burn-in of 5,000, thinned by 5. Panel A reports RMSE relative to CQR; values below one (in bold) indicate improvement. Panel B reports average empirical coverage of 95% credible/confidence intervals. Panel C reports the median calibrated learning rate $\hat{\eta}$ across replications; HAC uses the quadratic spectral kernel (Andrews (1991)). Panel D reports the true positive rate (TPR) and true negative rate (TNR) for variable selection based on median-probability thresholding (PIP > 0.5) for Bayesian methods. DGP error specifications: AR(1) with $\phi_\varepsilon = 0.85$; GARCH is GARCH(1,1) with Normal innovations ($\omega = 0.05$, $\alpha_1 = 0.10$, $\beta_1 = 0.85$); AR-GARCH and AR-GARCH- t_5 combine the AR(1) and GARCH structures; AR-GJR- t_5 adds asymmetric volatility ($\alpha_1 = 0.05$, $\gamma = 0.10$, $\beta_1 = 0.85$).

4 Empirical Application: Equity Premium Forecasting

This section applies the GBCQR framework to quarterly equity premium forecasting using the updated predictor dataset of Goyal et al. (2024), which extends the original Goyal and Welch (2008) database with additional variables and recent observations through 2024Q4. The application exploits the efficiency and robustness properties of CQR under the common-slope model: the composite quantile loss provides a more stable estimator of the conditional location than OLS when the error distribution has heavy tails, and the Bayesian framework further reduces forecast variability through posterior averaging. The dependent variable is the log excess return on the S&P 500 index. The predictor set comprises 31 series spanning valuation ratios, interest rates, macroeconomic indicators, variance and risk measures, and cross-sectional and technical indicators; Table 3 summarizes the full list and data availability¹¹. Each predictor is evaluated individually using an expanding-window recursive scheme, in which all available historical observations up to each forecast origin are used for estimation.

¹¹We excluded from the analysis all predictors that have look-ahead bias or full-sample dependence.

Two out-of-sample evaluation windows are considered. The baseline window presented here covers 1994Q4–2024Q4 (30 years), providing an extensive assessment of forecast performance across different market regimes. Under this evaluation window, 27 predictors enter with at least 50 estimation observations. Four predictors have shorter initial estimation samples: *crstd* (17 observations), *vp* and *vrp* (18 each), while *impvar* has no available data at the evaluation start (its effective sample begins in 1996Q2). Whenever the expanding estimation window contains fewer than 20 observations for a given predictor at a particular forecast origin, no forecast is produced for that period; the predictor effectively enters the evaluation window once sufficient estimation data have accumulated. A second evaluation window presented in the Supplementary Appendix covers 2004Q4–2024Q4 (20 years). In this configuration, 30 of the 31 predictors enter the evaluation period with at least 50 quarterly estimation observations; the sole exception is implied variance (34 estimation observations at the first forecast origin). Three variance-based predictors whose samples end before 2024Q4 deliver fewer than 80 out-of-sample forecasts: *vp* (69 forecasts, through 2021Q4), *impvar* (75, through 2023Q2), and *vrp* (77, through 2023Q4).

Following [Goyal and Welch \(2008\)](#), the forecasting exercise estimates univariate predictive regressions for the one-quarter-ahead equity premium

$$r_{t+1} = \alpha + \beta x_t + \varepsilon_{t+1},$$

where r_{t+1} denotes the equity premium and x_t is the lagged predictor. Parameters are re-estimated recursively at each forecast origin using an expanding window of all data available up to quarter t .

Five forecasting methods are compared: (i) the historical mean (HM), which serves as a standard no-predictability benchmark; (ii) ordinary least squares (OLS); (iii) Bayesian regression with Student- t errors and $\nu = 5$ degrees of freedom (BRT), which is a flexible alternative that accommodates fat-tailed return innovations; (iv) GBCQR with flat prior and calibrated learning rate (GB); and (v) GBCQR with flat prior and fixed learning rate $\eta = 1$ (GB_f)¹². The flat prior is used because the univariate predictive regressions involve only a single slope coefficient, making shrinkage unnecessary. Both GBCQR methods use the HAC-calibrated learning rate from Section 2.5, with the quadratic spectral kernel and the data-driven bandwidth of [Andrews \(1991\)](#), to account for serial dependence in the composite score¹³. The quantile grid is restricted to $\boldsymbol{\tau} = (0.10, 0.20, \dots, 0.90)^\top$, a coarser choice than the $K = 19$ grid used in the Monte Carlo, because the expanding estimation windows at early forecast origins contain very few observations for some variables, and tail levels such as $\tau = 0.05$ cannot be reliably identified from samples of this size—they contribute noise rather than signal to $\hat{J}_n$ and $\hat{I}_n$.

Predictors are standardized within each expanding estimation window before computing the calibrated learning rate. For all methods except HM, the point forecast is constructed as $\hat{r}_{t+1|t} = \hat{\alpha} + \hat{\beta}x_t$, where $(\hat{\alpha}, \hat{\beta})$ denote posterior means for the Bayesian methods, and least-

¹²The Supplementary Appendix also reports results for frequentist CQR. Its performance is similar to calibrated GBCQR, with the latter holding a small but consistent edge.

¹³The Supplementary Appendix compares HAC and non-HAC estimates of η across expanding windows for each predictor. Because equity premium residuals exhibit little serial correlation, the two estimates are close for most predictors; however, both are substantially below one.

Table 3: Predictor Variables

Variable	Description	Starting Date	Ending Date
<i>Panel A: Valuation Ratios</i>			
d/p	Log dividend-price ratio	1926Q1	2024Q4
d/y	Log dividend yield	1926Q1	2024Q4
e/p	Log earnings-price ratio	1926Q1	2024Q4
d/e	Log dividend-payout ratio	1926Q1	2024Q4
b/m	Book-to-market ratio	1926Q1	2024Q4
<i>Panel B: Interest Rates and Spreads</i>			
tbl	Treasury bill rate	1926Q1	2024Q4
lty	Long-term government bond yield	1926Q1	2024Q4
ltr	Long-term government bond return	1926Q1	2024Q4
tms	Term spread (lty – tbl)	1926Q1	2024Q4
dfy	Default yield spread (BAA – AAA)	1926Q1	2024Q4
dfr	Default return spread	1926Q1	2024Q4
crstd	Credit standard deviation	1990Q3	2024Q4
<i>Panel C: Macroeconomic Variables</i>			
infl	Inflation rate	1926Q1	2024Q4
ntis	Net equity issuance	1927Q1	2024Q4
svar	Stock return variance	1926Q1	2024Q4
i/k	Investment-to-capital ratio	1947Q2	2024Q4
govik	Government investment-to-capital	1947Q2	2024Q4
ndrbl	Net durable consumption growth	1958Q2	2024Q4
wtexas	West Texas oil price growth	1926Q2	2024Q4
<i>Panel D: Variance and Risk Measures</i>			
vp	Variance premium	1990Q2	2021Q4*
impvar	Implied variance	1996Q2	2023Q2*
vrp	Variance risk premium	1990Q2	2023Q4*
tail	Tail risk measure	1926Q1	2024Q4
skvw	Average stock skewness	1926Q1	2024Q4
<i>Panel E: Cross-Section and Technical</i>			
lzrt	Illiquidity measure	1926Q1	2024Q4
avgor	Average stock correlation	1926Q1	2024Q4
dtoy	Distance to Dow 52-week high	1926Q1	2024Q4
dtoat	Distance to Dow all-time high	1926Q1	2024Q4
rdsp	Stock return dispersion	1926Q4	2024Q4
ygap	Stock-bond yield gap	1953Q3	2024Q4
disag	Analyst disagreement	1982Q1	2024Q4

Notes: All data are obtained from the Goyal–Welch–Zafirov (2024) dataset, available at <https://sites.google.com/view/agoyal145>. The target variable is the quarterly equity premium, defined as the S&P 500 return inclusive of dividends minus the risk-free rate, available from 1926Q1. Starting dates denote the first quarter for which a predictive regression observation is available; for predictors with raw data preceding 1926, the effective start is determined by target variable availability. An asterisk (*) denotes predictors whose sample ends before 2024Q4.

squares estimates for OLS. Because the Gibbs posterior is defined through a loss function rather than a sampling model, no predictive distribution is directly available; point forecasts are therefore constructed by plugging in posterior means rather than integrating over a predictive density.

The forecasting performance is assessed with two general criteria, that is, statistical performance and economic value. Statistical performance is evaluated using the out-of-sample

R^2 of Campbell and Thompson (2008),

$$R_{\text{OOS}}^2 = 1 - \frac{\sum_{t=t_0}^{T-1} (r_{t+1} - \hat{r}_{t+1|t})^2}{\sum_{t=t_0}^{T-1} (r_{t+1} - \bar{r}_{t+1|t})^2},$$

where $\bar{r}_{t+1|t} = t^{-1} \sum_{s=1}^t r_s$ denotes the historical mean computed recursively using data available at the forecast origin. The R_{OOS}^2 measures the reduction in mean squared forecast error relative to the HM benchmark. Statistical significance is assessed via the Clark and West (2007) adjusted-MSE test, which evaluates the null of equal predictive ability against a one-sided alternative favouring the nesting model.

Economic evaluation follows Neely et al. (2014) and considers a mean-variance investor who allocates between the (risky) S&P 500 and risk-free Treasury bills. The optimal portfolio weight in the risky asset for a risk averse mean-variance investor is

$$w_t = \frac{1}{\gamma} \cdot \frac{\hat{r}_{t+1|t}}{\hat{\sigma}_{t+1|t}^2},$$

where γ denotes the relative risk aversion coefficient, and $\hat{\sigma}_{t+1|t}^2$ is a rolling-window estimate of return variance computed over the preceding 5 years. Portfolio weights are constrained to $w_t \in [0, 1.5]$. The resulting portfolio return is $R_{p,t+1} = r_{f,t+1} + w_t r_{t+1}$, where $r_{f,t+1}$ denotes the risk-free rate. Transaction costs of ϕ per unit of turnover yield net returns

$$R_{p,t+1}^{\text{net}} = R_{p,t+1} - \phi \times |w_t - w_{t-1}|.$$

The economic value of the strategy is summarized by three metrics. First, the certainty equivalent return (CER),

$$\text{CER} = \hat{\mu}_p - \frac{\gamma}{2} \hat{\sigma}_p^2,$$

where $\hat{\mu}_p$ and $\hat{\sigma}_p^2$ denote the sample mean and variance of net portfolio returns. A positive CER gain relative to the HM benchmark ($\Delta \text{CER} > 0$) indicates economically meaningful improvements in risk-adjusted performance. Second, the annualized Sharpe ratio (SR) of each strategy. Finally, portfolio turnover,

$$\text{TO} = \frac{1}{T - t_0} \sum_{t=t_0}^{T-1} |w_t - w_{t-1}|.$$

Lower TO implies more stable forecasts and reduced transaction costs. The baseline sets $\gamma = 3$ and $\phi = 0.5\%$, and the Supplementary Appendix reports results for the configurations ($\gamma \in \{3, 5\}$) and ($\phi \in \{0.0\%, 0.5\%, 1.0\%\}$).

Table 4 reports the out-of-sample forecasting performance over the thirty-year evaluation window (1994Q4–2024Q4). Consistent with the findings of Goyal and Welch (2008); Goyal et al. (2024), outperforming the historical mean (HM) benchmark in terms of R_{OOS}^2 remains challenging: the average R_{OOS}^2 is negative for all four methods, with OLS averaging -3.15% , BRT -0.46% , GBCQR -0.61% , and GBCQR_f -0.77% . Both BRT and GBCQR reduce the magnitude of underperformance by approximately 80–85% relative to OLS. GBCQR displays

positive R^2_{OOS} for a larger number of predictors than BRT (12 versus 10 out of 31). Turning to statistical significance, GBCQR achieves [Clark and West \(2007\)](#) rejections for four predictors (i/k , ndrbl, vp, avgcor), matching BRT. GBCQR_f attains three rejections, fewer than both GBCQR and BRT.

The economic evaluation reveals more pronounced differences across methods. OLS generates an average ΔCER of essentially zero (0.03 pp) relative to HM, indicating that conditioning on individual predictors via least squares provides negligible economic value on average. BRT delivers an average gain of 0.4 pp, while GBCQR achieves 0.7 pp, the highest among all methods. GBCQR_f averages 0.3 pp, below GBCQR but still positive and superior to OLS. The CER gains from GBCQR are non-negative for 23 of 31 predictors (74%), with eight variables generating improvements exceeding 1 pp. Among the predictors generating the largest CER gains are implied variance (3.4 pp), the investment-to-capital ratio (3.1 pp), the credit spread (2.7 pp), and the analyst disagreement index (1.8 pp). GBCQR improves over OLS in ΔCER for 21 of 31 predictors (68%). While GBCQR achieves consistently positive CER gains, these differences with HM are significant only in 6 cases.

The Sharpe ratios reinforce these findings. HM achieves an average Sharpe ratio of 0.33 and OLS 0.32, indicating that OLS-based forecasts offer negligible risk-adjusted improvement over the unconditional benchmark. BRT recovers to 0.35, while GBCQR attains 0.39, representing a 22% improvement over OLS and an 18% improvement over HM. GBCQR_f achieves 0.33, equal to HM and below both BRT and GBCQR. At the predictor level, GBCQR delivers the highest Sharpe ratio among all methods for 27 of 31 predictors (87% of cases). Relative to HM, GBCQR achieves a strictly superior Sharpe ratio for 27 of 31 predictors, with ties for two additional predictors (d/e and vrp) and only two instances of underperformance (govik and lzrt). The differences become even more striking once pairwise significance against HM is assessed via the Ledoit–Wolf (2008) circular block bootstrap: GBCQR delivers a statistically higher Sharpe ratio than HM for 13 of 31 predictors, more than twice as many as OLS, which achieves only 6 significant improvements.

CQR-based forecasts are more stable because of the bounded influence function of the check loss. This stability translates into smoother forecasts and lower portfolio rebalancing costs. GBCQR averages 11% quarterly turnover, compared to 13% for BRT and 16% for OLS. These turnover reductions reinforce the CER and Sharpe ratio gains documented above, since lower transaction costs directly improve net-of-cost portfolio performance. The economic gains from GBCQR do not primarily reflect large improvements in mean-squared forecast error relative to the historical mean. As in much of the equity-premium literature, statistical gains in R^2_{OOS} remain modest. The value of GBCQR lies instead in producing more stable predictive signals that, when mapped into constrained mean-variance portfolio weights, attenuate extreme reallocations and raise risk-adjusted returns net of transaction costs.

The comparison between GB and GB_f is informative here. Both use the same composite loss, and with the diffuse prior used in this application calibration affects the posterior mean only modestly. Nevertheless, GB achieves higher CER and Sharpe ratios, suggesting that even small differences in posterior concentration can translate into more stable portfolio decisions.

Figure 1 displays cumulative portfolio wealth for sixteen representative predictors. The

Table 4: Out-of-Sample Equity Premium Forecasting: $\gamma = 3$, TC = 0.50% (1994Q4–2024Q4)

	R^2_{OOS} (%)				ΔCER (%)				Sharpe Ratio				Turnover (%)			
	OLS	BRT	GB	GB_f	OLS	BRT	GB	GB_f	HM	OLS	BRT	GB	GB_f	OLS	BRT	GB
Panel A: Valuation Ratios																
d/p	-14.41	-0.23	0.79	-2.13	-0.03	0.06	0.75 [†]	0.49	0.31	0.32	0.30	0.35 [‡]	0.29	2	6	5
d/y	-18.07	-0.48	0.77	-2.78	-0.08	0.08	0.87	1.20	0.31	0.27	0.29	0.35	0.33	1	7	6
e/p	-3.70	-0.42	1.19	-1.64	0.87	-0.40	0.24	0.16	0.31	0.30	0.24	0.32	0.25	10	8	7
d/e	-0.59	-7.29	-6.40	-4.72	-0.21	-0.85	-0.87	-1.17	0.31	0.30	0.29	0.31	0.26	7	7	6
b/m	-16.95	-0.05	0.47	-1.09	-0.40	-0.06	0.73 [‡]	-0.33	0.31	0.34	0.31	0.37 [‡]	0.26	1	5	5
Panel B: Interest Rates and Spreads																
tbl	-0.38	-0.12	-0.78	-0.08	1.00	0.95	0.83	0.91	0.31	0.39 [†]	0.39 [†]	0.40 [†]	0.38 [†]	6	5	4
lty	0.65	0.67	-0.02	0.64	0.99 [†]	0.99	0.85	0.88 [†]	0.31	0.38 [‡]	0.39 [†]	0.40 [†]	0.37 [†]	5	5	4
ltr	0.41	0.77	0.05	0.77	-1.21	-0.79	0.00	-1.21	0.32	0.24	0.27	0.34	0.24	34	32	24
tms	-2.94	-1.39	-2.50	-1.39	-0.36	-0.14	0.17	-0.21	0.31	0.31	0.33	0.38	0.32	10	8	6
dfy	-2.52	-0.10	-0.46	-0.05	0.33	-1.01	-0.09	-0.89	0.31	0.32	0.26	0.34	0.26	9	9	6
dfr	-2.52	-0.84	-1.78	-0.98	-0.06	-0.39	-0.13	-0.57	0.32	0.30	0.30	0.35	0.28	17	14	9
crdstd	-0.95	-1.08	-0.30	-1.36	2.06	2.73	2.70	2.56	0.37	0.48	0.52	0.52	0.51	14	14	15
Panel C: Macroeconomic Variables																
infl	0.48	0.31	-0.31	0.32	0.74	0.36	0.80	0.25 [‡]	0.31	0.35	0.34 [‡]	0.39 [‡]	0.33 [‡]	23	6	5
ntis	-9.48	-2.37	-5.00	-2.68	1.32	0.63	1.24	0.52	0.31	0.42 [‡]	0.37 [†]	0.42 [‡]	0.36	8	6	4
svar	0.36	-0.67	-0.60	0.09	0.02	-0.26	0.46	-0.24	0.31	0.33	0.30	0.37 [†]	0.30	12	9	6
i/k	1.73 *	2.42 *	2.45 **	2.20 *	2.48 [†]	2.93 [†]	3.06 [†]	2.83 [†]	0.30	0.45 [†]	0.47 [†]	0.48 [‡]	0.47 [‡]	8	6	6
govik	-2.27	-0.81	-0.53	-1.21	-1.31	-0.53	-0.39	-0.85	0.30	0.17	0.26	0.27	0.22	7	6	6
ndrbl	0.66	0.76 *	0.73 *	0.79	0.14	0.22	0.24	0.15	0.25	0.31	0.33	0.33	0.32	22	23	22
wtexas	-2.29	-1.19	-1.90	-0.91	-2.11	-0.57	-0.27	-0.62	0.32	0.23	0.31	0.35	0.29	43	19	13
Panel D: Variance and Risk Measures																
vp	3.39 *	5.45 *	3.21 *	4.87 *	-2.02	-0.30	1.11	-2.11	0.41	0.31	0.41	0.49	0.31	31	21	15
impvar	-6.41	-3.11	-1.78	-5.51	1.48	2.81	3.40 [†]	1.93	0.67	0.75	0.80	0.81 [†]	0.76	21	23	21
vrp	-6.71	-1.08	-2.69	-2.74	-3.33	-0.79	-0.09	-2.66	0.35	0.11	0.31	0.35	0.18	51	49	47
tail	0.26	0.23	-0.87	0.21	0.64 [‡]	0.70 [‡]	0.86	0.51 [‡]	0.31	0.35 [‡]	0.36 [‡]	0.40 [‡]	0.35 [‡]	7	6	5
skvw	-1.45	-0.49	-0.46	-0.31	-1.23	-0.08	0.30	-0.10	0.31	0.22	0.33	0.37	0.32	29	18	12
Panel E: Cross-Section and Technical																
lzrt	-11.32	-1.92	-1.99	-2.88	-0.98	0.10	-0.15	1.06	0.32	0.21	0.29	0.31	0.34	30	21	22
avgcor	4.61 *	3.79 *	2.47 *	4.26 **	2.15 [‡]	2.16 [‡]	1.83 [†]	1.98 [‡]	0.32	0.43 [‡]	0.44 [‡]	0.45 [‡]	0.43 [‡]	21	12	8
dtoy	0.61	-3.84	-2.39	-2.23	0.09	0.41	-0.33	-0.14	0.32	0.32	0.33	0.33	0.30	8	15	11
dtoat	0.11	0.28	0.07	0.17	-0.28	0.27	0.79	0.00	0.32	0.28	0.34 [‡]	0.39 [‡]	0.31	9	5	4
ygap	-3.05	-1.76	-0.77	-3.05	0.63	0.70	0.45	0.93	0.28	0.27	0.30	0.32	0.30	9	8	8
rdsp	-3.30	0.59	0.05	0.44	-0.98	1.53 [†]	1.24 [†]	1.11 [†]	0.31	0.26	0.39 [†]	0.40 [‡]	0.37	10	13	8
disag	-1.73	-0.26	0.28	-0.79	0.48	1.95	1.82	1.57	0.28	0.25	0.36	0.36	0.33	22	18	16
Average	-3.15	-0.46	-0.61	-0.77	0.03	0.43	0.72	0.26	0.33	0.32	0.35	0.39	0.33	16	13	11

Notes: Out-of-sample forecasting results for the quarterly U.S. equity premium, 1994Q4 to 2024Q4. R^2_{OOS} is measured relative to the historical mean (HM) benchmark. ΔCER is the annualized certainty equivalent return gain (percentage points) relative to HM. Portfolio weights derive from mean-variance optimization with risk aversion $\gamma = 3$, transaction costs of 0.50%, and a 5-years rolling window for variance; weights are constrained to $[0.0, 1.5]$. Sharpe ratios are annualized; turnover is reported as a quarterly average. BRT: Bayesian regression with Student- t errors ($\nu = 5$). GB: generalized Bayesian CQR with flat prior and calibrated η . GB_f: same model with fixed $\eta = 1$. Bold entries denote positive R^2_{OOS} or ΔCER , the highest Sharpe ratio, or the lowest turnover among the reported methods. Stars on R^2_{OOS} (*, **, ***) denote rejection of equal MSPE against HM at the 10%, 5%, 1% level using the [Clark and West \(2007\)](#) MSPE-adjusted statistic with HAC standard errors. On ΔCER , † (‡) denotes rejection of $H_0 : \text{CER}^{(m)} \leq \text{CER}^{(\text{HM})}$ at the 10% (5%) level using a one-sided Diebold–Mariano test on the realised-utility differential. On Sharpe ratios, † (‡) denotes rejection of $H_0 : \text{SR}^{(m)} \leq \text{SR}^{(\text{HM})}$ at the 10% (5%) level using the one-sided [Ledoit and Wolf \(2008\)](#) HAC-studentised circular block bootstrap ($B = 4999$ replications, block length selected by the [Politis and White \(2004\)](#) automatic rule).

exercise begins with an initial investment of \$1 and tracks wealth accumulation through 2024Q4. For the majority of predictors shown, GBCQR-based portfolios outperform or match both the HM benchmark and OLS. In several cases, GBCQR portfolios accumulate terminal wealth approximately 50–100% greater than OLS by the end of the sample period. For instance, for book-to-market, the final wealth is almost \$6 for GBCQR and less than \$2 for OLS.

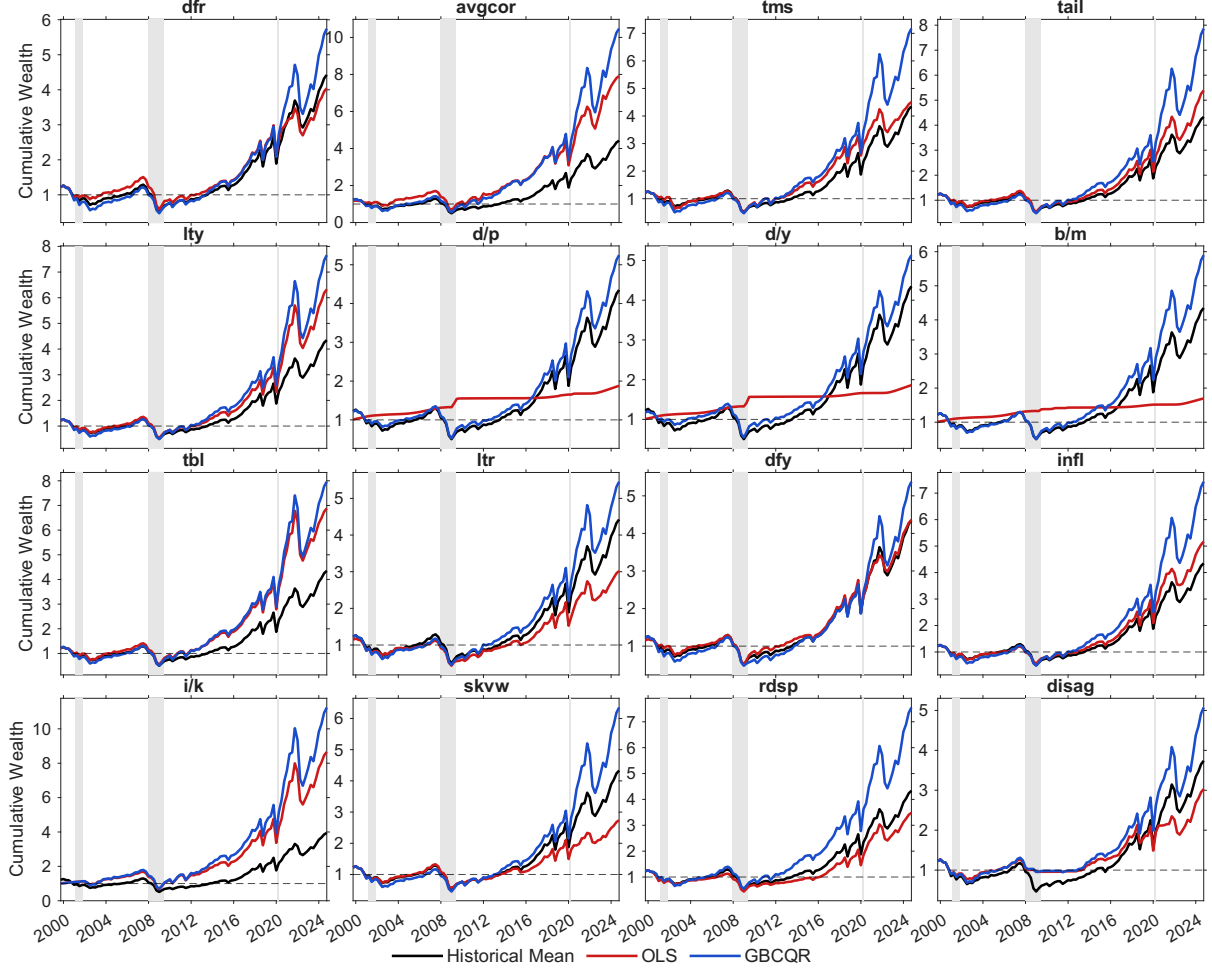


Figure 1: Cumulative Wealth for Selected Predictors from Mean-Variance Portfolio Allocation. The figure displays cumulative portfolio wealth (net of transaction costs) for sixteen selected predictors. Each panel compares three forecasting methods: Historical Mean (black), OLS (red), and GBCQR-Flat with calibrated η (blue). Shaded regions indicating NBER recession dates. Portfolio allocation follows the mean-variance framework with risk aversion $\gamma = 3$, weights constrained to $[0, 1.5]$, and transaction costs of 50 basis points. The horizontal dashed line indicates initial wealth of one.

Finally, we examine the forecasting performance using simultaneously a set of ten predictors¹⁴. The objective is to compare the GBCQR-SSL against OLS, where both methods include all ten predictors simultaneously. The Historical Mean serves again as the benchmark.

¹⁴We restrict the analysis to ten predictors for two reasons. First, some predictors in the full database are linear combinations or transformations of others. Second, restricting attention to predictors with longer historical coverage enables out-of-sample evaluation over an extended period. The list of predictors considered here is dividend-yield ratio, earnings-price ratio, book-to-market ratio, default yield spread, default return spread, long-term yield, illiquidity measure, stock return dispersion, net durable consumption growth, and investment-capital ratio.

Over the long sample (Figure 2, Panel A), only GBCQRs improve over HM: an initial investment of \$1 grows to over \$21.67 under GBCQR SSL and \$16.24 under GBCQR-SSL with fixed $\eta = 1$, compared to \$10.75 under OLS and \$13.73 under HM. The relative advantage of GBCQR-SSL is also observed over the recent subsample (Figure 2, Panel B). GBCQR-SSL terminal wealth reaches \$7.98 versus \$5.88 with OLS and \$3.26 for the HM forecast. To better understand the importance of each predictor in the multivariate setting, Figure 3 displays the posterior inclusion probabilities from the GBCQR-SSL model across out-of-sample forecast origins. Two predictors emerge as persistently selected. The book-to-market ratio maintains a PIP consistently above the inclusion threshold, near unity for most of the sample, and the dividend yield remains above the threshold throughout, with a brief softening around 2008–2010 before recovering. A second group of predictors exhibits intermittent inclusion. The long-term bond yield sustains high PIPs from the mid-1990s through the mid-2000s, drops below the threshold between 2008 and 2009, re-enters the active set around 2010–2017, and exhibits declining inclusion in the most recent years. The earnings-price ratio oscillates around the 0.50 boundary, entering sporadically after 2004. The default yield spread is selected only during isolated episodes around 1998–2002. The remaining five predictors stay well below the inclusion threshold throughout, indicating that the GBCQR-SSL model concentrates predictive content on a sparse and slowly evolving active set dominated by valuation ratios and, intermittently, by interest rate variables.

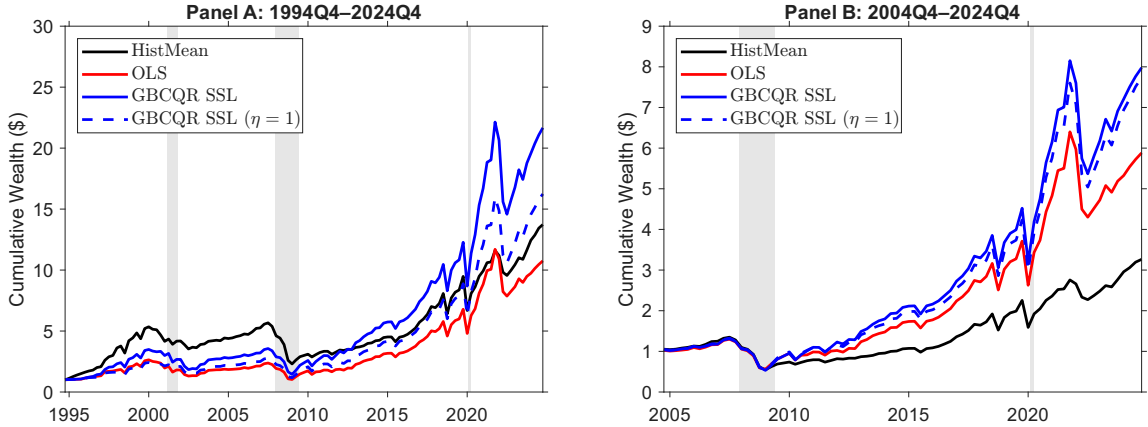


Figure 2: Cumulative Wealth from Mean-Variance Portfolio Allocation. This figure displays the cumulative wealth of a mean-variance investor who allocates between risky and risk-free assets based on equity premium forecasts. The methods compared are: Historical Mean (black), OLS (red), and GBCQR-SSL (blue). The dashed blue line is GBCQR-SSL with fixed $\eta = 1$. Variance is estimated using a 5-years rolling window. Risk aversion is set to $\gamma = 3$, and proportional transaction costs of 50 basis points are applied to portfolio rebalancing. Initial wealth is normalized to \$1.

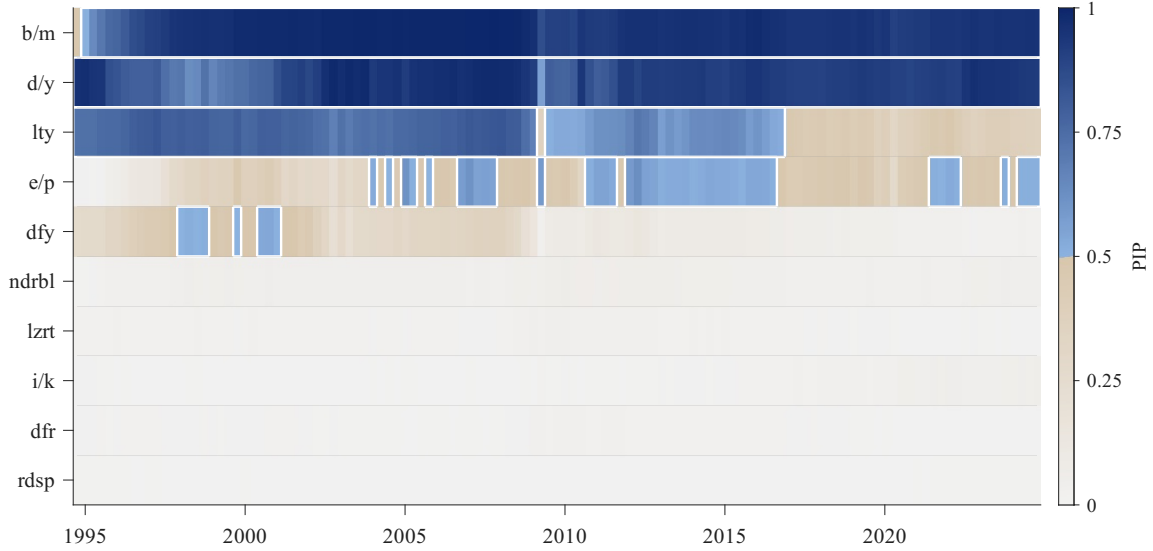


Figure 3: The heatmap displays posterior inclusion probabilities (PIPs) from the GBCQR-SSL model at each out-of-sample forecast origin. Predictors are ordered by time-averaged PIP (highest at top). The colormap shifts from warm gray (PIP < 0.5, excluded) to saturated blue (PIP $\geq$ 0.5, included); white rectangles demarcate contiguous periods of inclusion under the median-probability threshold of 0.5.

5 Concluding remarks

This paper develops Generalized Bayesian Composite Quantile Regression (GBCQR), a framework that exponentiates the composite quantile loss directly within the generalized Bayesian paradigm of [Bissiri et al. \(2016\)](#), avoiding the working-likelihood assumptions and latent-variable augmentation of ALD-based approaches. We propose a calibration rule for the learning rate based on coefficient-wise variance matching. It is closed form under flat priors, extends naturally to shrinkage and spike-and-slab LASSO priors, and delivers near-nominal frequentist coverage in cross-sectional settings while largely closing the undercoverage of i.i.d. intervals under serial dependence (the residual gap under high persistence mirrors the finite-sample bias of frequentist HAC inference and is not specific to the calibration strategy). Because the MCMC sampler only evaluates the composite loss rather than optimizing it, the approach achieves roughly two orders of magnitude faster computation than ALD-based Bayesian CQR, while also sidestepping the non-smooth optimization challenges of frequentist penalized CQR in moderate-to-high dimensions. Monte Carlo experiments confirm competitive estimation, reliable coverage, and robust variable selection across Gaussian, heavy-tailed, and contaminated error distributions.

An application to equity premium forecasting illustrates the practical relevance of the framework. While R^2_{OOS} improvements over the historical mean remain elusive, GBCQR delivers economically meaningful gains in certainty-equivalent returns and Sharpe ratios, together with lower portfolio turnover reflecting the stability of quantile-based coefficient estimates. The Supplementary Appendix demonstrates that these findings are robust across risk-aversion levels ($\gamma \in \{3, 5\}$), transaction costs ($\phi \in \{0\%, 0.5\%, 1.0\%\}$), and evaluation windows. Extensions to higher-dimensional predictors, alternative calibration targets, and formal asymptotic theory for

the calibrated Gibbs posterior remain promising directions for future work.

A Technical Details

A.1 Regularity conditions

The following conditions are sufficient for the baseline asymptotic theory of the CQR estimator under the linear location-shift model (2). Some Monte Carlo designs in Section 3 intentionally violate symmetry (DGP 5) or finite-moment assumptions (DGP 3, Cauchy) to study the robustness of GBCQR beyond the formal scope of these conditions.

Assumption 1. *The error density f is continuous, symmetric about zero, and strictly positive on its support. In particular, $f(F^{-1}(\tau_k)) > 0$ for each $k = 1, \dots, K$.*

Assumption 2. *The regressors $\{x_i\}_{i=1}^n$ are i.i.d. with $\mathbb{E}[x_i] = 0$, $\mathbb{E}[x_i x_i^\top] = C \succ 0$, and $\mathbb{E}[\|x_i\|^4] < \infty$. Furthermore, x_i is independent of ε_i .*

Assumption 3. *The error distribution satisfies $\mathbb{E}[|\varepsilon_i|^{2+\delta}] < \infty$ for some $\delta > 0$.*

Under Assumptions 1–3, the CQR estimator is root- n consistent and asymptotically normal with the sandwich covariance (4). The proof follows from the general M-estimator theory of Huber (1967) applied to the composite loss; see Zou and Yuan (2008) for the details specific to CQR.

It is worth clarifying which assumptions are needed for which purpose. The composite quantile score $s_i = \sum_{k=1}^K x_i(\tau_k - \mathbb{1}\{\varepsilon_i < \sigma F^{-1}(\tau_k)\})$ involves only bounded indicator functions, so $\mathbb{E}[\|s_i\|^2] \leq K^2 \mathbb{E}[\|x_i\|^2] < \infty$ under finite second moments of x_i . Consequently, the score CLT, the consistency of $\hat{I}_n$, and the Bahadur linearization all hold without finite error moments: the CQR estimator remains root- n consistent and asymptotically normal under Cauchy errors, as emphasized by Zou and Yuan (2008). Assumption 3 is invoked only in the asymptotic relative efficiency (ARE) comparisons with OLS in Section 2.1, which require $\text{Var}(\varepsilon_i) = \sigma^2 < \infty$. Assumption 1 is the binding distributional restriction for the curvature J in (5) to be well-defined and positive definite, and it excludes the skewed design (DGP 5) that appears as a robustness exercise in the simulations. The Cauchy design (DGP 3) satisfies Assumption 1 in full, since the Cauchy density is continuous, symmetric about zero, and strictly positive on $\mathbb{R}$; it violates Assumption 3 instead. As noted above, Assumption 3 is invoked only for the ARE comparisons with OLS, so neither assumption formally excludes the Cauchy from the root- n consistency and asymptotic normality results for the CQR estimator — which is precisely the robustness property emphasized by Zou and Yuan (2008).

A.2 Centering, profiling, and slope asymptotics

This section shows that the centering assumption $\mathbb{E}[x_i] = 0$ is a normalization, not a substantive restriction, and that profiling out the quantile-specific intercepts does not affect the first-order asymptotic distribution of $\hat{\beta}_n$.

Non-centered regressors. When $\mathbb{E}[x_i] = \mu \neq 0$, the joint sandwich covariance of the full parameter vector $\vartheta = (\theta_1, \dots, \theta_K, \beta^\top)^\top$ has non-zero cross-derivative blocks. The asymptotic covariance of $\hat{\beta}_n$ is then governed by the Schur complement

$$J_{\beta\beta\cdot\theta} = J_{\beta\beta} - J_{\beta\theta} J_{\theta\theta}^{-1} J_{\theta\beta}.$$

To verify the simplification, we compute the expected Hessian blocks of the smoothed composite loss at the true parameters. Let $f_k := f(F^{-1}(\tau_k))$ and $\mathbf{f} := (f_1, \dots, f_K)^\top$. Evaluating the per-observation expectations gives

$$J_{\beta\beta} = \frac{A_1}{\sigma} \mathbb{E}[x_i x_i^\top] = \frac{A_1}{\sigma} (\text{Var}(x_i) + \mu \mu^\top), \quad J_{\beta\theta_k} = \frac{f_k}{\sigma} \mu, \quad J_{\theta\theta} = \frac{1}{\sigma} \text{diag}(f_1, \dots, f_K),$$

where $A_1 = \sum_{k=1}^K f_k$ as defined in (6). Stacking the cross-derivatives across k yields the $p \times K$ block $J_{\beta\theta} = \sigma^{-1} \mu \mathbf{f}^\top$, and $J_{\theta\theta}$ is diagonal with inverse $J_{\theta\theta}^{-1} = \sigma \text{diag}(1/f_1, \dots, 1/f_K)$. The Schur-complement correction becomes

$$J_{\beta\theta} J_{\theta\theta}^{-1} J_{\theta\beta} = \frac{1}{\sigma} \mu \mathbf{f}^\top \text{diag}(1/f_1, \dots, 1/f_K) \mathbf{f} \mu^\top = \frac{1}{\sigma} \left(\sum_{k=1}^K f_k \right) \mu \mu^\top = \frac{A_1}{\sigma} \mu \mu^\top,$$

so that

$$J_{\beta\beta\cdot\theta} = J_{\beta\beta} - J_{\beta\theta} J_{\theta\theta}^{-1} J_{\theta\beta} = \frac{A_1}{\sigma} (\text{Var}(x_i) + \mu \mu^\top) - \frac{A_1}{\sigma} \mu \mu^\top = \frac{A_1}{\sigma} \text{Var}(x_i),$$

where $\text{Var}(x_i) = \mathbb{E}[x_i x_i^\top] - \mu \mu^\top$. The sandwich covariance becomes $\Sigma_{\text{CQR}} = \sigma^2 A_1^{-2} A_2 \text{Var}(x_i)^{-1}$. Under $\mathbb{E}[x_i] = 0$, the cross block $J_{\beta\theta}$ vanishes, the Jacobian is block diagonal in expectation, and the expression reduces to the centered case with $C = \mathbb{E}[x_i x_i^\top] = \text{Var}(x_i)$.

Profiling the intercepts. Under $E[x_i] = 0$, the score of $L_n(\beta, \theta)$ with respect to θ_k involves only $\mathbf{1}\{y_i - \theta_k - x_i^\top \beta < 0\}$, while the score with respect to β involves $x_i \cdot \mathbf{1}\{y_i - \theta_k - x_i^\top \beta < 0\}$. The composite quantile loss is not twice differentiable in the classical sense, so the second derivatives in this section are understood as derivatives of the population criterion $E[L_n(\beta, \theta)]$, which is smooth under Assumption 1: the expectation integrates the indicator against the error density f , and differentiating twice yields the density evaluated at the quantile locations. Under this convention, the normalised cross-derivative is

$$n^{-1} E \left[\frac{\partial^2 L_n}{\partial \theta_k \partial \beta} \right] = \sigma^{-1} f(F^{-1}(\tau_k)) E[x_i],$$

which vanishes when $E[x_i] = 0$. By the standard theory of profiled M -estimators (Murphy and Van der Vaart, 2000), applied at the population level, this block-diagonal structure implies that the profiled estimator $\hat{\beta}_n$ has the same first-order asymptotic distribution as the joint estimator. The kernel-smoothed sample estimator $\hat{J}_n$ defined in Appendix A.3 is the natural finite-sample analogue of this construction: integrating the indicator against the kernel $\Phi(\cdot/h_n)$ approximates the population integration against f as $h_n \rightarrow 0$.

A.3 Kernel smoothing for the curvature estimator

The population curvature of the composite quantile loss at the true parameter involves the error density f evaluated at the quantile levels. In the sample, the score of the k -th check function for observation i is

$$s_{ik}(\beta) = x_i(\tau_k - \mathbb{1}\{y_i - \theta_k - x_i^\top \beta < 0\}),$$

and the Hessian involves $f(0)$ -type quantities that cannot be computed from the indicator directly. Following [Powell \(1991\)](#), we replace $\mathbb{1}\{u < 0\}$ by $\Phi(-u/h_n)$ in the Hessian, yielding the kernel-smoothed curvature estimator

$$\hat{J}_n = \frac{1}{n} \sum_{k=1}^K \sum_{i=1}^n \frac{1}{h_n} \phi\left(\frac{y_i - \hat{\theta}_k - x_i^\top \hat{\beta}_n}{h_n}\right) x_i x_i^\top,$$

where ϕ is the standard normal density. We use the bandwidth $h_n = 0.9 \hat{\sigma} n^{-1/5}$, where $\hat{\sigma} = \text{MAD}(\hat{\varepsilon}_i)/0.675$ is a robust scale estimate from the median CQR residuals. Under $h_n \rightarrow 0$ and $nh_n \rightarrow \infty$, we have $\hat{J}_n \xrightarrow{p} J$.

The score covariance is estimated by the direct sample analogue:

$$\hat{I}_n = \frac{1}{n} \sum_{i=1}^n \hat{s}_i \hat{s}_i^\top, \quad \hat{s}_i = \sum_{k=1}^K x_i(\tau_k - \mathbb{1}\{y_i - \hat{\theta}_k - x_i^\top \hat{\beta}_n < 0\}).$$

This involves only indicator functions and requires no smoothing.

A.4 Properties of the calibration objective

Monotonicity of the posterior variance in η . Write $A(\eta) = \eta n \hat{J}_n + P$. Differentiating the matrix inverse gives

$$\frac{\partial}{\partial \eta} [A(\eta)^{-1}]_{jj} = -[A(\eta)^{-1}(n \hat{J}_n)A(\eta)^{-1}]_{jj}.$$

Since $\hat{J}_n \succ 0$ and $A(\eta) \succ 0$, the matrix $A(\eta)^{-1}(n \hat{J}_n)A(\eta)^{-1}$ is positive semidefinite, so each diagonal element is non-negative. It follows that $d_j(\eta) = n [A(\eta)^{-1}]_{jj}$ is strictly decreasing in η . Under the flat prior ($P = 0$), $d_j(\eta) = [\hat{J}_n^{-1}]_{jj}/\eta$ is continuous and strictly decreases from ∞ to 0 as η traverses $(0, \infty)$. At the left boundary, $d_j(\eta) \rightarrow \infty$ and the objective [\(11\)](#) diverges; at the right boundary, $d_j(\eta) \rightarrow 0$ and the objective approaches $|\mathcal{J}|$, the cardinality of the index set. By continuity and strict monotonicity, $d_j(\eta)$ crosses t_j at some interior point η_j^* , where the j -th term of [\(11\)](#) vanishes. The objective therefore dips strictly below $|\mathcal{J}|$ in the interior and attains its minimum there. Under proper priors, $\lim_{\eta \rightarrow 0^+} d_j(\eta) = n [P^{-1}]_{jj} < \infty$, so the objective need not diverge as $\eta \rightarrow 0^+$. An interior minimizer exists whenever $n [P^{-1}]_{jj} > t_j$ for at least one $j \in \mathcal{J}$.

Closed-form solution under the flat prior. Under $P = 0$ with $\mathcal{J} = \{1, \dots, p\}$, the calibration objective (11) becomes

$$f(\eta) = \sum_{j=1}^p \left(\frac{r_j}{\eta} - 1 \right)^2.$$

Differentiating and setting $f'(\eta) = 0$ yields the closed-form solution

$$\eta^* = \frac{\sum_{j=1}^p r_j^2}{\sum_{j=1}^p r_j}.$$

The second derivative at η^* is $f''(\eta^*) = 2 \sum r_j / (\eta^*)^3 > 0$, confirming that η^* is a minimum.

A.5 Time series assumptions

In time series applications, the i.i.d. assumption is replaced by conditions ensuring ergodicity. The following are convenient sufficient conditions that support the HAC-based calibration of Section 2.5. They are stronger than strictly necessary: the mutual independence of $\{x_t\}$ and $\{\varepsilon_t\}$ simplifies the factorization of lagged score covariances, but in the empirical application of Section 4 the predictors are lagged, so the operative requirement is predeterminedness rather than full independence.

Assumption 4 (Mixing condition). *The joint process $\{(x_t, \varepsilon_t)\}_{t=1}^n$ is strictly stationary and α -mixing with mixing coefficients satisfying $\alpha(h) = O(h^{-a})$ for some $a > 2$. Furthermore, $\{x_t\}$ and $\{\varepsilon_t\}$ are mutually independent.*

Assumption 5 (Higher-order moments). *$\mathbb{E}[\|x_t\|^{4+\delta}] < \infty$ and $\mathbb{E}[|\varepsilon_t|^{2+\delta}] < \infty$ for some $\delta > 0$.*

Under Assumptions 4–5, the long-run score covariance $\Omega = \sum_{h=-\infty}^{\infty} \text{Cov}(s_t, s_{t+h})$ is well-defined, and the HAC estimator $\hat{\Omega}_n$ of Section 2.5 is consistent. As in the cross-sectional case, the finite error moment condition in Assumption 5 can be relaxed for the CQR score CLT itself, since the composite quantile score is bounded by $K\|x_t\|$.

A.6 Asymptotic calibration

We establish that under the maintained model and flat priors, the diagonal variance-matching calibration of Section 2.4 produces asymptotically valid marginal credible intervals. The result combines the Gibbs posterior Bernstein–von Mises theory of Syring and Martin (2022) with the structural algebra of the CQR sandwich covariance under the location-shift model (2).

Proposition 1. *Suppose Assumptions 1–2 hold, the data $\{(y_i, x_i)\}_{i=1}^n$ are i.i.d., the prior is flat ($P = 0$), and the kernel-smoothed curvature estimator uses bandwidth $h_n \rightarrow 0$ with $nh_n \rightarrow \infty$. Let J and I denote the population curvature and score covariance from (5), let $\Sigma_{\text{CQR}} = J^{-1}IJ^{-1}$ denote the population sandwich covariance from (4), and define the population variance ratios*

$$\bar{r}_j := \frac{[J^{-1}]_{jj}}{[\Sigma_{\text{CQR}}]_{jj}}, \quad j = 1, \dots, p,$$

and the population learning rate

$$\bar{\eta} := \frac{\sum_{j=1}^p \bar{r}_j^2}{\sum_{j=1}^p \bar{r}_j}.$$

Then:

- (i) (Gibbs posterior Bernstein–von Mises.) *The Gibbs posterior $\pi_n^{(\bar{\eta})}(\beta \mid y) \propto \exp\{-\bar{\eta}L_n(\beta)\}$ satisfies, in total variation,*

$$\|\pi_n^{(\bar{\eta})}(\cdot \mid y) - \mathcal{N}(\hat{\beta}_n, (\bar{\eta}nJ)^{-1})\|_{\text{TV}} \xrightarrow{P} 0.$$

- (ii) (Calibration consistency.) $\hat{\eta}^* \xrightarrow{P} \bar{\eta}$, and for each $j = 1, \dots, p$,

$$n[V_{\text{post}}^F(\hat{\eta}^*)]_{jj} \xrightarrow{P} \frac{[J^{-1}]_{jj}}{\bar{\eta}}.$$

- (iii) (Exact marginal calibration under the maintained model.) *Under the location-shift model (2) with centered regressors and any positive definite design covariance C , the variance ratios are constant across coordinates,*

$$\bar{r}_j = \frac{A_1}{\sigma A_2} \quad \text{for every } j = 1, \dots, p,$$

so $\bar{\eta} = A_1/(\sigma A_2)$ and the diagonal calibration matches the frequentist sandwich exactly:

$$\kappa_j := \frac{[J^{-1}]_{jj}/\bar{\eta}}{[\Sigma_{\text{CQR}}]_{jj}} = 1 \quad \text{for every } j.$$

Combined with parts (i) and (ii), the calibrated posterior marginal for β_j is asymptotically $\mathcal{N}(\hat{\beta}_{n,j}, [\Sigma_{\text{CQR}}]_{jj}/n)$, matching the frequentist sampling distribution coordinate by coordinate.

- (iv) (Robustness bound under variation in $\bar{r}_j$.) *When the population variance ratios are not constant across coordinates, the residual coverage distortion satisfies*

$$|\kappa_j - 1| \leq \frac{\bar{r}_{\max} - \bar{r}_{\min}}{\bar{r}_{\min}},$$

where $\bar{r}_{\max}$ and $\bar{r}_{\min}$ are the largest and smallest population variance ratios.

The result concerns marginal (coordinate-wise) coverage only; joint coverage of all p coordinates simultaneously cannot in general be controlled by a scalar learning rate.

Proof sketch. Part (i): Gibbs posterior Bernstein–von Mises.

Let $L(\beta) := \mathbb{E}[L_n(\beta)/n]$ denote the population profiled risk. Under Assumptions 1–2, L is twice differentiable at β^* , with

$$\nabla L(\beta^*) = 0, \quad \nabla^2 L(\beta^*) = J,$$

where the gradient vanishes by first-order optimality of β^* as the population CQR minimizer, and the Hessian is obtained by dominated convergence through the quantile indicators (see

Koenker 2005, Chapter 4). Correspondingly, the total sample loss satisfies $\nabla^2 L_n(\beta^*) = nJ$ in expectation, consistent with the normalization $\hat{J}_n = n^{-1}\nabla^2 L_n$ adopted in Section 2.3. The sample loss admits the expansion

$$L_n(\beta) = L_n(\beta^*) - (\beta - \beta^*)^\top \sum_{i=1}^n s_i(\beta^*) + \frac{n}{2}(\beta - \beta^*)^\top J(\beta - \beta^*) + o_p(n\|\beta - \beta^*\|^2),$$

uniformly over $\|\beta - \beta^*\| = O_p(n^{-1/2})$, where $s_i(\beta^*) = \sum_{k=1}^K x_i(\tau_k - \mathbb{1}\{\varepsilon_i < \sigma F^{-1}(\tau_k)\})$ is the composite quantile score. Although the check loss is non-differentiable, this expansion is the Bahadur-type representation for composite quantile M-estimators: the piecewise-linear sample gradient is replaced by the population-level score at β^* , with the non-smoothness absorbed into the o_p remainder. The single-quantile version is Theorem 4.1 in Koenker (2005); the extension to the composite loss is immediate because $s_i(\beta^*)$ is a finite sum of individual quantile scores.

From the first-order condition of the CQR estimator and the expansion in Step 1,

$$\sqrt{n}(\hat{\beta}_n - \beta^*) = J^{-1} \cdot \frac{1}{\sqrt{n}} \sum_{i=1}^n s_i(\beta^*) + o_p(1).$$

Since $s_i(\beta^*)$ is bounded by $K\|x_i\|$, the Lindeberg–Feller central limit theorem applies under Assumption 2 alone (without requiring finite error moments), yielding

$$\sqrt{n}(\hat{\beta}_n - \beta^*) \xrightarrow{d} \mathcal{N}(0, J^{-1} I J^{-1}).$$

Consider the log-posterior kernel $\ell_n(\beta) := -\bar{\eta} L_n(\beta)$. Substituting the expansion in Step 1 and re-centering at $\hat{\beta}_n$ —which absorbs the linear score term into the quadratic component because $\hat{\beta}_n - \beta^* = J^{-1} n^{-1} \sum_i s_i(\beta^*) + o_p(n^{-1/2})$ by Step 2—yields

$$\ell_n(\beta) = \ell_n(\hat{\beta}_n) - \frac{\bar{\eta}n}{2}(\beta - \hat{\beta}_n)^\top J(\beta - \hat{\beta}_n) + o_p(n\|\beta - \hat{\beta}_n\|^2),$$

uniformly over neighbourhoods of radius $O(n^{-1/2})$. The Gibbs posterior is therefore locally equivalent to a Gaussian density with mean $\hat{\beta}_n$ and covariance $(\bar{\eta}nJ)^{-1}$. The total-variation conclusion follows from Theorem 2 of Syring and Martin (2022), whose conditions are satisfied by the CQR loss under Assumptions 1–2: quadratic minorization of the population risk at β^* , boundedness of the score, and the Bahadur representation of Step 1. Profiling out the quantile-specific intercepts does not affect the first-order expansion: under $\mathbb{E}[x_i] = 0$, slope and intercept scores are orthogonal in expectation, and the Schur complement argument in Appendix A.2 implies that the asymptotic distribution of $\hat{\beta}_n$ is unchanged.

Part (ii): Calibration consistency. The sample variance ratios are $r_j = [\hat{J}_n^{-1}]_{jj} / [\hat{J}_n^{-1} \hat{I}_n \hat{J}_n^{-1}]_{jj}$. Under Assumptions 1–2, the kernel-smoothed curvature estimator satisfies $\hat{J}_n \xrightarrow{P} J$ under $h_n \rightarrow 0$ and $nh_n \rightarrow \infty$ (Appendix A.3), and $\hat{I}_n \xrightarrow{P} I$ by the law of large numbers for the sample score covariance. By the continuous mapping theorem, $r_j \xrightarrow{P} \bar{r}_j$ for each j , and since $\hat{\eta}^*$ is a continuous function of $(r_1, \dots, r_p)$ via the closed form (14), $\hat{\eta}^* \xrightarrow{P} \bar{\eta}$. The marginal variance convergence follows from $n[V_{\text{post}}^F(\hat{\eta}^*)]_{jj} = [\hat{J}_n^{-1}]_{jj} / \hat{\eta}^* \xrightarrow{P} [J^{-1}]_{jj} / \bar{\eta}$.

Part (iii): Exact marginal calibration under the maintained model. Under the location-shift

model with centered regressors, $J = (A_1/\sigma)C$ and $I = A_2C$ by (5). Both matrices are scalar multiples of the same design covariance C , so

$$\Sigma_{\text{CQR}} = J^{-1} I J^{-1} = \frac{\sigma^2 A_2}{A_1^2} C^{-1}.$$

Taking diagonals,

$$[J^{-1}]_{jj} = \frac{\sigma}{A_1} [C^{-1}]_{jj}, \quad [\Sigma_{\text{CQR}}]_{jj} = \frac{\sigma^2 A_2}{A_1^2} [C^{-1}]_{jj},$$

and the variance ratio

$$\bar{r}_j = \frac{(\sigma/A_1)[C^{-1}]_{jj}}{(\sigma^2 A_2/A_1^2)[C^{-1}]_{jj}} = \frac{A_1}{\sigma A_2}$$

is constant in j regardless of the structure of C : the $[C^{-1}]_{jj}$ factor cancels because Σ_{CQR} and J^{-1} share the same coordinate-wise dependence on the design. Consequently $\bar{\eta} = A_1/(\sigma A_2)$ and $\kappa_j = \bar{r}_j/\bar{\eta} = 1$ for every j . Combining with Parts (i) and (ii), the calibrated posterior marginal for β_j is asymptotically $\mathcal{N}(\hat{\beta}_{n,j}, [\Sigma_{\text{CQR}}]_{jj}/n)$, matching the frequentist sampling distribution from (4).

Part (iv): Robustness bound. When $\bar{r}_j$ varies across j , write

$$\bar{\eta} = \sum_{\ell=1}^p \bar{r}_\ell \cdot \frac{\bar{r}_\ell}{\sum_m \bar{r}_m},$$

which exhibits $\bar{\eta}$ as a weighted average of the $\bar{r}_\ell$ with non-negative weights summing to one. Therefore $\bar{\eta} \in [\bar{r}_{\min}, \bar{r}_{\max}]$, and consequently $\kappa_j = \bar{r}_j/\bar{\eta} \in [\bar{r}_{\min}/\bar{r}_{\max}, \bar{r}_{\max}/\bar{r}_{\min}]$. The distance from unity is maximized on the upper branch because $\bar{r}_{\max}/\bar{r}_{\min} - 1 \geq 1 - \bar{r}_{\min}/\bar{r}_{\max}$ whenever $\bar{r}_{\max} > \bar{r}_{\min} > 0$, giving

$$|\kappa_j - 1| \leq \frac{\bar{r}_{\max}}{\bar{r}_{\min}} - 1 = \frac{\bar{r}_{\max} - \bar{r}_{\min}}{\bar{r}_{\min}}. \quad \square$$

Under the maintained location-shift model, $J = (A_1/\sigma)C$ and $I = A_2C$ are proportional, so $I = cJ$ with $c = \sigma A_2/A_1$. Any scalar criterion respecting this proportionality returns the same population learning rate. Diagonal matching gives $\bar{\eta} = A_1/(\sigma A_2)$ by Proposition 1(iii); the loss-likelihood-bootstrap trace rule of Lyddon et al. (2019) gives

$$\eta^{\text{LLB}} = \frac{\text{tr}(J I^{-1} J)}{\text{tr}(J)} = \frac{(A_1^2/(\sigma^2 A_2)) \text{tr}(C)}{(A_1/\sigma) \text{tr}(C)} = \frac{A_1}{\sigma A_2} = \bar{\eta},$$

and determinant and eigenvalue-matching rules collapse identically. Proposition 1(iii) is therefore a recoverability check. The diagonal criterion becomes operationally distinct from scalar alternatives only when proportionality fails, as under informative priors or serial dependence.

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Supplementary Appendix: Generalized Bayesian Composite Quantile Regression with an Application to Equity Premium Forecasting

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Abstract

This supplement provides additional results supporting the main paper. It contains: (i) Monte Carlo robustness exercises varying the Toeplitz correlation parameter and sample size; (ii) a Monte Carlo comparison with all methods set to $K = 5$ quantile levels, verifying that the main findings are not driven by the asymmetric quantile grid; (iii) Monte Carlo simulations under time-series error structures with varying persistence; (iv) the time-series behaviour of the calibrated learning rate $\hat{\eta}$ across predictors and expanding estimation windows; (v) out-of-sample equity premium forecasting results under alternative risk aversion coefficients and transaction cost levels; (vi) a shorter 20-year evaluation window (2004Q4–2024Q4); and (vii) a direct comparison of calibrated GBCQR with frequentist CQR.

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1 Monte Carlo: Robustness to Correlation and Sample Size

This section extends the baseline Monte Carlo experiment by varying the Toeplitz correlation parameter ($\rho \in \{0.25, 0.50, 0.75\}$) and the sample size ($n \in \{100, 200\}$). Tables 1–6 report results across the six configurations. The RMSE panels additionally decompose accuracy into active and inactive coefficient subsets.

Three findings stand out. First, the RMSE rankings are remarkably stable across all (n, ρ) configurations. The calibrated GBCQR variants dominate under every non-Gaussian DGP, with average relative RMSE in the range 0.95–0.97; the fixed- η variants achieve slightly lower RMSE (0.94–0.95) at the cost of severe coverage distortion. BR collapses under Cauchy errors regardless of n or ρ , with relative RMSE exceeding 5 in every case.

Second, the RMSE decomposition reveals no meaningful difference between active and inactive coefficients: the relative accuracy gains of GBCQR are uniform across the two subsets, confirming that the method does not trade off signal estimation against noise shrinkage.

Third, coverage and computation are equally invariant. Calibrated GBCQR maintains near-nominal 95% coverage (averaging 0.95) across all settings, while BCQR-ALD systematically undercovers (0.94 or below). The computational advantage of GBCQR is stark: it runs at roughly the same cost as BR, while BCQR-ALD is 58–124 $\times$ slower. Variable selection via GBCQR-SSL achieves perfect TPR (100%) throughout, with TNR declining modestly as ρ increases — a natural consequence of stronger collinearity between active and inactive covariates.

Table 1: Monte Carlo Results: $n = 100$, $\rho = 0.25$

Method	Normal	t_3	Contam.	Cauchy	Skewed	Avg.
<i>Panel A: RMSE (all coefficients, relative to CQR)</i>						
GBCQR-Flat	1.007	0.972	0.957	0.919	0.973	0.966
GBCQR-Flat ($\eta = 1$)	1.011	0.964	0.951	0.862	0.953	0.948
GBCQR-Normal	1.006	0.972	0.956	0.918	0.971	0.965
GBCQR-Normal ($\eta = 1$)	1.012	0.964	0.951	0.862	0.952	0.948
BCQR-ALD	1.115	1.003	0.955	0.718	0.900	0.938
BRT	1.020	0.956	0.929	0.910	1.021	0.967
BR	0.990	1.266	1.434	8.268	1.110	2.614
<i>Panel B: RMSE (active coefficients, relative to CQR)</i>						
GBCQR-Flat	1.007	0.973	0.957	0.916	0.973	0.965
GBCQR-Flat ($\eta = 1$)	1.013	0.966	0.950	0.858	0.954	0.948
GBCQR-Normal	1.007	0.973	0.956	0.916	0.971	0.965
GBCQR-Normal ($\eta = 1$)	1.013	0.965	0.949	0.858	0.952	0.948
BCQR-ALD	1.085	0.998	0.936	0.761	0.902	0.937
BRT	1.021	0.955	0.930	0.900	1.017	0.964
BR	0.986	1.284	1.423	8.468	1.111	2.655
<i>Panel C: RMSE (inactive coefficients, relative to CQR)</i>						
GBCQR-Flat	1.006	0.972	0.957	0.921	0.973	0.966
GBCQR-Flat ($\eta = 1$)	1.010	0.963	0.953	0.864	0.952	0.949
GBCQR-Normal	1.005	0.971	0.957	0.919	0.971	0.965
GBCQR-Normal ($\eta = 1$)	1.010	0.963	0.952	0.864	0.952	0.948
BCQR-ALD	1.133	1.006	0.966	0.692	0.898	0.939
BRT	1.019	0.956	0.929	0.916	1.023	0.969
BR	0.993	1.255	1.441	8.147	1.109	2.589
<i>Panel D: Coverage of 95% Intervals</i>						
GBCQR-Flat	0.954	0.943	0.945	0.958	0.957	0.951
GBCQR-Flat ($\eta = 1$)	0.558	0.531	0.537	0.528	0.549	0.541
GBCQR-Normal	0.953	0.943	0.946	0.956	0.957	0.951
GBCQR-Normal ($\eta = 1$)	0.560	0.530	0.539	0.528	0.547	0.541
BCQR-ALD	0.921	0.933	0.937	0.968	0.925	0.937
CQR	0.950	0.944	0.947	0.958	0.951	0.950
BRT	0.946	0.953	0.963	0.986	0.956	0.961
BR	0.954	0.945	0.951	0.960	0.954	0.953
<i>Panel E: Computation Time (relative to BR, $K = 5$)</i>						
GBCQR-Flat	1.07	1.06	1.07	1.07	1.06	1.07
GBCQR-Normal	1.07	1.07	1.08	1.08	1.07	1.08
BCQR-ALD	58.05	57.99	58.05	57.89	58.03	58.00
BRT	2.05	2.05	2.05	2.05	2.05	2.05
<i>Panel F: Variable Selection (TPR, TNR)</i>						
GBCQR-SSL	(100, 99)	(100, 100)	(100, 100)	(100, 96)	(100, 99)	(100, 99)
BR-SSL	(100, 100)	(99, 100)	(98, 100)	(2, 100)	(99, 100)	(80, 100)
CQR-LASSO	(100, 100)	(100, 100)	(100, 100)	(83, 100)	(100, 100)	(97, 100)

Notes: Results based on 500 Monte Carlo replications. The design considers $p = 8$ covariates with Toeplitz correlation $\Sigma_{jk} = \rho^{|j-k|}$ and true coefficients $\beta^* = (3, 1.5, 0, 0, 2, 0, 0, 0)^\top$; active set $\mathcal{A} = \{1, 2, 5\}$, inactive set $\mathcal{A}^c = \{3, 4, 6, 7, 8\}$. All methods except BCQR-ALD use $K = 19$ equally spaced quantile levels; BCQR-ALD uses $K = 5$ due to its prohibitive $O(K)$ latent-variable augmentation cost. RMSE panels report values relative to frequentist CQR; bold entries indicate values below unity (superior accuracy). Coverage panels report empirical coverage of nominal 95% credible/confidence intervals. Computation times are reported at $K = 5$ for cross-method comparability. “ $\eta = 1$ ” denotes GBCQR variants with the learning rate fixed at unity (no calibration). TPR: fraction of active coefficients correctly identified; TNR: fraction of inactive coefficients correctly excluded. GBCQR-SSL and BR-SSL select via $\text{PIP}_j > 0.5$; CQR-LASSO selects if $|\hat{\beta}_j| > 0$. The final column reports the arithmetic mean across all DGPs.

Table 2: Monte Carlo Results: $n = 100$, $\rho = 0.50$

Method	Normal	t_3	Contam.	Cauchy	Skewed	Avg.
<i>Panel A: RMSE (all coefficients, relative to CQR)</i>						
GBCQR-Flat	1.008	0.974	0.951	0.921	0.976	0.966
GBCQR-Flat ($\eta = 1$)	1.014	0.967	0.944	0.864	0.957	0.949
GBCQR-Normal	1.008	0.973	0.950	0.918	0.973	0.964
GBCQR-Normal ($\eta = 1$)	1.014	0.966	0.944	0.864	0.957	0.949
BCQR-ALD	1.134	1.002	0.938	0.737	0.888	0.940
BRT	1.022	0.960	0.920	0.915	1.021	0.967
BR	0.987	1.267	1.481	7.902	1.105	2.548
<i>Panel B: RMSE (active coefficients, relative to CQR)</i>						
GBCQR-Flat	1.010	0.975	0.951	0.918	0.975	0.966
GBCQR-Flat ($\eta = 1$)	1.016	0.969	0.944	0.859	0.956	0.949
GBCQR-Normal	1.010	0.975	0.950	0.916	0.972	0.965
GBCQR-Normal ($\eta = 1$)	1.016	0.969	0.944	0.858	0.956	0.949
BCQR-ALD	1.111	0.977	0.929	0.741	0.927	0.937
BRT	1.023	0.962	0.918	0.900	1.018	0.964
BR	0.983	1.241	1.474	8.143	1.109	2.590
<i>Panel C: RMSE (inactive coefficients, relative to CQR)</i>						
GBCQR-Flat	1.007	0.973	0.951	0.922	0.976	0.966
GBCQR-Flat ($\eta = 1$)	1.013	0.965	0.944	0.867	0.958	0.949
GBCQR-Normal	1.006	0.972	0.950	0.919	0.974	0.964
GBCQR-Normal ($\eta = 1$)	1.013	0.965	0.943	0.867	0.957	0.949
BCQR-ALD	1.148	1.018	0.943	0.735	0.867	0.942
BRT	1.021	0.959	0.920	0.924	1.022	0.969
BR	0.989	1.282	1.485	7.758	1.104	2.524
<i>Panel D: Coverage of 95% Intervals</i>						
GBCQR-Flat	0.953	0.949	0.949	0.963	0.957	0.954
GBCQR-Flat ($\eta = 1$)	0.550	0.544	0.551	0.518	0.547	0.542
GBCQR-Normal	0.953	0.948	0.947	0.959	0.953	0.952
GBCQR-Normal ($\eta = 1$)	0.549	0.542	0.552	0.517	0.550	0.542
BCQR-ALD	0.916	0.934	0.942	0.967	0.931	0.938
CQR	0.952	0.955	0.950	0.958	0.956	0.954
BRT	0.948	0.958	0.964	0.988	0.957	0.963
BR	0.956	0.951	0.947	0.966	0.950	0.954
<i>Panel E: Computation Time (relative to BR, $K = 5$)</i>						
GBCQR-Flat	1.07	1.06	1.06	1.06	1.06	1.06
GBCQR-Normal	1.07	1.07	1.07	1.07	1.07	1.07
BCQR-ALD	58.02	58.00	58.05	57.87	57.96	57.98
BRT	2.04	2.04	2.04	2.04	2.04	2.04
<i>Panel F: Variable Selection (TPR, TNR)</i>						
GBCQR-SSL	(100, 96)	(100, 99)	(100, 100)	(100, 92)	(100, 96)	(100, 97)
BR-SSL	(100, 100)	(98, 100)	(98, 100)	(2, 100)	(99, 100)	(79, 100)
CQR-LASSO	(100, 100)	(100, 100)	(100, 100)	(80, 100)	(100, 100)	(96, 100)

Notes: See notes to Table 1. Results based on 500 Monte Carlo replications.

Table 3: Monte Carlo Results: $n = 100$, $\rho = 0.75$

Method	Normal	t_3	Contam.	Cauchy	Skewed	Avg.
<i>Panel A: RMSE (all coefficients, relative to CQR)</i>						
GBCQR-Flat	1.006	0.973	0.953	0.925	0.975	0.967
GBCQR-Flat ($\eta = 1$)	1.012	0.965	0.945	0.865	0.953	0.948
GBCQR-Normal	1.007	0.971	0.953	0.914	0.971	0.963
GBCQR-Normal ($\eta = 1$)	1.013	0.966	0.945	0.865	0.953	0.948
BCQR-ALD	1.132	0.993	0.962	0.759	0.888	0.947
BRT	1.017	0.955	0.919	0.910	1.018	0.964
BR	0.987	1.322	1.440	5.878	1.108	2.147
<i>Panel B: RMSE (active coefficients, relative to CQR)</i>						
GBCQR-Flat	1.005	0.975	0.953	0.922	0.974	0.966
GBCQR-Flat ($\eta = 1$)	1.010	0.968	0.945	0.857	0.950	0.946
GBCQR-Normal	1.005	0.974	0.952	0.908	0.971	0.962
GBCQR-Normal ($\eta = 1$)	1.010	0.968	0.946	0.857	0.950	0.946
BCQR-ALD	1.117	0.983	0.969	0.745	0.884	0.940
BRT	1.015	0.961	0.919	0.897	1.014	0.961
BR	0.992	1.317	1.449	6.229	1.116	2.221
<i>Panel C: RMSE (inactive coefficients, relative to CQR)</i>						
GBCQR-Flat	1.007	0.973	0.954	0.926	0.976	0.967
GBCQR-Flat ($\eta = 1$)	1.014	0.964	0.945	0.870	0.955	0.950
GBCQR-Normal	1.008	0.969	0.953	0.917	0.971	0.964
GBCQR-Normal ($\eta = 1$)	1.015	0.964	0.945	0.870	0.955	0.950
BCQR-ALD	1.141	0.999	0.958	0.767	0.890	0.951
BRT	1.018	0.952	0.919	0.917	1.019	0.965
BR	0.984	1.325	1.434	5.673	1.103	2.104
<i>Panel D: Coverage of 95% Intervals</i>						
GBCQR-Flat	0.952	0.946	0.953	0.963	0.954	0.954
GBCQR-Flat ($\eta = 1$)	0.566	0.545	0.554	0.526	0.538	0.546
GBCQR-Normal	0.946	0.940	0.953	0.958	0.947	0.949
GBCQR-Normal ($\eta = 1$)	0.569	0.544	0.553	0.522	0.537	0.545
BCQR-ALD	0.919	0.943	0.940	0.964	0.929	0.939
CQR	0.952	0.945	0.962	0.959	0.954	0.954
BRT	0.946	0.957	0.969	0.988	0.957	0.963
BR	0.950	0.953	0.958	0.966	0.954	0.956
<i>Panel E: Computation Time (relative to BR, $K = 5$)</i>						
GBCQR-Flat	1.07	1.06	1.06	1.07	1.06	1.07
GBCQR-Normal	1.07	1.07	1.07	1.07	1.07	1.07
BCQR-ALD	58.04	58.03	58.04	57.91	57.99	58.00
BRT	2.05	2.05	2.05	2.05	2.05	2.05
<i>Panel F: Variable Selection (TPR, TNR)</i>						
GBCQR-SSL	(100, 84)	(100, 94)	(100, 96)	(100, 79)	(100, 87)	(100, 88)
BR-SSL	(100, 100)	(96, 100)	(96, 100)	(2, 100)	(97, 100)	(78, 100)
CQR-LASSO	(98, 98)	(100, 100)	(99, 100)	(75, 100)	(99, 100)	(94, 99)

Notes: See notes to Table 1. Results based on 500 Monte Carlo replications.

Table 4: Monte Carlo Results: $n = 200$, $\rho = 0.25$

Method	Normal	t_3	Contam.	Cauchy	Skewed	Avg.
<i>Panel A: RMSE (all coefficients, relative to CQR)</i>						
GBCQR-Flat	1.011	0.966	0.953	0.883	0.956	0.954
GBCQR-Flat ($\eta = 1$)	1.014	0.962	0.950	0.854	0.945	0.945
GBCQR-Normal	1.009	0.966	0.952	0.883	0.956	0.953
GBCQR-Normal ($\eta = 1$)	1.014	0.962	0.949	0.854	0.944	0.945
BCQR-ALD	1.129	0.983	0.934	0.709	0.868	0.925
BRT	1.025	0.952	0.925	0.896	1.013	0.962
BR	0.987	1.292	1.433	12.400	1.130	3.448
<i>Panel B: RMSE (active coefficients, relative to CQR)</i>						
GBCQR-Flat	1.011	0.971	0.949	0.885	0.956	0.954
GBCQR-Flat ($\eta = 1$)	1.015	0.968	0.945	0.856	0.944	0.945
GBCQR-Normal	1.010	0.972	0.949	0.884	0.954	0.954
GBCQR-Normal ($\eta = 1$)	1.014	0.968	0.945	0.856	0.944	0.945
BCQR-ALD	1.129	0.999	0.950	0.732	0.850	0.932
BRT	1.026	0.961	0.920	0.907	1.014	0.966
BR	0.988	1.265	1.439	12.688	1.128	3.502
<i>Panel C: RMSE (inactive coefficients, relative to CQR)</i>						
GBCQR-Flat	1.011	0.963	0.955	0.883	0.956	0.953
GBCQR-Flat ($\eta = 1$)	1.013	0.959	0.953	0.853	0.945	0.945
GBCQR-Normal	1.009	0.962	0.954	0.882	0.957	0.953
GBCQR-Normal ($\eta = 1$)	1.013	0.959	0.952	0.853	0.945	0.944
BCQR-ALD	1.129	0.974	0.924	0.695	0.879	0.920
BRT	1.025	0.946	0.928	0.888	1.012	0.960
BR	0.987	1.309	1.429	12.222	1.131	3.416
<i>Panel D: Coverage of 95% Intervals</i>						
GBCQR-Flat	0.953	0.948	0.947	0.953	0.953	0.951
GBCQR-Flat ($\eta = 1$)	0.575	0.561	0.562	0.527	0.560	0.557
GBCQR-Normal	0.951	0.949	0.946	0.949	0.951	0.949
GBCQR-Normal ($\eta = 1$)	0.576	0.564	0.563	0.528	0.561	0.558
BCQR-ALD	0.910	0.940	0.944	0.971	0.932	0.939
CQR	0.953	0.949	0.945	0.944	0.946	0.947
BRT	0.944	0.957	0.959	0.984	0.953	0.959
BR	0.952	0.949	0.949	0.952	0.948	0.950
<i>Panel E: Computation Time (relative to BR, $K = 5$)</i>						
GBCQR-Flat	1.47	1.46	1.46	1.47	1.46	1.46
GBCQR-Normal	1.49	1.49	1.49	1.49	1.49	1.49
BCQR-ALD	108.63	108.49	108.51	107.93	108.45	108.40
BRT	2.66	2.66	2.66	2.66	2.66	2.66
<i>Panel F: Variable Selection (TPR, TNR)</i>						
GBCQR-SSL	(100, 99)	(100, 100)	(100, 100)	(100, 97)	(100, 99)	(100, 99)
BR-SSL	(100, 100)	(100, 100)	(100, 100)	(2, 100)	(100, 100)	(80, 100)
CQR-LASSO	(100, 100)	(100, 100)	(100, 100)	(94, 100)	(100, 100)	(99, 100)

Notes: See notes to Table 1. Results based on 500 Monte Carlo replications.

Table 5: Monte Carlo Results: $n = 200$, $\rho = 0.50$

Method	Normal	t_3	Contam.	Cauchy	Skewed	Avg.
<i>Panel A: RMSE (all coefficients, relative to CQR)</i>						
GBCQR-Flat	1.012	0.971	0.950	0.882	0.956	0.954
GBCQR-Flat ($\eta = 1$)	1.014	0.966	0.948	0.853	0.944	0.945
GBCQR-Normal	1.011	0.968	0.949	0.881	0.956	0.953
GBCQR-Normal ($\eta = 1$)	1.013	0.966	0.948	0.853	0.944	0.945
BCQR-ALD	1.105	1.006	0.961	0.708	0.851	0.926
BRT	1.027	0.957	0.918	0.890	1.018	0.962
BR	0.983	1.309	1.471	11.290	1.128	3.236
<i>Panel B: RMSE (active coefficients, relative to CQR)</i>						
GBCQR-Flat	1.012	0.973	0.956	0.882	0.957	0.956
GBCQR-Flat ($\eta = 1$)	1.014	0.967	0.953	0.852	0.946	0.946
GBCQR-Normal	1.011	0.969	0.954	0.880	0.957	0.954
GBCQR-Normal ($\eta = 1$)	1.013	0.968	0.953	0.853	0.946	0.947
BCQR-ALD	1.114	1.007	0.953	0.713	0.840	0.925
BRT	1.027	0.959	0.926	0.892	1.016	0.964
BR	0.984	1.312	1.453	11.576	1.131	3.291
<i>Panel C: RMSE (inactive coefficients, relative to CQR)</i>						
GBCQR-Flat	1.012	0.970	0.947	0.882	0.955	0.953
GBCQR-Flat ($\eta = 1$)	1.014	0.964	0.945	0.854	0.943	0.944
GBCQR-Normal	1.011	0.968	0.947	0.881	0.955	0.952
GBCQR-Normal ($\eta = 1$)	1.013	0.964	0.945	0.853	0.943	0.944
BCQR-ALD	1.100	1.004	0.966	0.704	0.858	0.926
BRT	1.028	0.955	0.914	0.889	1.020	0.961
BR	0.981	1.307	1.481	11.119	1.126	3.203
<i>Panel D: Coverage of 95% Intervals</i>						
GBCQR-Flat	0.947	0.952	0.953	0.960	0.956	0.954
GBCQR-Flat ($\eta = 1$)	0.565	0.565	0.562	0.562	0.562	0.563
GBCQR-Normal	0.945	0.952	0.953	0.957	0.955	0.952
GBCQR-Normal ($\eta = 1$)	0.563	0.566	0.561	0.558	0.561	0.562
BCQR-ALD	0.907	0.934	0.945	0.976	0.926	0.938
CQR	0.942	0.951	0.952	0.953	0.946	0.949
BRT	0.936	0.960	0.965	0.990	0.955	0.961
BR	0.944	0.951	0.954	0.959	0.951	0.952
<i>Panel E: Computation Time (relative to BR, $K = 5$)</i>						
GBCQR-Flat	1.22	1.21	1.21	1.22	1.21	1.21
GBCQR-Normal	1.22	1.22	1.22	1.23	1.22	1.22
BCQR-ALD	124.45	124.38	124.56	123.62	124.34	124.27
BRT	2.91	2.91	2.91	2.91	2.91	2.91
<i>Panel F: Variable Selection (TPR, TNR)</i>						
GBCQR-SSL	(100, 96)	(100, 99)	(100, 100)	(100, 95)	(100, 98)	(100, 98)
BR-SSL	(100, 100)	(100, 100)	(100, 100)	(3, 100)	(100, 100)	(81, 100)
CQR-LASSO	(100, 100)	(100, 100)	(100, 100)	(93, 100)	(100, 100)	(99, 100)

Notes: See notes to Table 1. Results based on 500 Monte Carlo replications.

Table 6: Monte Carlo Results: $n = 200$, $\rho = 0.75$

Method	Normal	t_3	Contam.	Cauchy	Skewed	Avg.
<i>Panel A: RMSE (all coefficients, relative to CQR)</i>						
GBCQR-Flat	1.010	0.969	0.948	0.888	0.959	0.955
GBCQR-Flat ($\eta = 1$)	1.012	0.965	0.947	0.857	0.948	0.946
GBCQR-Normal	1.011	0.967	0.949	0.883	0.958	0.954
GBCQR-Normal ($\eta = 1$)	1.012	0.964	0.946	0.856	0.948	0.945
BCQR-ALD	1.117	1.001	0.977	0.713	0.850	0.932
BRT	1.025	0.955	0.918	0.900	1.015	0.963
BR	0.987	1.306	1.463	8.944	1.123	2.765
<i>Panel B: RMSE (active coefficients, relative to CQR)</i>						
GBCQR-Flat	1.008	0.973	0.950	0.887	0.958	0.955
GBCQR-Flat ($\eta = 1$)	1.011	0.969	0.947	0.856	0.946	0.946
GBCQR-Normal	1.010	0.970	0.949	0.883	0.955	0.953
GBCQR-Normal ($\eta = 1$)	1.009	0.968	0.946	0.855	0.947	0.945
BCQR-ALD	1.135	1.017	0.992	0.715	0.878	0.947
BRT	1.023	0.959	0.921	0.897	1.017	0.963
BR	0.991	1.287	1.467	9.388	1.124	2.852
<i>Panel C: RMSE (inactive coefficients, relative to CQR)</i>						
GBCQR-Flat	1.012	0.966	0.947	0.888	0.960	0.955
GBCQR-Flat ($\eta = 1$)	1.013	0.962	0.946	0.858	0.948	0.945
GBCQR-Normal	1.011	0.966	0.949	0.884	0.959	0.954
GBCQR-Normal ($\eta = 1$)	1.014	0.962	0.946	0.857	0.948	0.945
BCQR-ALD	1.107	0.992	0.968	0.712	0.835	0.923
BRT	1.027	0.953	0.917	0.903	1.014	0.963
BR	0.985	1.318	1.460	8.684	1.122	2.714
<i>Panel D: Coverage of 95% Intervals</i>						
GBCQR-Flat	0.955	0.946	0.951	0.957	0.951	0.952
GBCQR-Flat ($\eta = 1$)	0.590	0.555	0.564	0.534	0.552	0.559
GBCQR-Normal	0.949	0.944	0.951	0.951	0.943	0.947
GBCQR-Normal ($\eta = 1$)	0.587	0.555	0.566	0.531	0.552	0.558
BCQR-ALD	0.917	0.929	0.929	0.970	0.927	0.935
CQR	0.950	0.948	0.947	0.949	0.943	0.947
BRT	0.943	0.955	0.969	0.987	0.946	0.960
BR	0.951	0.945	0.946	0.967	0.941	0.950
<i>Panel E: Computation Time (relative to BR, $K = 5$)</i>						
GBCQR-Flat	1.21	1.21	1.21	1.21	1.21	1.21
GBCQR-Normal	1.22	1.22	1.22	1.22	1.22	1.22
BCQR-ALD	124.13	123.97	124.09	123.18	123.94	123.86
BRT	2.88	2.88	2.88	2.88	2.88	2.88
<i>Panel F: Variable Selection (TPR, TNR)</i>						
GBCQR-SSL	(100, 87)	(100, 96)	(100, 98)	(100, 84)	(100, 88)	(100, 91)
BR-SSL	(100, 100)	(99, 100)	(100, 100)	(1, 100)	(100, 100)	(80, 100)
CQR-LASSO	(100, 100)	(100, 100)	(100, 100)	(85, 100)	(100, 100)	(97, 100)

Notes: See notes to Table [1](#). Results based on 500 Monte Carlo replications.

2 Effective Sample Size Diagnostics

This section reports sampling efficiency across the cross-sectional data-generating processes of the Monte Carlo Section. For each replication and each method, the effective sample size (ESS) is computed for every slope coefficient and normalized by wall-clock time. Table 7 reports the minimum ESS per second across slopes (Panel A) and the mean ESS per second (Panel B), averaged over 30 Monte Carlo replications. The minimum is the headline metric, since posterior inference is bounded by the worst-mixing coordinate.

Two stylized facts emerge. First, the random-walk Metropolis–Hastings sampler underlying GBCQR-Flat and GBCQR-Normal mixes uniformly well: minimum ESS per second lies between 830 and 905 across all five error distributions, and the gap between the minimum and the mean is modest (roughly a factor of 1.3), indicating that no individual slope coefficient is a binding constraint on inference. Efficiency is essentially invariant to the tail behaviour of the error distribution, confirming that the direct posterior update on the composite quantile loss avoids the degeneracies that typically afflict samplers under heavy tails.

Second, BCQR-ALD is one to three orders of magnitude less efficient than GBCQR across every DGP. Under Normal errors the minimum ESS per second is 1.9 for BCQR-ALD against 904.6 for GBCQR-Flat, a 476-fold gap; under t_3 errors the gap is 117-fold; under the skewed DGP it is 213-fold. This reflects the $O(nK)$ latent-variable augmentation required by the asymmetric-Laplace representation, which inflates per-iteration cost and, through the coupling of the latent mixing weights with the slope block, slows mixing of the coefficients themselves. The Gaussian benchmarks BR and BRT, which do not target the composite quantile loss, are reported for reference and achieve substantially higher ESS per second as expected from their conjugate Gibbs updates.

3 Monte Carlo: Equal Quantile Grid Across Methods

In the paper, GBCQR methods use $K = 19$ equally spaced quantile levels while BCQR-ALD uses $K = 5$. This simplification is due to computational complexity: BCQR-ALD is orders of magnitude slower, so using $K = 19$ is impractical. To verify that this asymmetry does not drive the results, Tables 8–9 repeat the Monte Carlo comparison with all methods set to $K = 5$ quantile levels $\tau \in \{0.1, 0.25, 0.5, 0.75, 0.9\}$.

Table 7: Sampling efficiency across data-generating processes

Method	Normal	t_3	Cauchy	Mixture	Skewed
<i>Panel A: Minimum ESS per second</i>					
GBCQR-Flat	904.6	880.8	829.7	877.7	893.8
GBCQR-Normal	892.5	878.9	859.4	852.4	842.2
BCQR-ALD	1.9	7.5	62.4	8.0	4.2
BR	25708.1	25311.9	25870.2	25670.1	25728.7
BRT	8028.5	8124.9	8036.7	8035.6	8156.1
<i>Panel B: Mean ESS per second</i>					
GBCQR-Flat	1153.6	1136.9	1115.9	1125.8	1156.1
GBCQR-Normal	1133.2	1126.8	1117.2	1123.6	1122.1
BCQR-ALD	4.1	17.9	88.0	17.8	10.0
BR	27202.7	27164.0	27270.1	27255.8	27257.8
BRT	8558.1	8571.4	8554.9	8545.4	8576.5

Notes: Effective sample size (ESS) per parameter. Panel A reports the minimum ESS across the eight slope coefficients, divided by wall-clock time, and averaged over 50 Monte Carlo replications. Panel B replaces the minimum with the mean across slopes. Baseline design: $n = 200$, $\rho = 0.50$; MCMC chains retain 10000 draws after burn-in and thinning. Higher values indicate greater sampling efficiency per unit of computation.

The qualitative conclusions are unchanged. GBCQR-Flat uniformly improves upon CQR under non-Gaussian errors (average relative RMSE of 0.97–0.98) while matching it under normality, and maintains near-nominal coverage across all DGPs and configurations. BCQR-ALD achieves the lowest RMSE overall—particularly under Cauchy errors—but exhibits undercoverage in some DGPs, a pattern that persists across sample sizes and correlation levels.

4 Monte Carlo Time-Series: Robustness to Persistent Errors and Sample Sizes

This section extends the Monte Carlo considering an error persistence parameter ϕ_ε that varies across $\{0.25, 0.50, 0.85\}$, and also different sample sizes $n \in \{100, 200\}$. Tables [10–15](#) present the six configurations.

Three findings emerge. First, the RMSE rankings are qualitatively unchanged relative to the iid baseline: calibrated GBCQR variants match or beat CQR under every non-Gaussian DGP, with gains concentrated in the heavy-tailed specifications (AR-GARCH- t , AR-GJR- t). BR dominates under Gaussian innovations but deteriorates sharply under heavy tails. Increasing ϕ_ε compresses the

Table 8: Monte Carlo Results: $n = 100$, $K = 5$

Method	Normal	t_3	Contam.	Cauchy	Skewed	Avg.
$\rho = 0.25$						
<i>RMSE (relative to CQR)</i>						
GBCQR-Flat	1.005	0.987	0.960	0.961	0.968	0.976
BCQR-ALD	1.114	0.994	0.939	0.669	0.871	0.917
Bayesian Reg.	0.988	1.269	1.405	6.788	1.087	2.307
<i>Coverage of 95% Intervals</i>						
GBCQR-Flat	0.951	0.944	0.943	0.953	0.948	0.948
BCQR-ALD	0.921	0.933	0.937	0.968	0.925	0.937
CQR	0.950	0.943	0.948	0.955	0.949	0.949
Bayesian Reg.	0.952	0.945	0.955	0.962	0.951	0.953
$\rho = 0.50$						
<i>RMSE (relative to CQR)</i>						
GBCQR-Flat	1.002	0.984	0.956	0.962	0.969	0.975
BCQR-ALD	1.111	0.988	0.937	0.660	0.854	0.910
Bayesian Reg.	0.990	1.247	1.423	5.931	1.090	2.136
<i>Coverage of 95% Intervals</i>						
GBCQR-Flat	0.946	0.945	0.945	0.948	0.947	0.946
BCQR-ALD	0.916	0.934	0.942	0.967	0.931	0.938
CQR	0.947	0.952	0.949	0.951	0.943	0.948
Bayesian Reg.	0.949	0.948	0.947	0.964	0.948	0.951
$\rho = 0.75$						
<i>RMSE (relative to CQR)</i>						
GBCQR-Flat	1.005	0.985	0.964	0.964	0.970	0.978
BCQR-ALD	1.096	1.000	0.952	0.662	0.868	0.916
Bayesian Reg.	0.984	1.251	1.416	4.861	1.091	1.921
<i>Coverage of 95% Intervals</i>						
GBCQR-Flat	0.944	0.948	0.950	0.948	0.949	0.948
BCQR-ALD	0.919	0.943	0.940	0.964	0.929	0.939
CQR	0.944	0.949	0.952	0.947	0.948	0.948
Bayesian Reg.	0.947	0.953	0.956	0.957	0.953	0.953

Table 9: Monte Carlo Results: $n = 200$, $K = 5$

Method	Normal	t_3	Contam.	Cauchy	Skewed	Avg.
$\rho = 0.25$						
<i>RMSE (relative to CQR)</i>						
GBCQR-Flat	1.005	0.980	0.956	0.926	0.953	0.964
BCQR-ALD	1.134	0.969	0.945	0.668	0.853	0.914
Bayesian Reg.	0.991	1.272	1.460	6.122	1.116	2.192
<i>Coverage of 95% Intervals</i>						
GBCQR-Flat	0.954	0.949	0.956	0.950	0.957	0.953
BCQR-ALD	0.910	0.940	0.944	0.971	0.932	0.939
CQR	0.945	0.946	0.950	0.948	0.949	0.948
Bayesian Reg.	0.952	0.947	0.955	0.968	0.950	0.954
$\rho = 0.50$						
<i>RMSE (relative to CQR)</i>						
GBCQR-Flat	1.006	0.979	0.955	0.920	0.953	0.963
BCQR-ALD	1.111	0.991	0.953	0.648	0.852	0.911
Bayesian Reg.	0.988	1.291	1.448	10.126	1.118	2.994
<i>Coverage of 95% Intervals</i>						
GBCQR-Flat	0.950	0.954	0.953	0.959	0.959	0.955
BCQR-ALD	0.907	0.934	0.945	0.976	0.926	0.938
CQR	0.941	0.951	0.953	0.954	0.952	0.950
Bayesian Reg.	0.947	0.951	0.955	0.966	0.953	0.954
$\rho = 0.75$						
<i>RMSE (relative to CQR)</i>						
GBCQR-Flat	1.004	0.980	0.955	0.924	0.957	0.964
BCQR-ALD	1.122	0.988	0.957	0.655	0.877	0.920
Bayesian Reg.	0.991	1.289	1.444	7.388	1.113	2.445
<i>Coverage of 95% Intervals</i>						
GBCQR-Flat	0.954	0.949	0.949	0.949	0.960	0.952
BCQR-ALD	0.917	0.929	0.929	0.970	0.927	0.934
CQR	0.950	0.944	0.944	0.942	0.952	0.946
Bayesian Reg.	0.952	0.946	0.945	0.969	0.957	0.954

RMSE differentials toward unity, as serial correlation in the errors inflates estimation uncertainty uniformly across methods.

Second, coverage is the dimension most affected by serial dependence. Under low persistence ($\phi_\varepsilon = 0.25$), calibrated GBCQR achieves approximately 90% coverage — below the nominal 95% but comparable to all competitors. As ϕ_ε rises to 0.85, all IID-based intervals collapse: coverage drops to roughly 66% for CQR and 68% for GBCQR. The HAC-calibrated variants recover substantially, reaching 78–85% at $\phi_\varepsilon = 0.85$, consistently outperforming both CQR (HAC) and all non-GB methods. The calibrated learning rate $\hat{\eta}$ reflects this: the HAC estimate falls from ≈ 0.16 at low persistence to ≈ 0.06 – 0.08 at $\phi_\varepsilon = 0.85$, as the kernel absorbs the serial dependence that the IID calibration ignores.

Third, variable selection via GB-SSL maintains perfect TPR (100%) across all configurations, though TNR declines at $\phi_\varepsilon = 0.85$ — particularly for DGPs combining AR errors with GARCH volatility, where persistent serial correlation blurs the distinction between active and inactive coefficients. Doubling the sample size from $n = 100$ to $n = 200$ largely restores TNR for moderate persistence but cannot fully overcome the identification challenge at $\phi_\varepsilon = 0.85$.

Table 10: Monte Carlo Results (Time Series): $n = 100$, $\phi_\varepsilon = 0.25$

Method	AR(1)	GARCH	AR-GARCH	AR-GARCH- t	AR-GJR- t	Avg.
<i>Panel A: RMSE (relative to CQR)</i>						
GB-Flat	1.009	1.011	1.005	0.980	0.978	0.997
GB-Flat (HAC)	1.008	1.014	1.005	0.980	0.979	0.997
GB-Flat ($\eta = 1$)	1.014	1.018	1.007	0.972	0.975	0.997
GB-Normal	1.008	1.012	1.004	0.977	0.979	0.996
GB-Normal (HAC)	1.008	1.013	1.003	0.978	0.980	0.997
GB-Normal ($\eta = 1$)	1.013	1.017	1.007	0.972	0.975	0.997
CQR (HAC)	1.000	1.000	1.000	1.000	1.000	1.000
BR	0.992	0.993	1.000	1.095	1.089	1.034
BRT	1.017	1.028	1.014	0.969	0.969	0.999
BCQR-ALD	1.075	1.119	1.073	0.996	1.015	1.056
<i>Panel B: Coverage of 95% Intervals</i>						
GB-Flat	0.882	0.946	0.899	0.894	0.885	0.901
GB-Flat (HAC)	0.886	0.930	0.905	0.898	0.893	0.902
GB-Flat ($\eta = 1$)	0.460	0.561	0.477	0.469	0.470	0.487
GB-Normal	0.878	0.943	0.897	0.889	0.883	0.898
GB-Normal (HAC)	0.885	0.927	0.905	0.896	0.893	0.901
GB-Normal ($\eta = 1$)	0.460	0.565	0.478	0.466	0.471	0.488
CQR	0.852	0.932	0.875	0.855	0.851	0.873
CQR (HAC)	0.868	0.915	0.888	0.871	0.866	0.881
BR	0.891	0.955	0.913	0.903	0.901	0.913
BRT	0.883	0.942	0.902	0.903	0.893	0.904
BCQR-ALD	0.852	0.920	0.877	0.880	0.873	0.881
<i>Panel C: Calibrated $\hat{\eta}$ (median)</i>						
GB-Flat	0.164	0.164	0.163	0.158	0.160	0.162
GB-Flat (HAC)	0.157	0.185	0.157	0.153	0.150	0.161
GB-Normal	0.166	0.166	0.166	0.159	0.162	0.164
GB-Normal (HAC)	0.160	0.189	0.159	0.154	0.152	0.163
<i>Panel D: Variable Selection (TPR, TNR)</i>						
GB-SSL	(100, 97)	(100, 99)	(100, 97)	(100, 98)	(100, 98)	(100, 98)
GB-SSL (HAC)	(100, 97)	(100, 99)	(100, 97)	(100, 98)	(100, 98)	(100, 98)
GB-SSL ($\eta = 1$)	(100, 99)	(100, 99)	(100, 98)	(100, 99)	(100, 99)	(100, 99)
BR-SSL	(100, 98)	(100, 99)	(100, 98)	(99, 99)	(100, 98)	(100, 99)
CQR-LASSO	(96, 91)	(95, 92)	(95, 91)	(96, 93)	(96, 93)	(96, 92)

Notes: Results based on 500 Monte Carlo replications. The design considers $p = 8$ covariates with Toeplitz correlation $\Sigma_{jk} = \rho^{|j-k|}$, $\rho = 0.50$, and covariate persistence $\phi_x = 0.85$. True coefficients are $\beta^* = (3, 1.5, 0, 0, 2, 0, 0, 0)^\top$; active set $\mathcal{A} = \{1, 2, 5\}$, inactive set $\mathcal{A}^c = \{3, 4, 6, 7, 8\}$. All methods except BCQR-ALD use $K = 19$ equally spaced quantile levels; BCQR-ALD uses $K = 5$ due to its prohibitive $O(K)$ latent-variable augmentation cost. Panel A reports RMSE relative to frequentist CQR; bold entries indicate values below unity. Panel B reports empirical coverage of nominal 95% intervals. Panel C reports the median calibrated learning rate $\hat{\eta}$; HAC uses the quadratic spectral kernel. “ $\eta = 1$ ” denotes variants with fixed learning rate (no calibration). Panel D reports TPR and TNR for variable selection. GB-SSL and BR-SSL select via $\text{PIP}_j > 0.5$; CQR-LASSO selects if $|\hat{\beta}_j| > 0$. DGP error specifications: AR(1) with persistence ϕ_ε ; GARCH is GARCH(1,1) with Normal innovations; AR-GARCH and AR-GARCH- t combine AR(1) and GARCH structures; AR-GJR- t adds asymmetric volatility. The final column reports the arithmetic mean across all DGPs.

Table 11: Monte Carlo Results (Time Series): $n = 100$, $\phi_\varepsilon = 0.50$

Method	AR(1)	GARCH	AR-GARCH	AR-GARCH- t	AR-GJR- t	Avg.
<i>Panel A: RMSE (relative to CQR)</i>						
GB-Flat	1.009	1.009	1.002	0.988	0.985	0.999
GB-Flat (HAC)	1.009	1.009	1.001	0.989	0.987	0.999
GB-Flat ($\eta = 1$)	1.011	1.012	1.002	0.983	0.981	0.998
GB-Normal	1.007	1.009	1.001	0.986	0.984	0.997
GB-Normal (HAC)	1.007	1.008	0.998	0.988	0.986	0.997
GB-Normal ($\eta = 1$)	1.011	1.012	1.002	0.983	0.981	0.998
CQR (HAC)	1.000	1.000	1.000	1.000	1.000	1.000
BR	0.992	0.994	1.007	1.049	1.071	1.023
BRT	1.016	1.019	1.005	0.984	0.978	1.000
BCQR-ALD	1.053	1.111	1.034	1.002	0.998	1.039
<i>Panel B: Coverage of 95% Intervals</i>						
GB-Flat	0.794	0.932	0.812	0.805	0.814	0.831
GB-Flat (HAC)	0.846	0.914	0.863	0.857	0.865	0.869
GB-Flat ($\eta = 1$)	0.391	0.541	0.402	0.384	0.408	0.425
GB-Normal	0.788	0.935	0.806	0.799	0.813	0.828
GB-Normal (HAC)	0.844	0.912	0.859	0.851	0.863	0.866
GB-Normal ($\eta = 1$)	0.389	0.540	0.401	0.383	0.406	0.424
CQR	0.762	0.915	0.788	0.763	0.776	0.801
CQR (HAC)	0.834	0.899	0.847	0.834	0.846	0.852
BR	0.808	0.941	0.817	0.816	0.824	0.841
BRT	0.797	0.934	0.807	0.815	0.824	0.835
BCQR-ALD	0.778	0.914	0.794	0.791	0.810	0.817
<i>Panel C: Calibrated $\hat{\eta}$ (median)</i>						
GB-Flat	0.168	0.163	0.163	0.157	0.160	0.162
GB-Flat (HAC)	0.133	0.183	0.128	0.124	0.125	0.139
GB-Normal	0.170	0.165	0.166	0.159	0.162	0.165
GB-Normal (HAC)	0.134	0.186	0.130	0.126	0.127	0.141
<i>Panel D: Variable Selection (TPR, TNR)</i>						
GB-SSL	(100, 95)	(100, 99)	(100, 92)	(100, 94)	(100, 96)	(100, 95)
GB-SSL (HAC)	(100, 95)	(100, 99)	(100, 92)	(100, 94)	(100, 96)	(100, 95)
GB-SSL ($\eta = 1$)	(100, 97)	(100, 99)	(100, 95)	(100, 96)	(100, 97)	(100, 97)
BR-SSL	(100, 95)	(100, 99)	(100, 96)	(99, 96)	(99, 96)	(100, 97)
CQR-LASSO	(96, 92)	(95, 94)	(94, 91)	(95, 94)	(95, 93)	(95, 93)

Notes: See notes to Table [10](#). Results based on 500 Monte Carlo replications.

Table 12: Monte Carlo Results (Time Series): $n = 100$, $\phi_\varepsilon = 0.85$

Method	AR(1)	GARCH	AR-GARCH	AR-GARCH- t	AR-GJR- t	Avg.
<i>Panel A: RMSE (relative to CQR)</i>						
GB-Flat	1.001	1.013	0.997	0.993	0.994	1.000
GB-Flat (HAC)	1.001	1.012	0.997	0.995	0.996	1.000
GB-Flat ($\eta = 1$)	1.001	1.016	0.995	0.991	0.991	0.999
GB-Normal	1.001	1.010	0.996	0.992	0.992	0.998
GB-Normal (HAC)	1.000	1.012	0.996	0.994	0.992	0.999
GB-Normal ($\eta = 1$)	1.001	1.016	0.995	0.991	0.991	0.999
CQR (HAC)	1.000	1.000	1.000	1.000	1.000	1.000
BR	1.001	0.990	1.012	1.024	1.028	1.011
BRT	1.005	1.025	0.996	0.992	0.991	1.002
BCQR-ALD	0.994	1.115	0.977	0.952	0.950	0.998
<i>Panel B: Coverage of 95% Intervals</i>						
GB-Flat	0.612	0.940	0.622	0.620	0.625	0.684
GB-Flat (HAC)	0.754	0.924	0.756	0.748	0.758	0.788
GB-Flat ($\eta = 1$)	0.282	0.560	0.273	0.264	0.265	0.329
GB-Normal	0.608	0.938	0.616	0.614	0.616	0.678
GB-Normal (HAC)	0.749	0.921	0.748	0.738	0.754	0.782
GB-Normal ($\eta = 1$)	0.284	0.555	0.273	0.264	0.266	0.328
CQR	0.583	0.919	0.607	0.596	0.596	0.660
CQR (HAC)	0.752	0.902	0.763	0.762	0.760	0.788
BR	0.626	0.951	0.623	0.625	0.621	0.689
BRT	0.620	0.940	0.622	0.617	0.626	0.685
BCQR-ALD	0.630	0.916	0.626	0.618	0.631	0.684
<i>Panel C: Calibrated $\hat{\eta}$ (median)</i>						
GB-Flat	0.173	0.164	0.162	0.161	0.159	0.164
GB-Flat (HAC)	0.101	0.187	0.095	0.093	0.092	0.114
GB-Normal	0.174	0.166	0.167	0.165	0.163	0.167
GB-Normal (HAC)	0.102	0.190	0.098	0.095	0.095	0.116
<i>Panel D: Variable Selection (TPR, TNR)</i>						
GB-SSL	(100, 89)	(100, 99)	(98, 63)	(99, 67)	(98, 67)	(99, 77)
GB-SSL (HAC)	(100, 90)	(100, 99)	(98, 63)	(99, 67)	(98, 67)	(99, 77)
GB-SSL ($\eta = 1$)	(100, 91)	(100, 100)	(98, 68)	(99, 72)	(98, 72)	(99, 81)
BR-SSL	(100, 82)	(100, 99)	(95, 87)	(96, 87)	(95, 87)	(97, 88)
CQR-LASSO	(97, 87)	(96, 93)	(84, 81)	(87, 82)	(85, 82)	(90, 85)

Notes: See notes to Table [10](#). Results based on 500 Monte Carlo replications.

Table 13: Monte Carlo Results (Time Series): $n = 200$, $\phi_\varepsilon = 0.25$

Method	AR(1)	GARCH	AR-GARCH	AR-GARCH- t	AR-GJR- t	Avg.
<i>Panel A: RMSE (relative to CQR)</i>						
GB-Flat	1.010	1.011	1.004	0.973	0.966	0.993
GB-Flat (HAC)	1.010	1.011	1.004	0.973	0.968	0.993
GB-Flat ($\eta = 1$)	1.012	1.014	1.006	0.972	0.963	0.993
GB-Normal	1.010	1.012	1.005	0.973	0.966	0.993
GB-Normal (HAC)	1.010	1.013	1.004	0.974	0.969	0.994
GB-Normal ($\eta = 1$)	1.012	1.014	1.005	0.972	0.963	0.993
CQR (HAC)	1.000	1.000	1.000	1.000	1.000	1.000
BR	0.992	0.999	1.008	1.128	1.128	1.051
BRT	1.021	1.025	1.010	0.965	0.958	0.996
BCQR-ALD	1.079	1.130	1.079	1.013	0.977	1.056
<i>Panel B: Coverage of 95% Intervals</i>						
GB-Flat	0.899	0.948	0.891	0.890	0.882	0.902
GB-Flat (HAC)	0.921	0.938	0.912	0.917	0.907	0.919
GB-Flat ($\eta = 1$)	0.489	0.556	0.476	0.453	0.481	0.491
GB-Normal	0.897	0.946	0.888	0.890	0.877	0.900
GB-Normal (HAC)	0.920	0.937	0.913	0.913	0.906	0.918
GB-Normal ($\eta = 1$)	0.486	0.557	0.474	0.450	0.480	0.489
CQR	0.891	0.941	0.879	0.873	0.857	0.888
CQR (HAC)	0.919	0.934	0.911	0.904	0.890	0.912
BR	0.902	0.954	0.889	0.896	0.888	0.906
BRT	0.892	0.942	0.881	0.896	0.889	0.900
BCQR-ALD	0.851	0.913	0.848	0.864	0.874	0.870
<i>Panel C: Calibrated $\hat{\eta}$ (median)</i>						
GB-Flat	0.167	0.164	0.165	0.157	0.157	0.162
GB-Flat (HAC)	0.142	0.176	0.142	0.133	0.134	0.145
GB-Normal	0.169	0.165	0.166	0.158	0.158	0.163
GB-Normal (HAC)	0.143	0.178	0.143	0.134	0.135	0.146
<i>Panel D: Variable Selection (TPR, TNR)</i>						
GB-SSL	(100, 99)	(100, 100)	(100, 99)	(100, 99)	(100, 99)	(100, 99)
GB-SSL (HAC)	(100, 99)	(100, 100)	(100, 99)	(100, 99)	(100, 99)	(100, 99)
GB-SSL ($\eta = 1$)	(100, 100)	(100, 100)	(100, 100)	(100, 100)	(100, 100)	(100, 100)
BR-SSL	(100, 99)	(100, 100)	(100, 99)	(100, 99)	(100, 99)	(100, 99)
CQR-LASSO	(100, 99)	(100, 99)	(100, 99)	(100, 99)	(100, 99)	(100, 99)

Notes: See notes to Table [10](#). Results based on 500 Monte Carlo replications.

Table 14: Monte Carlo Results (Time Series): $n = 200$, $\phi_\varepsilon = 0.50$

Method	AR(1)	GARCH	AR-GARCH	AR-GARCH- t	AR-GJR- t	Avg.
<i>Panel A: RMSE (relative to CQR)</i>						
GB-Flat	1.008	1.014	0.997	0.974	0.975	0.994
GB-Flat (HAC)	1.007	1.013	0.997	0.978	0.978	0.994
GB-Flat ($\eta = 1$)	1.010	1.015	0.998	0.973	0.972	0.994
GB-Normal	1.007	1.012	0.997	0.975	0.975	0.993
GB-Normal (HAC)	1.007	1.014	0.997	0.976	0.977	0.994
GB-Normal ($\eta = 1$)	1.010	1.015	0.997	0.973	0.973	0.994
CQR (HAC)	1.000	1.000	1.000	1.000	1.000	1.000
BR	0.993	0.993	1.018	1.095	1.108	1.041
BRT	1.015	1.029	0.999	0.966	0.966	0.995
BCQR-ALD	1.060	1.105	1.038	0.985	0.983	1.034
<i>Panel B: Coverage of 95% Intervals</i>						
GB-Flat	0.807	0.947	0.794	0.808	0.800	0.831
GB-Flat (HAC)	0.889	0.939	0.892	0.896	0.885	0.900
GB-Flat ($\eta = 1$)	0.388	0.569	0.390	0.385	0.385	0.423
GB-Normal	0.802	0.946	0.790	0.806	0.797	0.828
GB-Normal (HAC)	0.887	0.935	0.888	0.895	0.883	0.898
GB-Normal ($\eta = 1$)	0.387	0.569	0.391	0.385	0.385	0.423
CQR	0.797	0.940	0.779	0.782	0.774	0.815
CQR (HAC)	0.880	0.934	0.883	0.887	0.878	0.892
BR	0.809	0.953	0.793	0.813	0.802	0.834
BRT	0.795	0.941	0.788	0.817	0.805	0.829
BCQR-ALD	0.773	0.924	0.759	0.786	0.790	0.806
<i>Panel C: Calibrated $\hat{\eta}$ (median)</i>						
GB-Flat	0.167	0.163	0.162	0.159	0.157	0.162
GB-Flat (HAC)	0.105	0.174	0.105	0.101	0.099	0.117
GB-Normal	0.169	0.165	0.164	0.160	0.158	0.163
GB-Normal (HAC)	0.106	0.176	0.106	0.102	0.100	0.118
<i>Panel D: Variable Selection (TPR, TNR)</i>						
GB-SSL	(100, 97)	(100, 100)	(100, 95)	(100, 97)	(100, 97)	(100, 97)
GB-SSL (HAC)	(100, 97)	(100, 100)	(100, 95)	(100, 97)	(100, 97)	(100, 97)
GB-SSL ($\eta = 1$)	(100, 98)	(100, 100)	(100, 97)	(100, 98)	(100, 98)	(100, 98)
BR-SSL	(100, 97)	(100, 100)	(100, 98)	(100, 98)	(100, 98)	(100, 98)
CQR-LASSO	(100, 98)	(100, 100)	(100, 98)	(100, 98)	(100, 98)	(100, 98)

Notes: See notes to Table [10](#). Results based on 500 Monte Carlo replications.

Table 15: Monte Carlo Results (Time Series): $n = 200$, $\phi_\varepsilon = 0.85$

Method	AR(1)	GARCH	AR-GARCH	AR-GARCH- t	AR-GJR- t	Avg.
<i>Panel A: RMSE (relative to CQR)</i>						
GB-Flat	1.004	1.012	0.996	0.983	0.986	0.996
GB-Flat (HAC)	1.004	1.012	0.997	0.990	0.992	0.999
GB-Flat ($\eta = 1$)	1.005	1.016	0.996	0.982	0.985	0.997
GB-Normal	1.004	1.012	0.995	0.983	0.986	0.996
GB-Normal (HAC)	1.002	1.013	0.995	0.988	0.991	0.998
GB-Normal ($\eta = 1$)	1.005	1.016	0.996	0.981	0.985	0.997
CQR (HAC)	1.000	1.000	1.000	1.000	1.000	1.000
BR	0.996	0.997	1.015	1.071	1.052	1.026
BRT	1.011	1.027	0.997	0.978	0.984	0.999
BCQR-ALD	1.012	1.132	1.003	0.946	0.952	1.009
<i>Panel B: Coverage of 95% Intervals</i>						
GB-Flat	0.591	0.943	0.586	0.596	0.591	0.661
GB-Flat (HAC)	0.823	0.927	0.825	0.843	0.834	0.850
GB-Flat ($\eta = 1$)	0.255	0.566	0.255	0.258	0.257	0.318
GB-Normal	0.587	0.940	0.577	0.587	0.583	0.655
GB-Normal (HAC)	0.822	0.929	0.820	0.835	0.827	0.847
GB-Normal ($\eta = 1$)	0.258	0.567	0.254	0.259	0.256	0.319
CQR	0.579	0.935	0.582	0.592	0.584	0.654
CQR (HAC)	0.829	0.925	0.839	0.851	0.835	0.856
BR	0.592	0.952	0.577	0.590	0.596	0.661
BRT	0.581	0.935	0.572	0.590	0.590	0.654
BCQR-ALD	0.571	0.903	0.562	0.589	0.587	0.643
<i>Panel C: Calibrated $\hat{\eta}$ (median)</i>						
GB-Flat	0.168	0.164	0.158	0.156	0.157	0.161
GB-Flat (HAC)	0.063	0.177	0.059	0.058	0.058	0.083
GB-Normal	0.170	0.166	0.163	0.160	0.161	0.164
GB-Normal (HAC)	0.063	0.178	0.060	0.060	0.059	0.084
<i>Panel D: Variable Selection (TPR, TNR)</i>						
GB-SSL	(100, 88)	(100, 100)	(100, 63)	(100, 69)	(100, 67)	(100, 77)
GB-SSL (HAC)	(100, 88)	(100, 100)	(100, 63)	(100, 69)	(100, 68)	(100, 77)
GB-SSL ($\eta = 1$)	(100, 90)	(100, 100)	(100, 67)	(100, 73)	(100, 72)	(100, 80)
BR-SSL	(100, 84)	(100, 100)	(100, 89)	(98, 89)	(99, 87)	(99, 90)
CQR-LASSO	(100, 93)	(100, 99)	(96, 86)	(96, 88)	(97, 86)	(98, 90)

Notes: See notes to Table [10](#). Results based on 500 Monte Carlo replications.

5 Empirical Illustration: Time-Series Behaviour of η

This section examines the behaviour of the calibrated learning rate η across expanding estimation windows for each of the 31 predictors. We report estimates obtained with both the standard (non-HAC) and the HAC score covariance estimators, as described in the Methodology Section.

Figures [1-2](#) plot the estimated η over time, while Table [16](#) reports summary statistics. Two findings stand out. First, the non-HAC and HAC estimates are remarkably similar for the vast majority of predictors: the two paths nearly overlap for variables such as d/p, b/m, svar, and skvw, and differ only modestly for others (e.g., ntis, govik, tail). This close agreement reflects the near-zero serial correlation in the score contributions, which are driven primarily by the equity

premium residuals. Because equity premium innovations generally exhibits low autocorrelation, the check-function component of the score process inherits this property, limiting the scope for the HAC correction to alter the long-run variance estimate—even when the predictor itself is highly persistent.

Second, the calibrated η is substantially below one for nearly all predictors and estimation windows, with typical values in the range 0.2–0.5. Only two variance-based predictors—the variance premium (vp) and implied variance (impvar)—occasionally exceed the $\eta = 1$ threshold, and even then only during short episodes in relatively small samples. This finding reinforces the conclusion that the standard Gibbs posterior with $\eta = 1$ systematically over-weights the composite quantile loss, producing posteriors that are too concentrated relative to the frequentist sampling distribution, and that learning-rate calibration is empirically important.



Figure 1: Calibrated learning rate η across expanding estimation windows. The blue line shows the non-HAC estimate, the red line shows the HAC estimate, and the horizontal black line marks $\eta = 1$. The out-of-sample period begins in 1994Q4.

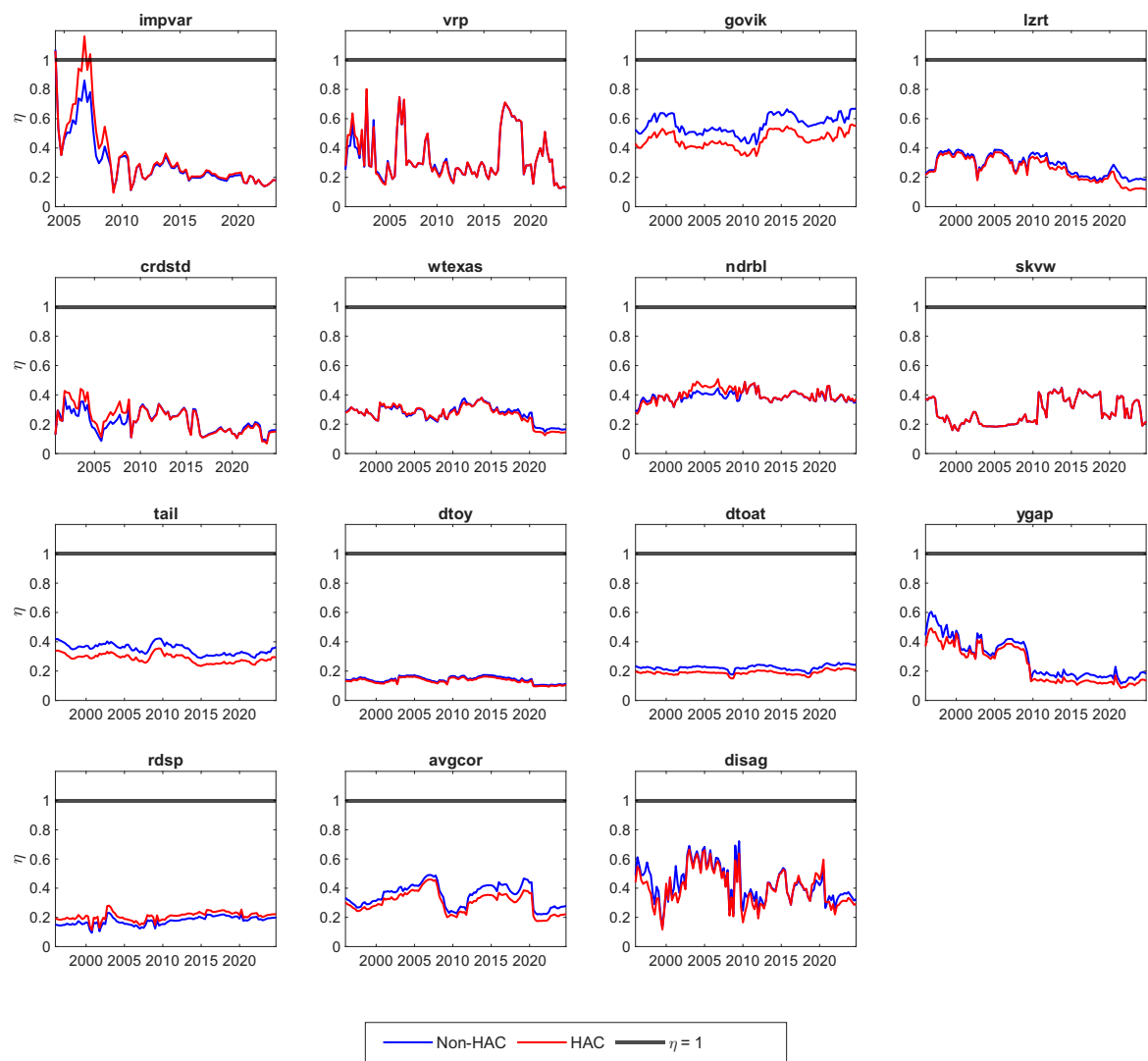


Figure 2: Calibrated learning rate η across expanding estimation windows. See notes to Figure 1.

Table 16: Summary statistics of calibrated η across expanding windows.

Predictor	Non-HAC				HAC			
	Mean	Std	Min	Max	Mean	Std	Min	Max
d/p	0.203	0.019	0.166	0.259	0.203	0.020	0.166	0.259
d/y	0.252	0.029	0.190	0.307	0.283	0.034	0.206	0.341
e/p	0.373	0.062	0.299	0.496	0.305	0.056	0.229	0.414
d/e	0.193	0.091	0.086	0.381	0.175	0.075	0.083	0.350
b/m	0.228	0.025	0.175	0.274	0.233	0.027	0.178	0.305
tbl	0.334	0.028	0.279	0.413	0.283	0.025	0.232	0.344
lty	0.382	0.041	0.271	0.464	0.326	0.035	0.229	0.394
ltr	0.342	0.062	0.231	0.540	0.361	0.075	0.225	0.589
tms	0.399	0.045	0.307	0.497	0.362	0.045	0.274	0.469
dfy	0.123	0.023	0.083	0.165	0.118	0.023	0.079	0.165
dfr	0.265	0.069	0.127	0.536	0.270	0.065	0.133	0.515
infl	0.358	0.068	0.214	0.449	0.345	0.065	0.207	0.434
ntis	0.172	0.053	0.100	0.256	0.230	0.092	0.107	0.409
svar	0.221	0.060	0.127	0.346	0.214	0.059	0.125	0.333
i/k	0.337	0.075	0.218	0.512	0.326	0.077	0.209	0.513
vp	0.446	0.308	0.077	2.024	0.467	0.311	0.090	2.104
impvar	0.304	0.186	0.112	1.070	0.340	0.239	0.096	1.162
vrp	0.335	0.160	0.127	0.746	0.339	0.166	0.127	0.802
govik	0.559	0.062	0.424	0.670	0.456	0.053	0.345	0.561
lzrt	0.281	0.070	0.171	0.390	0.255	0.078	0.112	0.373
crdstd	0.215	0.074	0.075	0.385	0.230	0.090	0.068	0.441
wtexas	0.274	0.057	0.148	0.381	0.267	0.063	0.124	0.381
ndrbl	0.383	0.044	0.237	0.488	0.393	0.057	0.227	0.508
skvw	0.287	0.087	0.154	0.449	0.287	0.087	0.155	0.447
tail	0.354	0.036	0.289	0.423	0.291	0.031	0.235	0.355
dtoy	0.143	0.020	0.100	0.174	0.134	0.020	0.093	0.163
dtoat	0.225	0.017	0.177	0.261	0.192	0.015	0.150	0.221
ygap	0.298	0.142	0.114	0.605	0.250	0.131	0.083	0.502
rdsp	0.175	0.027	0.094	0.231	0.208	0.028	0.113	0.280
avgcor	0.349	0.075	0.221	0.491	0.305	0.072	0.174	0.460
disag	0.420	0.131	0.112	0.722	0.392	0.132	0.100	0.665

Notes: The table reports mean, standard deviation, minimum, and maximum of the calibrated learning rate η computed at each expanding estimation window. Non-HAC uses the standard score covariance $\hat{I} = n^{-1} \sum_i g_i g_i'$; HAC uses the Quadratic Spectral kernel with the data-driven bandwidth of Andrews (1991). The out-of-sample period begins in 1994Q4.

6 Robustness: Risk Aversion and Transaction Costs

This section examines the sensitivity of the out-of-sample results to the investor’s risk aversion coefficient ($\gamma \in \{3, 5\}$) and proportional transaction costs ($\text{TC} \in \{0\%, 0.50\%, 1.00\%\}$). Tables 17–22 report the full set of nine configurations. Notice that R_{OOS}^2 is mechanically invariant to (γ, TC) , so we focus the discussion on the economic gains.

First, GB’s Sharpe ratio advantage is robust across all (γ, TC) configurations. At every risk aversion level, GB achieves the highest average annualized Sharpe ratio, indicating that its risk-adjusted performance is not an artefact of the baseline calibration.

Second, GB dominates on ΔCER at moderate risk aversion. At the baseline $\gamma = 3$ (Table 18), GB averages 0.72 pp — consistent with the main results. At $\gamma = 5$, OLS delivers the highest average ΔCER , benefiting from more aggressive positions that the lower variance penalty permits, but GB retains the best Sharpe ratio.

Finally, GB’s turnover advantage persists throughout: it consistently records the lowest average quarterly turnover. This explains why transaction costs have a relatively muted effect on GB’s economic metrics compared with OLS and BRT, whose higher turnover amplifies the cost drag. For instance, raising TC from 0% to 1% at $\gamma = 3$ reduces GB’s average ΔCER by only 0.22 pp (from 0.83 to 0.61), versus 0.41 pp for OLS and 0.30 pp for BRT.

Table 17: Out-of-Sample Equity Premium Forecasting: $\gamma = 3$, TC = 0.00% (1994Q4–2024Q4)

	R^2_{OOS} (%)				ΔCER (%)				Sharpe Ratio					Turnover (%)		
	OLS	BRT	GB	GB_f	OLS	BRT	GB	GB_f	HM	OLS	BRT	GB	GB_f	OLS	BRT	GB
<i>Panel A: Valuation Ratios</i>																
d/p	-14.41	-0.23	0.79	-2.13	-0.10	0.09	0.75	0.53	0.32	0.33	0.30	0.36	0.30	2	6	5
d/y	-18.07	-0.48	0.77	-2.78	-0.16	0.13	0.88	1.25	0.32	0.27	0.30	0.36	0.35	1	7	6
e/p	-3.70	-0.42	1.19	-1.64	0.97	-0.32	0.29	0.26	0.32	0.32	0.25	0.33	0.27	10	8	7
d/e	-0.59	-7.29	-6.40	-4.72	-0.18	-0.80	-0.83	-1.12	0.32	0.30	0.30	0.32	0.27	7	7	6
b/m	-16.95	-0.05	0.47	-1.09	-0.48	-0.05	0.73	-0.28	0.32	0.36	0.32	0.37	0.27	1	5	5
<i>Panel B: Interest Rates and Spreads</i>																
tbl	-0.38	-0.12	-0.78	-0.08	1.00	0.96	0.82	0.91	0.32	0.39	0.40	0.41	0.39	6	5	4
lty	0.65	0.67	-0.02	0.64	0.98	0.98	0.82	0.87	0.32	0.38	0.39	0.40	0.38	5	5	4
ltr	0.41	0.77	0.05	0.77	-0.67	-0.28	0.36	-0.67	0.32	0.27	0.31	0.37	0.27	34	32	24
tms	-2.94	-1.39	-2.50	-1.39	-0.26	-0.09	0.18	-0.16	0.32	0.32	0.34	0.39	0.33	10	8	6
dfy	-2.52	-0.10	-0.46	-0.05	0.40	-0.93	-0.07	-0.83	0.32	0.33	0.27	0.35	0.27	9	9	6
dfr	-2.52	-0.84	-1.78	-0.98	0.16	-0.22	-0.08	-0.41	0.32	0.32	0.32	0.36	0.30	17	14	9
crdstd	-0.95	-1.08	-0.30	-1.36	2.22	2.88	2.87	2.72	0.38	0.50	0.54	0.54	0.53	14	14	15
<i>Panel C: Macroeconomic Variables</i>																
infl	0.48	0.31	-0.31	0.32	1.09	0.37	0.79	0.27	0.32	0.37	0.35	0.40	0.34	23	6	5
ntis	-9.48	-2.37	-5.00	-2.68	1.38	0.63	1.23	0.54	0.32	0.42	0.38	0.43	0.37	8	6	4
svar	0.36	-0.67	-0.60	0.09	0.15	-0.17	0.47	-0.18	0.32	0.34	0.31	0.38	0.31	12	9	6
i/k	1.73*	2.42*	2.45**	2.20*	2.54	2.94	3.07	2.84	0.30	0.45	0.48	0.49	0.47	8	6	6
govik	-2.27	-0.81	-0.53	-1.21	-1.26	-0.51	-0.37	-0.82	0.30	0.18	0.26	0.28	0.23	7	6	6
ndrbl	0.66	0.76*	0.73*	0.79	0.48	0.55	0.57	0.49	0.26	0.33	0.35	0.35	0.34	22	23	22
wtexas	-2.29	-1.19	-1.90	-0.91	-1.37	-0.32	-0.12	-0.37	0.32	0.27	0.32	0.36	0.31	43	19	13
<i>Panel D: Variance and Risk Measures</i>																
vp	3.39*	5.45*	3.21*	4.87*	-1.55	0.00	1.28	-1.61	0.41	0.34	0.44	0.50	0.35	31	21	15
impvar	-6.41	-3.11	-1.78	-5.51	1.75	3.11	3.68	2.34	0.68	0.77	0.82	0.84	0.79	21	23	21
vrp	-6.71	-1.08	-2.69	-2.74	-2.40	0.08	0.74	-1.70	0.35	0.18	0.36	0.40	0.24	51	49	47
tail	0.26	0.23	-0.87	0.21	0.66	0.70	0.85	0.51	0.32	0.36	0.37	0.41	0.35	7	6	5
skvw	-1.45	-0.49	-0.46	-0.31	-0.74	0.15	0.42	0.07	0.32	0.25	0.35	0.38	0.33	29	18	12
<i>Panel E: Cross-Section and Technical</i>																
lzrt	-11.32	-1.92	-1.99	-2.88	-0.52	0.40	0.18	1.37	0.32	0.25	0.32	0.34	0.37	30	21	22
avgcor	4.61*	3.79*	2.47*	4.26**	2.46	2.29	1.89	2.18	0.32	0.45	0.46	0.46	0.44	21	12	8
dtoy	0.61	-3.84	-2.39	-2.23	0.15	0.61	-0.21	0.00	0.32	0.33	0.34	0.34	0.31	8	15	11
dtoat	0.11	0.28	0.07	0.17	-0.21	0.26	0.76	0.00	0.32	0.29	0.35	0.40	0.32	9	5	4
ygap	-3.05	-1.76	-0.77	-3.05	0.71	0.75	0.49	1.01	0.28	0.29	0.31	0.33	0.31	9	8	8
rdsp	-3.30	0.59	0.05	0.44	-0.89	1.67	1.30	1.21	0.32	0.27	0.41	0.41	0.38	10	13	8
disag	-1.73	-0.26	0.28	-0.79	0.80	2.19	2.04	1.82	0.29	0.28	0.38	0.38	0.35	22	18	16
Average	-3.15	-0.46	-0.61	-0.77	0.23	0.58	0.83	0.42	0.33	0.34	0.37	0.40	0.35	16	13	11

Notes: Out-of-sample forecasting results for the quarterly U.S. equity premium, 1994Q4 to 2024Q4. R^2_{OOS} is measured relative to the historical mean (HM) benchmark. ΔCER is the annualized certainty equivalent return gain (percentage points) relative to HM. Portfolio weights derive from mean-variance optimization with risk aversion $\gamma = 3$, transaction costs of 0.00%, and a 5-years rolling window for variance; weights are constrained to $[0.0, 1.5]$. Sharpe ratios are annualized; turnover is reported as a quarterly average. BRT: Bayesian regression with Student- t errors ($\nu = 5$). GB: generalized Bayesian CQR with flat prior and calibrated η . GB_f : same model with fixed $\eta = 1$. Bold entries denote positive R^2_{OOS} or ΔCER , the highest Sharpe ratio, or the lowest turnover among the reported methods. *, **, ***: 10%, 5%, 1% significance.

Table 18: Out-of-Sample Equity Premium Forecasting: $\gamma = 3$, TC = 0.50% (1994Q4–2024Q4)

	R^2_{OOS} (%)				ΔCER (%)				Sharpe Ratio					Turnover (%)		
	OLS	BRT	GB	GB_f	OLS	BRT	GB	GB_f	HM	OLS	BRT	GB	GB_f	OLS	BRT	GB
<i>Panel A: Valuation Ratios</i>																
d/p	-14.41	-0.23	0.79	-2.13	-0.03	0.06	0.75	0.49	0.31	0.32	0.30	0.35	0.29	2	6	5
d/y	-18.07	-0.48	0.77	-2.78	-0.08	0.08	0.87	1.20	0.31	0.27	0.29	0.35	0.33	1	7	6
e/p	-3.70	-0.42	1.19	-1.64	0.87	-0.40	0.24	0.16	0.31	0.30	0.24	0.32	0.25	10	8	7
d/e	-0.59	-7.29	-6.40	-4.72	-0.21	-0.85	-0.87	-1.17	0.31	0.30	0.29	0.31	0.26	7	7	6
b/m	-16.95	-0.05	0.47	-1.09	-0.40	-0.06	0.73	-0.33	0.31	0.34	0.31	0.37	0.26	1	5	5
<i>Panel B: Interest Rates and Spreads</i>																
tbl	-0.38	-0.12	-0.78	-0.08	1.00	0.95	0.83	0.91	0.31	0.39	0.39	0.40	0.38	6	5	4
lty	0.65	0.67	-0.02	0.64	0.99	0.99	0.85	0.88	0.31	0.38	0.39	0.40	0.37	5	5	4
ltr	0.41	0.77	0.05	0.77	-1.21	-0.79	0.00	-1.21	0.32	0.24	0.27	0.34	0.24	34	32	24
tms	-2.94	-1.39	-2.50	-1.39	-0.36	-0.14	0.17	-0.21	0.31	0.31	0.33	0.38	0.32	10	8	6
dfy	-2.52	-0.10	-0.46	-0.05	0.33	-1.01	-0.09	-0.89	0.31	0.32	0.26	0.34	0.26	9	9	6
dfr	-2.52	-0.84	-1.78	-0.98	-0.06	-0.39	-0.13	-0.57	0.32	0.30	0.30	0.35	0.28	17	14	9
crdstd	-0.95	-1.08	-0.30	-1.36	2.06	2.73	2.70	2.56	0.37	0.48	0.52	0.52	0.51	14	14	15
<i>Panel C: Macroeconomic Variables</i>																
infl	0.48	0.31	-0.31	0.32	0.74	0.36	0.80	0.25	0.31	0.35	0.34	0.39	0.33	23	6	5
ntis	-9.48	-2.37	-5.00	-2.68	1.32	0.63	1.24	0.52	0.31	0.42	0.37	0.42	0.36	8	6	4
svar	0.36	-0.67	-0.60	0.09	0.02	-0.26	0.46	-0.24	0.31	0.33	0.30	0.37	0.30	12	9	6
i/k	1.73*	2.42*	2.45**	2.20*	2.48	2.93	3.06	2.83	0.30	0.45	0.47	0.48	0.47	8	6	6
govik	-2.27	-0.81	-0.53	-1.21	-1.31	-0.53	-0.39	-0.85	0.30	0.17	0.26	0.27	0.22	7	6	6
ndrbl	0.66	0.76*	0.73*	0.79	0.14	0.22	0.24	0.15	0.25	0.31	0.33	0.33	0.32	22	23	22
wtexas	-2.29	-1.19	-1.90	-0.91	-2.11	-0.57	-0.27	-0.62	0.32	0.23	0.31	0.35	0.29	43	19	13
<i>Panel D: Variance and Risk Measures</i>																
vp	3.39*	5.45*	3.21*	4.87*	-2.02	-0.30	1.11	-2.11	0.41	0.31	0.41	0.49	0.31	31	21	15
impvar	-6.41	-3.11	-1.78	-5.51	1.48	2.81	3.40	1.93	0.67	0.75	0.80	0.81	0.76	21	23	21
vrp	-6.71	-1.08	-2.69	-2.74	-3.33	-0.79	-0.09	-2.66	0.35	0.11	0.31	0.35	0.18	51	49	47
tail	0.26	0.23	-0.87	0.21	0.64	0.70	0.86	0.51	0.31	0.35	0.36	0.40	0.35	7	6	5
skvw	-1.45	-0.49	-0.46	-0.31	-1.23	-0.08	0.30	-0.10	0.31	0.22	0.33	0.37	0.32	29	18	12
<i>Panel E: Cross-Section and Technical</i>																
lzrt	-11.32	-1.92	-1.99	-2.88	-0.98	0.10	-0.15	1.06	0.32	0.21	0.29	0.31	0.34	30	21	22
avgcor	4.61*	3.79*	2.47*	4.26**	2.15	2.16	1.83	1.98	0.32	0.43	0.44	0.45	0.43	21	12	8
dtoy	0.61	-3.84	-2.39	-2.23	0.09	0.41	-0.33	-0.14	0.32	0.32	0.33	0.33	0.30	8	15	11
dtoat	0.11	0.28	0.07	0.17	-0.28	0.27	0.79	0.00	0.32	0.28	0.34	0.39	0.31	9	5	4
ygap	-3.05	-1.76	-0.77	-3.05	0.63	0.70	0.45	0.93	0.28	0.27	0.30	0.32	0.30	9	8	8
rdsp	-3.30	0.59	0.05	0.44	-0.98	1.53	1.24	1.11	0.31	0.26	0.39	0.40	0.37	10	13	8
disag	-1.73	-0.26	0.28	-0.79	0.48	1.95	1.82	1.57	0.28	0.25	0.36	0.36	0.33	22	18	16
Average	-3.15	-0.46	-0.61	-0.77	0.03	0.43	0.72	0.26	0.33	0.32	0.35	0.39	0.33	16	13	11

Notes: Out-of-sample forecasting results for the quarterly U.S. equity premium, 1994Q4 to 2024Q4. R^2_{OOS} is measured relative to the historical mean (HM) benchmark. ΔCER is the annualized certainty equivalent return gain (percentage points) relative to HM. Portfolio weights derive from mean-variance optimization with risk aversion $\gamma = 3$, transaction costs of 0.50%, and a 5-years rolling window for variance; weights are constrained to $[0.0, 1.5]$. Sharpe ratios are annualized; turnover is reported as a quarterly average. BRT: Bayesian regression with Student- t errors ($\nu = 5$). GB: generalized Bayesian CQR with flat prior and calibrated η . GB_f : same model with fixed $\eta = 1$. Bold entries denote positive R^2_{OOS} or ΔCER , the highest Sharpe ratio, or the lowest turnover among the reported methods. *, **, ***: 10%, 5%, 1% significance.

Table 19: Out-of-Sample Equity Premium Forecasting: $\gamma = 3$, TC = 1.00% (1994Q4–2024Q4)

	R^2_{OOS} (%)				ΔCER (%)				Sharpe Ratio					Turnover (%)		
	OLS	BRT	GB	GB_f	OLS	BRT	GB	GB_f	HM	OLS	BRT	GB	GB_f	OLS	BRT	GB
<i>Panel A: Valuation Ratios</i>																
d/p	-14.41	-0.23	0.79	-2.13	0.04	0.04	0.76	0.44	0.31	0.31	0.29	0.35	0.27	2	6	5
d/y	-18.07	-0.48	0.77	-2.78	-0.00	0.03	0.86	1.15	0.31	0.27	0.28	0.35	0.32	1	7	6
e/p	-3.70	-0.42	1.19	-1.64	0.78	-0.48	0.19	0.07	0.31	0.28	0.23	0.31	0.24	10	8	7
d/e	-0.59	-7.29	-6.40	-4.72	-0.24	-0.91	-0.90	-1.22	0.31	0.29	0.29	0.31	0.25	7	7	6
b/m	-16.95	-0.05	0.47	-1.09	-0.32	-0.07	0.74	-0.38	0.31	0.33	0.30	0.36	0.25	1	5	5
<i>Panel B: Interest Rates and Spreads</i>																
tbl	-0.38	-0.12	-0.78	-0.08	0.99	0.95	0.85	0.91	0.31	0.38	0.39	0.40	0.38	6	5	4
lty	0.65	0.67	-0.02	0.64	1.00	1.01	0.88	0.89	0.31	0.38	0.38	0.40	0.37	5	5	4
ltr	0.41	0.77	0.05	0.77	-1.76	-1.30	-0.36	-1.75	0.31	0.20	0.24	0.32	0.20	34	32	24
tms	-2.94	-1.39	-2.50	-1.39	-0.46	-0.20	0.15	-0.27	0.31	0.30	0.32	0.38	0.31	10	8	6
dfy	-2.52	-0.10	-0.46	-0.05	0.26	-1.09	-0.11	-0.95	0.31	0.31	0.25	0.34	0.25	9	9	6
dfr	-2.52	-0.84	-1.78	-0.98	-0.28	-0.56	-0.19	-0.73	0.31	0.28	0.29	0.34	0.27	17	14	9
crdstd	-0.95	-1.08	-0.30	-1.36	1.90	2.58	2.52	2.41	0.36	0.46	0.50	0.50	0.49	14	14	15
<i>Panel C: Macroeconomic Variables</i>																
infl	0.48	0.31	-0.31	0.32	0.39	0.36	0.80	0.23	0.31	0.32	0.34	0.39	0.32	23	6	5
ntis	-9.48	-2.37	-5.00	-2.68	1.27	0.62	1.26	0.50	0.31	0.41	0.37	0.42	0.36	8	6	4
svar	0.36	-0.67	-0.60	0.09	-0.12	-0.36	0.46	-0.30	0.31	0.31	0.29	0.37	0.29	12	9	6
i/k	1.73*	2.42*	2.45**	2.20*	2.42	2.91	3.06	2.81	0.29	0.44	0.47	0.48	0.46	8	6	6
govik	-2.27	-0.81	-0.53	-1.21	-1.36	-0.56	-0.41	-0.89	0.29	0.16	0.25	0.26	0.22	7	6	6
ndrbl	0.66	0.76*	0.73*	0.79	-0.20	-0.11	-0.09	-0.19	0.25	0.28	0.31	0.31	0.30	22	23	22
wtexas	-2.29	-1.19	-1.90	-0.91	-2.86	-0.83	-0.41	-0.86	0.31	0.19	0.29	0.34	0.28	43	19	13
<i>Panel D: Variance and Risk Measures</i>																
vp	3.39*	5.45*	3.21*	4.87*	-2.50	-0.61	0.93	-2.62	0.40	0.28	0.39	0.47	0.28	31	21	15
impvar	-6.41	-3.11	-1.78	-5.51	1.20	2.51	3.11	1.52	0.66	0.73	0.78	0.79	0.73	21	23	21
vrp	-6.71	-1.08	-2.69	-2.74	-4.26	-1.67	-0.93	-3.62	0.34	0.05	0.25	0.30	0.12	51	49	47
tail	0.26	0.23	-0.87	0.21	0.63	0.70	0.88	0.51	0.31	0.35	0.36	0.40	0.34	7	6	5
skvw	-1.45	-0.49	-0.46	-0.31	-1.72	-0.32	0.17	-0.28	0.31	0.18	0.32	0.36	0.30	29	18	12
<i>Panel E: Cross-Section and Technical</i>																
lzrt	-11.32	-1.92	-1.99	-2.88	-1.44	-0.21	-0.48	0.75	0.31	0.17	0.27	0.29	0.31	30	21	22
avgcor	4.61*	3.79*	2.47*	4.26**	1.84	2.03	1.77	1.78	0.31	0.41	0.43	0.44	0.41	21	12	8
dtoy	0.61	-3.84	-2.39	-2.23	0.03	0.21	-0.45	-0.28	0.31	0.31	0.31	0.31	0.28	8	15	11
dtoat	0.11	0.28	0.07	0.17	-0.36	0.29	0.81	-0.00	0.31	0.27	0.34	0.39	0.31	9	5	4
ygap	-3.05	-1.76	-0.77	-3.05	0.56	0.64	0.41	0.85	0.27	0.26	0.29	0.31	0.29	9	8	8
rdsp	-3.30	0.59	0.05	0.44	-1.07	1.38	1.19	1.00	0.31	0.25	0.38	0.40	0.36	10	13	8
disag	-1.73	-0.26	0.28	-0.79	0.15	1.71	1.60	1.32	0.27	0.22	0.33	0.34	0.30	22	18	16
Average	-3.15	-0.46	-0.61	-0.77	-0.18	0.28	0.61	0.09	0.32	0.30	0.34	0.38	0.32	16	13	11

Notes: Out-of-sample forecasting results for the quarterly U.S. equity premium, 1994Q4 to 2024Q4. R^2_{OOS} is measured relative to the historical mean (HM) benchmark. ΔCER is the annualized certainty equivalent return gain (percentage points) relative to HM. Portfolio weights derive from mean-variance optimization with risk aversion $\gamma = 3$, transaction costs of 1.00%, and a 5-years rolling window for variance; weights are constrained to $[0.0, 1.5]$. Sharpe ratios are annualized; turnover is reported as a quarterly average. BRT: Bayesian regression with Student- t errors ($\nu = 5$). GB: generalized Bayesian CQR with flat prior and calibrated η . GB_f : same model with fixed $\eta = 1$. Bold entries denote positive R^2_{OOS} or ΔCER , the highest Sharpe ratio, or the lowest turnover among the reported methods. *, **, ***: 10%, 5%, 1% significance.

Table 20: Out-of-Sample Equity Premium Forecasting: $\gamma = 5$, TC = 0.00% (1994Q4–2024Q4)

	R^2_{OOS} (%)				ΔCER (%)				Sharpe Ratio					Turnover (%)		
	OLS	BRT	GB	GB_f	OLS	BRT	GB	GB_f	HM	OLS	BRT	GB	GB_f	OLS	BRT	GB
<i>Panel A: Valuation Ratios</i>																
d/p	-14.41	-0.23	0.79	-2.13	2.58	1.45	0.54	2.79	0.22	0.33	0.26	0.26	0.29	1	6	7
d/y	-18.07	-0.48	0.77	-2.78	2.54	2.07	1.13	3.14	0.22	0.27	0.28	0.27	0.32	1	6	7
e/p	-3.70	-0.42	1.19	-1.64	2.87	1.80	0.35	2.41	0.22	0.27	0.24	0.23	0.24	7	8	8
d/e	-0.59	-7.29	-6.40	-4.72	0.15	-1.69	-1.84	-1.38	0.22	0.22	0.19	0.22	0.17	9	8	7
b/m	-16.95	-0.05	0.47	-1.09	2.35	0.30	0.13	2.17	0.22	0.36	0.23	0.26	0.27	1	7	6
<i>Panel B: Interest Rates and Spreads</i>																
tbl	-0.38	-0.12	-0.78	-0.08	0.19	0.21	0.12	0.14	0.22	0.29	0.30	0.33	0.28	6	6	5
lty	0.65	0.67	-0.02	0.64	0.24	0.24	0.16	0.17	0.22	0.27	0.29	0.33	0.26	6	6	5
ltr	0.41	0.77	0.05	0.77	0.50	0.36	-0.48	0.41	0.22	0.23	0.24	0.27	0.22	31	30	24
tms	-2.94	-1.39	-2.50	-1.39	0.24	0.19	-0.51	0.31	0.22	0.24	0.26	0.31	0.25	9	8	7
dfy	-2.52	-0.10	-0.46	-0.05	0.79	-0.26	-0.60	-0.34	0.22	0.27	0.20	0.25	0.18	8	9	7
dfr	-2.52	-0.84	-1.78	-0.98	-0.35	-0.38	-0.40	-0.31	0.22	0.20	0.22	0.29	0.20	15	13	10
crdstd	-0.95	-1.08	-0.30	-1.36	2.04	2.31	2.33	2.27	0.27	0.40	0.42	0.42	0.42	15	15	15
<i>Panel C: Macroeconomic Variables</i>																
infl	0.48	0.31	-0.31	0.32	1.64	-0.15	-0.17	0.10	0.22	0.32	0.24	0.30	0.23	21	6	6
ntis	-9.48	-2.37	-5.00	-2.68	-0.35	-0.40	-0.31	-0.35	0.22	0.33	0.27	0.34	0.26	9	6	6
svar	0.36	-0.67	-0.60	0.09	-0.17	-0.43	-0.22	-0.18	0.22	0.25	0.20	0.29	0.20	12	9	7
i/k	1.73*	2.42*	2.45**	2.20*	3.46	2.79	2.67	2.84	0.21	0.47	0.47	0.46	0.46	8	8	8
govik	-2.27	-0.81	-0.53	-1.21	1.16	0.47	0.25	0.91	0.21	0.19	0.21	0.22	0.22	6	7	7
ndrbl	0.66	0.76*	0.73*	0.79	0.10	-0.97	-0.99	-0.69	0.22	0.33	0.33	0.33	0.33	19	20	20
wtexas	-2.29	-1.19	-1.90	-0.91	-0.70	-0.35	-0.66	-0.13	0.22	0.22	0.24	0.28	0.22	33	16	14
<i>Panel D: Variance and Risk Measures</i>																
vp	3.39*	5.45*	3.21*	4.87*	-1.26	-0.72	-0.14	-1.52	0.29	0.27	0.37	0.43	0.28	28	28	22
impvar	-6.41	-3.11	-1.78	-5.51	1.51	2.16	2.45	1.82	0.60	0.71	0.76	0.78	0.73	17	20	21
vrp	-6.71	-1.08	-2.69	-2.74	-0.86	-0.18	-0.27	-0.53	0.25	0.12	0.27	0.30	0.19	40	47	47
tail	0.26	0.23	-0.87	0.21	0.25	0.05	-0.04	0.17	0.22	0.25	0.26	0.32	0.24	7	6	5
skvw	-1.45	-0.49	-0.46	-0.31	0.54	-0.48	-0.35	-0.33	0.22	0.18	0.25	0.29	0.23	24	18	12
<i>Panel E: Cross-Section and Technical</i>																
lzrt	-11.32	-1.92	-1.99	-2.88	1.88	2.13	1.47	2.91	0.22	0.26	0.31	0.32	0.34	21	16	18
avgcor	4.61*	3.79*	2.47*	4.26**	0.88	0.92	1.02	0.66	0.22	0.32	0.34	0.40	0.30	21	12	11
dtoy	0.61	-3.84	-2.39	-2.23	0.26	1.42	-0.05	0.99	0.22	0.23	0.29	0.27	0.25	8	14	11
dtoat	0.11	0.28	0.07	0.17	1.09	-0.07	-0.17	0.19	0.22	0.22	0.24	0.30	0.22	8	6	5
ygap	-3.05	-1.76	-0.77	-3.05	1.88	1.11	-0.07	1.81	0.21	0.28	0.27	0.25	0.29	7	8	8
rdsp	-3.30	0.59	0.05	0.44	-0.82	1.17	0.59	1.05	0.22	0.16	0.30	0.32	0.28	11	11	8
disag	-1.73	-0.26	0.28	-0.79	1.64	2.53	2.32	2.24	0.20	0.21	0.31	0.32	0.28	18	17	15
Average	-3.15	-0.46	-0.61	-0.77	0.85	0.57	0.27	0.77	0.23	0.28	0.29	0.32	0.28	14	13	12

Notes: Out-of-sample forecasting results for the quarterly U.S. equity premium, 1994Q4 to 2024Q4. R^2_{OOS} is measured relative to the historical mean (HM) benchmark. ΔCER is the annualized certainty equivalent return gain (percentage points) relative to HM. Portfolio weights derive from mean-variance optimization with risk aversion $\gamma = 5$, transaction costs of 0.00%, and a 5-years rolling window for variance; weights are constrained to $[0.0, 1.5]$. Sharpe ratios are annualized; turnover is reported as a quarterly average. BRT: Bayesian regression with Student- t errors ($\nu = 5$). GB: generalized Bayesian CQR with flat prior and calibrated η . GB_f : same model with fixed $\eta = 1$. Bold entries denote positive R^2_{OOS} or ΔCER , the highest Sharpe ratio, or the lowest turnover among the reported methods. *, **, ***: 10%, 5%, 1% significance.

Table 21: Out-of-Sample Equity Premium Forecasting: $\gamma = 5$, TC = 0.50% (1994Q4–2024Q4)

	R^2_{OOS} (%)				ΔCER (%)				Sharpe Ratio					Turnover (%)		
	OLS	BRT	GB	GB _f	OLS	BRT	GB	GB _f	HM	OLS	BRT	GB	GB _f	OLS	BRT	GB
<i>Panel A: Valuation Ratios</i>																
d/p	-14.41	-0.23	0.79	-2.13	2.71	1.45	0.52	2.82	0.21	0.32	0.25	0.25	0.28	1	6	7
d/y	-18.07	-0.48	0.77	-2.78	2.68	2.08	1.12	3.16	0.21	0.27	0.27	0.26	0.31	1	6	7
e/p	-3.70	-0.42	1.19	-1.64	2.88	1.77	0.31	2.40	0.21	0.25	0.22	0.22	0.22	7	8	8
d/e	-0.59	-7.29	-6.40	-4.72	0.12	-1.75	-1.87	-1.43	0.21	0.21	0.18	0.21	0.16	9	8	7
b/m	-16.95	-0.05	0.47	-1.09	2.49	0.29	0.12	2.18	0.21	0.34	0.22	0.25	0.26	1	7	6
<i>Panel B: Interest Rates and Spreads</i>																
tbl	-0.38	-0.12	-0.78	-0.08	0.21	0.23	0.17	0.15	0.21	0.28	0.29	0.33	0.27	6	6	5
lty	0.65	0.67	-0.02	0.64	0.25	0.26	0.20	0.18	0.21	0.26	0.28	0.32	0.26	6	6	5
ltr	0.41	0.77	0.05	0.77	0.04	-0.08	-0.78	-0.04	0.21	0.18	0.20	0.24	0.18	31	30	24
tms	-2.94	-1.39	-2.50	-1.39	0.20	0.18	-0.51	0.29	0.21	0.23	0.25	0.30	0.24	9	8	7
dfy	-2.52	-0.10	-0.46	-0.05	0.77	-0.33	-0.61	-0.38	0.21	0.26	0.19	0.24	0.17	8	9	7
dfr	-2.52	-0.84	-1.78	-0.98	-0.50	-0.51	-0.46	-0.43	0.21	0.18	0.21	0.28	0.19	15	13	10
crdstd	-0.95	-1.08	-0.30	-1.36	1.89	2.17	2.18	2.14	0.26	0.38	0.40	0.40	0.40	15	15	15
<i>Panel C: Macroeconomic Variables</i>																
infl	0.48	0.31	-0.31	0.32	1.37	-0.16	-0.15	0.08	0.21	0.29	0.23	0.30	0.22	21	6	6
ntis	-9.48	-2.37	-5.00	-2.68	-0.37	-0.39	-0.27	-0.35	0.21	0.32	0.26	0.33	0.25	9	6	6
svar	0.36	-0.67	-0.60	0.09	-0.25	-0.50	-0.23	-0.22	0.21	0.23	0.19	0.28	0.19	12	9	7
i/k	1.73*	2.42*	2.45**	2.20*	3.45	2.76	2.63	2.81	0.20	0.46	0.45	0.45	0.45	8	8	8
govik	-2.27	-0.81	-0.53	-1.21	1.18	0.46	0.24	0.92	0.20	0.18	0.20	0.21	0.20	6	7	7
ndrbl	0.66	0.76*	0.73*	0.79	-0.17	-1.28	-1.28	-0.99	0.21	0.31	0.31	0.31	0.31	19	20	20
wtexas	-2.29	-1.19	-1.90	-0.91	-1.19	-0.52	-0.78	-0.28	0.21	0.18	0.22	0.26	0.21	33	16	14
<i>Panel D: Variance and Risk Measures</i>																
vp	3.39*	5.45*	3.21*	4.87*	-1.55	-1.02	-0.35	-1.87	0.27	0.24	0.34	0.41	0.25	28	28	22
impvar	-6.41	-3.11	-1.78	-5.51	1.34	1.93	2.18	1.53	0.59	0.69	0.73	0.75	0.70	17	20	21
vrp	-6.71	-1.08	-2.69	-2.74	-1.50	-0.95	-1.04	-1.27	0.24	0.06	0.21	0.24	0.12	40	47	47
tail	0.26	0.23	-0.87	0.21	0.24	0.06	0.01	0.17	0.21	0.24	0.25	0.32	0.23	7	6	5
skvw	-1.45	-0.49	-0.46	-0.31	0.18	-0.65	-0.43	-0.48	0.21	0.14	0.23	0.28	0.21	24	18	12
<i>Panel E: Cross-Section and Technical</i>																
lzrt	-11.32	-1.92	-1.99	-2.88	1.67	1.98	1.27	2.78	0.21	0.23	0.28	0.30	0.31	21	16	18
avgcor	4.61*	3.79*	2.47*	4.26**	0.62	0.84	0.98	0.49	0.21	0.29	0.32	0.39	0.28	21	12	11
dtoy	0.61	-3.84	-2.39	-2.23	0.24	1.28	-0.13	0.89	0.21	0.22	0.27	0.26	0.24	8	14	11
dtoat	0.11	0.28	0.07	0.17	1.06	-0.06	-0.14	0.19	0.21	0.21	0.23	0.29	0.21	8	6	5
ygap	-3.05	-1.76	-0.77	-3.05	1.87	1.08	-0.11	1.78	0.20	0.27	0.25	0.23	0.27	7	8	8
rdsp	-3.30	0.59	0.05	0.44	-0.91	1.07	0.56	0.99	0.21	0.14	0.28	0.31	0.26	11	11	8
disag	-1.73	-0.26	0.28	-0.79	1.41	2.35	2.18	2.05	0.19	0.18	0.28	0.30	0.25	18	17	15
Average	-3.15	-0.46	-0.61	-0.77	0.72	0.45	0.18	0.65	0.22	0.26	0.27	0.31	0.26	14	13	12

Notes: Out-of-sample forecasting results for the quarterly U.S. equity premium, 1994Q4 to 2024Q4. R^2_{OOS} is measured relative to the historical mean (HM) benchmark. ΔCER is the annualized certainty equivalent return gain (percentage points) relative to HM. Portfolio weights derive from mean-variance optimization with risk aversion $\gamma = 5$, transaction costs of 0.50%, and a 5-years rolling window for variance; weights are constrained to $[0.0, 1.5]$. Sharpe ratios are annualized; turnover is reported as a quarterly average. BRT: Bayesian regression with Student- t errors ($\nu = 5$). GB: generalized Bayesian CQR with flat prior and calibrated η . GB_f: same model with fixed $\eta = 1$. Bold entries denote positive R^2_{OOS} or ΔCER , the highest Sharpe ratio, or the lowest turnover among the reported methods. *, **, ***: 10%, 5%, 1% significance.

Table 22: Out-of-Sample Equity Premium Forecasting: $\gamma = 5$, $TC = 1.00\%$ (1994Q4–2024Q4)

	R^2_{OOS} (%)				ΔCER (%)				Sharpe Ratio					Turnover (%)		
	OLS	BRT	GB	GB_f	OLS	BRT	GB	GB_f	HM	OLS	BRT	GB	GB_f	OLS	BRT	GB
<i>Panel A: Valuation Ratios</i>																
d/p	-14.41	-0.23	0.79	-2.13	2.84	1.46	0.51	2.85	0.20	0.31	0.24	0.24	0.27	1	6	7
d/y	-18.07	-0.48	0.77	-2.78	2.82	2.09	1.11	3.19	0.20	0.27	0.26	0.25	0.29	1	6	7
e/p	-3.70	-0.42	1.19	-1.64	2.88	1.75	0.27	2.38	0.20	0.22	0.21	0.21	0.21	7	8	8
d/e	-0.59	-7.29	-6.40	-4.72	0.08	-1.81	-1.90	-1.48	0.20	0.20	0.17	0.20	0.15	9	8	7
b/m	-16.95	-0.05	0.47	-1.09	2.63	0.27	0.11	2.19	0.20	0.33	0.21	0.24	0.25	1	7	6
<i>Panel B: Interest Rates and Spreads</i>																
tbl	-0.38	-0.12	-0.78	-0.08	0.22	0.25	0.21	0.16	0.20	0.27	0.28	0.32	0.26	6	6	5
lty	0.65	0.67	-0.02	0.64	0.26	0.28	0.25	0.19	0.20	0.26	0.27	0.32	0.25	6	6	5
ltr	0.41	0.77	0.05	0.77	-0.42	-0.51	-1.08	-0.50	0.20	0.14	0.16	0.21	0.14	31	30	24
tms	-2.94	-1.39	-2.50	-1.39	0.17	0.17	-0.50	0.28	0.20	0.22	0.24	0.29	0.22	9	8	7
dfy	-2.52	-0.10	-0.46	-0.05	0.76	-0.41	-0.63	-0.43	0.20	0.25	0.17	0.24	0.16	8	9	7
dfr	-2.52	-0.84	-1.78	-0.98	-0.66	-0.64	-0.53	-0.54	0.20	0.16	0.19	0.26	0.17	15	13	10
crdstd	-0.95	-1.08	-0.30	-1.36	1.74	2.03	2.04	2.00	0.25	0.36	0.38	0.38	0.38	15	15	15
<i>Panel C: Macroeconomic Variables</i>																
infl	0.48	0.31	-0.31	0.32	1.11	-0.16	-0.12	0.07	0.20	0.26	0.22	0.29	0.21	21	6	6
ntis	-9.48	-2.37	-5.00	-2.68	-0.39	-0.38	-0.23	-0.34	0.20	0.31	0.25	0.33	0.24	9	6	6
svar	0.36	-0.67	-0.60	0.09	-0.33	-0.57	-0.24	-0.26	0.20	0.22	0.18	0.27	0.18	12	9	7
i/k	1.73*	2.42*	2.45**	2.20*	3.44	2.73	2.60	2.78	0.19	0.45	0.44	0.44	0.44	8	8	8
govik	-2.27	-0.81	-0.53	-1.21	1.20	0.46	0.23	0.94	0.19	0.17	0.19	0.19	0.19	6	7	7
ndrbl	0.66	0.76*	0.73*	0.79	-0.43	-1.58	-1.58	-1.30	0.20	0.28	0.29	0.29	0.29	19	20	20
wtexas	-2.29	-1.19	-1.90	-0.91	-1.69	-0.68	-0.90	-0.43	0.20	0.14	0.20	0.25	0.19	33	16	14
<i>Panel D: Variance and Risk Measures</i>																
vp	3.39*	5.45*	3.21*	4.87*	-1.86	-1.33	-0.57	-2.23	0.26	0.21	0.32	0.39	0.22	28	28	22
impvar	-6.41	-3.11	-1.78	-5.51	1.17	1.69	1.91	1.24	0.58	0.67	0.71	0.73	0.67	17	20	21
vrp	-6.71	-1.08	-2.69	-2.74	-2.15	-1.72	-1.82	-2.01	0.22	0.01	0.15	0.18	0.06	40	47	47
tail	0.26	0.23	-0.87	0.21	0.24	0.07	0.05	0.17	0.20	0.24	0.24	0.31	0.23	7	6	5
skvw	-1.45	-0.49	-0.46	-0.31	-0.18	-0.83	-0.50	-0.63	0.20	0.10	0.21	0.26	0.19	24	18	12
<i>Panel E: Cross-Section and Technical</i>																
lzrt	-11.32	-1.92	-1.99	-2.88	1.45	1.83	1.07	2.65	0.20	0.20	0.26	0.27	0.29	21	16	18
avgcor	4.61*	3.79*	2.47*	4.26**	0.35	0.77	0.94	0.32	0.20	0.27	0.31	0.38	0.26	21	12	11
dtoy	0.61	-3.84	-2.39	-2.23	0.21	1.15	-0.21	0.80	0.20	0.21	0.25	0.25	0.22	8	14	11
dtoat	0.11	0.28	0.07	0.17	1.02	-0.06	-0.11	0.19	0.20	0.20	0.22	0.28	0.20	8	6	5
ygap	-3.05	-1.76	-0.77	-3.05	1.87	1.04	-0.15	1.76	0.19	0.25	0.24	0.22	0.26	7	8	8
rdsp	-3.30	0.59	0.05	0.44	-1.00	0.97	0.54	0.93	0.20	0.13	0.27	0.30	0.25	11	11	8
disag	-1.73	-0.26	0.28	-0.79	1.18	2.17	2.04	1.87	0.18	0.15	0.25	0.28	0.22	18	17	15
Average	-3.15	-0.46	-0.61	-0.77	0.60	0.34	0.09	0.54	0.21	0.24	0.26	0.29	0.24	14	13	12

Notes: Out-of-sample forecasting results for the quarterly U.S. equity premium, 1994Q4 to 2024Q4. R^2_{OOS} is measured relative to the historical mean (HM) benchmark. ΔCER is the annualized certainty equivalent return gain (percentage points) relative to HM. Portfolio weights derive from mean-variance optimization with risk aversion $\gamma = 5$, transaction costs of 1.00%, and a 5-years rolling window for variance; weights are constrained to $[0.0, 1.5]$. Sharpe ratios are annualized; turnover is reported as a quarterly average. BRT: Bayesian regression with Student- t errors ($\nu = 5$). GB: generalized Bayesian CQR with flat prior and calibrated η . GB_f : same model with fixed $\eta = 1$. Bold entries denote positive R^2_{OOS} or ΔCER , the highest Sharpe ratio, or the lowest turnover among the reported methods. *, **, ***: 10%, 5%, 1% significance.

7 Robustness: Different Evaluation Window (2004Q4–2024Q4)

This section repeats the out-of-sample exercise over a shorter 20-year evaluation window (2004Q4–2024Q4). Tables 23–28 report results for $\gamma \in \{3, 5\}$ and $TC \in \{0\%, 0.50\%, 1.00\%\}$.

The main findings from the 30-year window are reinforced and, in several dimensions, strengthened. GB dominates on ΔCER at $\gamma = 3$ and $\gamma = 5$ across all transaction cost levels, with average gains that are substantially larger than in the baseline. At $\gamma = 3$ with $TC = 0.50\%$ (Table 24), GB averages 1.22 pp — nearly double the 0.72 pp obtained over the 30-year window. Even at $\gamma = 5$, GB remains the only method with positive average ΔCER across all TC levels.

GB’s Sharpe ratio advantage is unanimous: it achieves the highest average annualized Sharpe ratio in every one of the nine configurations. GB also retains the lowest average turnover throughout, confirming that its economic gains are not driven by aggressive rebalancing. Overall, the shorter evaluation window — which places greater weight on crisis and post-crisis dynamics — amplifies the advantages of the generalized Bayesian approach documented in the main text.

Table 23: Out-of-Sample Equity Premium Forecasting: $\gamma = 3$, TC = 0.00% (2004Q4–2024Q4)

	R^2_{OOS} (%)				ΔCER (%)				Sharpe Ratio				Turnover (%)			
	OLS	BRT	GB	GB_f	OLS	BRT	GB	GB_f	HM	OLS	BRT	GB	GB_f	OLS	BRT	GB
<i>Panel A: Valuation Ratios</i>																
d/p	-11.21	-0.06	0.98**	-1.82	-8.24	-0.89	0.36	-2.56	0.75	0.45	0.71	0.77	0.64	1	4	4
d/y	-13.57	-0.28	0.81*	-2.37	-8.42	-1.38	-0.01	-2.92	0.75	0.26	0.69	0.76	0.63	0	5	5
e/p	-5.01	-1.35	0.21	-2.88	-3.65	-1.72	0.61	-2.50	0.75	0.60	0.67	0.77	0.65	9	7	6
d/e	-0.57	-10.54	-7.46	-7.19	-0.27	0.17	1.05	-0.25	0.75	0.75	0.74	0.77	0.73	6	6	6
b/m	-9.69	0.45	1.41**	-0.53	-8.17	0.07	1.14	-1.66	0.75	0.51	0.76	0.80	0.68	1	4	4
<i>Panel B: Interest Rates and Spreads</i>																
tbl	0.14	0.31	-0.43	0.33	0.93	1.26	1.85	0.91	0.75	0.77	0.78	0.80	0.77	5	4	3
lty	1.32	1.27	0.60	1.21	1.16	1.47	2.06	1.03	0.75	0.79	0.80	0.82	0.79	4	4	3
ltr	0.60	1.21	1.06	1.00	-0.47	0.27	1.72	-0.55	0.76	0.74	0.77	0.82	0.73	31	30	23
tms	-3.09	-1.33	-2.04	-1.47	-0.11	0.13	1.81	-0.25	0.75	0.75	0.74	0.80	0.73	7	6	5
dfy	-1.43	-0.93	-0.27	-0.39	-0.10	-0.07	1.75	-0.02	0.75	0.77	0.73	0.81	0.75	9	7	5
dfr	-4.37	-0.87	-1.48	-1.22	-1.44	-0.24	1.16	-0.80	0.76	0.68	0.73	0.79	0.71	13	13	10
crdstd	-4.82	-4.09	-3.39	-4.33	-2.02	-1.06	-1.13	-1.26	0.71	0.58	0.63	0.63	0.62	15	14	15
<i>Panel C: Macroeconomic Variables</i>																
infl	0.89	0.54	0.30	0.48	0.72	0.76	2.27	0.37	0.75	0.79	0.79	0.84	0.77	18	5	5
ntis	-12.41	-2.66	-5.74	-3.06	3.52	2.13	3.36	1.81	0.75	0.91	0.85	0.88	0.84	8	5	4
svar	0.36	-0.43	0.23	0.35	0.61	0.48	2.18	0.51	0.75	0.79	0.77	0.83	0.78	13	7	5
i/k	5.02**	5.41**	5.43**	5.32**	1.13	2.03	2.26	1.84	0.74	0.75	0.79	0.80	0.78	6	5	5
govik	-2.45	-0.54	-0.20	-1.16	-2.64	-0.85	-0.55	-1.54	0.74	0.59	0.69	0.70	0.66	5	5	5
ndrbl	2.06	2.94*	2.77*	2.85*	0.75	2.15	2.11	1.80	0.69	0.71	0.78	0.77	0.76	17	18	17
wtexas	-2.76	-1.24	-1.69	-1.10	-2.50	-0.59	0.52	-0.86	0.76	0.60	0.70	0.75	0.69	37	17	12
<i>Panel D: Variance and Risk Measures</i>																
vp	2.44	7.25*	6.80*	5.67	-1.49	0.91	2.42	-1.39	0.82	0.73	0.83	0.89	0.73	40	26	17
impvar	-6.32	-3.13	-1.89	-5.44	0.52	1.88	2.06	0.84	0.64	0.68	0.74	0.74	0.69	20	22	21
vrp	-13.97	-4.82	-4.69	-7.79	-5.02	-0.60	0.39	-3.12	0.68	0.37	0.65	0.70	0.50	53	50	47
tail	-0.28	0.15	-0.65	-0.03	0.62	1.06	2.49	0.51	0.75	0.78	0.80	0.84	0.77	6	5	4
skvw	-0.15	-2.15	-1.55	-1.55	-2.07	0.83	1.70	0.26	0.75	0.65	0.77	0.81	0.75	32	16	11
<i>Panel E: Cross-Section and Technical</i>																
lzrt	-16.96	-3.15	-3.02	-4.56	-6.18	-2.26	-0.96	-3.28	0.76	0.40	0.62	0.68	0.56	41	22	20
avgcor	3.63	3.95	2.37	4.22	1.38	2.43	3.09	1.73	0.76	0.82	0.86	0.87	0.84	21	12	8
dtoy	0.69	-3.95	-2.00	-2.14	0.22	-0.77	1.14	-0.50	0.76	0.78	0.70	0.78	0.72	8	14	8
dtoat	0.26	0.51	0.73	0.25	-1.23	0.57	2.04	-0.27	0.76	0.72	0.78	0.83	0.75	9	4	4
ygap	-4.90	-4.45	-3.89	-5.97	-1.24	0.24	1.46	-0.71	0.71	0.64	0.72	0.76	0.67	6	6	6
rdsp	0.18	-1.48	-1.17	-1.01	-0.83	0.76	2.00	0.30	0.75	0.71	0.79	0.83	0.77	10	11	7
disag	-4.62	-3.60	-2.39	-4.00	-3.61	-1.96	-1.26	-2.49	0.74	0.54	0.63	0.67	0.60	25	18	15
Average	-3.26	-0.87	-0.65	-1.24	-1.55	0.23	1.33	-0.48	0.75	0.66	0.74	0.79	0.71	15	12	10

Notes: Out-of-sample forecasting results for the quarterly U.S. equity premium, 2004Q4 to 2024Q4. R^2_{OOS} is measured relative to the historical mean (HM) benchmark. ΔCER is the annualized certainty equivalent return gain (percentage points) relative to HM. Portfolio weights derive from mean-variance optimization with risk aversion $\gamma = 3$, transaction costs of 0.00%, and a 5-years rolling window for variance; weights are constrained to $[0.0, 1.5]$. Sharpe ratios are annualized; turnover is reported as a quarterly average. BRT: Bayesian regression with Student- t errors ($\nu = 5$). GB: generalized Bayesian CQR with flat prior and calibrated η . GB_f : same model with fixed $\eta = 1$. Bold entries denote positive R^2_{OOS} or ΔCER , the highest Sharpe ratio, or the lowest turnover among the reported methods. *, **, ***: 10%, 5%, 1% significance.

Table 24: Out-of-Sample Equity Premium Forecasting: $\gamma = 3$, TC = 0.50% (2004Q4–2024Q4)

	R^2_{OOS} (%)				ΔCER (%)				Sharpe Ratio				Turnover (%)			
	OLS	BRT	GB	GB_f	OLS	BRT	GB	GB_f	HM	OLS	BRT	GB	GB_f	OLS	BRT	GB
<i>Panel A: Valuation Ratios</i>																
d/p	-11.21	-0.06	0.98**	-1.82	-8.18	-0.90	0.37	-2.58	0.75	0.41	0.71	0.77	0.63	1	4	4
d/y	-13.57	-0.28	0.81*	-2.37	-8.34	-1.40	-0.02	-2.96	0.75	0.26	0.68	0.75	0.62	0	5	5
e/p	-5.01	-1.35	0.21	-2.88	-3.76	-1.79	0.58	-2.57	0.75	0.58	0.66	0.77	0.63	9	7	6
d/e	-0.57	-10.54	-7.46	-7.19	-0.32	0.14	1.02	-0.28	0.75	0.74	0.73	0.77	0.72	6	6	6
b/m	-9.69	0.45	1.41**	-0.53	-8.11	0.07	1.16	-1.69	0.75	0.48	0.75	0.80	0.67	1	4	4
<i>Panel B: Interest Rates and Spreads</i>																
tbl	0.14	0.31	-0.43	0.33	0.93	1.26	1.87	0.91	0.75	0.76	0.78	0.80	0.77	5	4	3
lty	1.32	1.27	0.60	1.21	1.16	1.48	2.08	1.04	0.75	0.79	0.80	0.82	0.78	4	4	3
ltr	0.60	1.21	1.06	1.00	-1.01	-0.25	1.34	-1.12	0.75	0.70	0.73	0.80	0.69	31	30	23
tms	-3.09	-1.33	-2.04	-1.47	-0.17	0.08	1.80	-0.30	0.75	0.74	0.74	0.80	0.72	7	6	5
dfy	-1.43	-0.93	-0.27	-0.39	-0.20	-0.12	1.75	-0.04	0.75	0.76	0.73	0.81	0.74	9	7	5
dfr	-4.37	-0.87	-1.48	-1.22	-1.60	-0.42	1.05	-0.98	0.75	0.66	0.72	0.78	0.69	13	13	10
crdstd	-4.82	-4.09	-3.39	-4.33	-2.22	-1.25	-1.33	-1.45	0.71	0.56	0.62	0.61	0.61	15	14	15
<i>Panel C: Macroeconomic Variables</i>																
infl	0.89	0.54	0.30	0.48	0.46	0.75	2.25	0.34	0.75	0.77	0.78	0.83	0.77	18	5	5
ntis	-12.41	-2.66	-5.74	-3.06	3.45	2.13	3.37	1.79	0.75	0.90	0.84	0.88	0.83	8	5	4
svar	0.36	-0.43	0.23	0.35	0.44	0.43	2.17	0.48	0.75	0.77	0.76	0.83	0.77	13	7	5
i/k	5.02**	5.41**	5.43**	5.32**	1.08	2.02	2.26	1.82	0.73	0.75	0.79	0.80	0.78	6	5	5
govik	-2.45	-0.54	-0.20	-1.16	-2.67	-0.86	-0.56	-1.56	0.73	0.58	0.68	0.70	0.65	5	5	5
ndrbl	2.06	2.94*	2.77*	2.85*	0.51	1.90	1.86	1.55	0.68	0.69	0.76	0.75	0.74	17	18	17
wtexas	-2.76	-1.24	-1.69	-1.10	-3.11	-0.82	0.37	-1.08	0.75	0.57	0.69	0.74	0.68	37	17	12
<i>Panel D: Variance and Risk Measures</i>																
vp	2.44	7.25*	6.80*	5.67	-2.20	0.46	2.16	-2.15	0.81	0.68	0.80	0.87	0.68	40	26	17
impvar	-6.32	-3.13	-1.89	-5.44	0.26	1.58	1.77	0.44	0.64	0.66	0.71	0.72	0.66	20	22	21
vrp	-13.97	-4.82	-4.69	-7.79	-5.99	-1.48	-0.42	-4.13	0.68	0.30	0.59	0.65	0.43	53	50	47
tail	-0.28	0.15	-0.65	-0.03	0.61	1.06	2.49	0.51	0.75	0.77	0.79	0.84	0.77	6	5	4
skvw	-0.15	-2.15	-1.55	-1.55	-2.63	0.62	1.58	0.09	0.75	0.60	0.75	0.80	0.74	32	16	11
<i>Panel E: Cross-Section and Technical</i>																
lzrt	-16.96	-3.15	-3.02	-4.56	-6.82	-2.56	-1.22	-3.65	0.75	0.36	0.60	0.67	0.54	41	22	20
avgcor	3.63	3.95	2.37	4.22	1.02	2.27	3.01	1.48	0.75	0.79	0.84	0.86	0.82	21	12	8
dtoy	0.69	-3.95	-2.00	-2.14	0.14	-0.92	1.07	-0.58	0.75	0.77	0.68	0.78	0.71	8	14	8
dtoat	0.26	0.51	0.73	0.25	-1.33	0.58	2.04	-0.28	0.75	0.70	0.78	0.83	0.74	9	4	4
ygap	-4.90	-4.45	-3.89	-5.97	-1.29	0.22	1.45	-0.77	0.71	0.63	0.71	0.76	0.66	6	6	6
rdsp	0.18	-1.48	-1.17	-1.01	-0.94	0.65	1.97	0.22	0.75	0.69	0.78	0.82	0.76	10	11	7
disag	-4.62	-3.60	-2.39	-4.00	-4.02	-2.22	-1.46	-2.77	0.74	0.50	0.61	0.65	0.58	25	18	15
Average	-3.26	-0.87	-0.65	-1.24	-1.77	0.09	1.22	-0.65	0.74	0.64	0.73	0.78	0.70	15	12	10

Notes: Out-of-sample forecasting results for the quarterly U.S. equity premium, 2004Q4 to 2024Q4. R^2_{OOS} is measured relative to the historical mean (HM) benchmark. ΔCER is the annualized certainty equivalent return gain (percentage points) relative to HM. Portfolio weights derive from mean-variance optimization with risk aversion $\gamma = 3$, transaction costs of 0.50%, and a 5-years rolling window for variance; weights are constrained to $[0.0, 1.5]$. Sharpe ratios are annualized; turnover is reported as a quarterly average. BRT: Bayesian regression with Student- t errors ($\nu = 5$). GB: generalized Bayesian CQR with flat prior and calibrated η . GB_f : same model with fixed $\eta = 1$. Bold entries denote positive R^2_{OOS} or ΔCER , the highest Sharpe ratio, or the lowest turnover among the reported methods. *, **, ***: 10%, 5%, 1% significance.

Table 25: Out-of-Sample Equity Premium Forecasting: $\gamma = 3$, TC = 1.00% (2004Q4–2024Q4)

	R^2_{OOS} (%)				ΔCER (%)				Sharpe Ratio				Turnover (%)			
	OLS	BRT	GB	GB_f	OLS	BRT	GB	GB_f	HM	OLS	BRT	GB	GB_f	OLS	BRT	GB
<i>Panel A: Valuation Ratios</i>																
d/p	-11.21	-0.06	0.98**	-1.82	-8.13	-0.92	0.37	-2.60	0.74	0.36	0.70	0.76	0.62	1	4	4
d/y	-13.57	-0.28	0.81*	-2.37	-8.25	-1.42	-0.03	-2.99	0.74	0.26	0.68	0.74	0.61	0	5	5
e/p	-5.01	-1.35	0.21	-2.88	-3.88	-1.86	0.55	-2.65	0.74	0.56	0.65	0.76	0.62	9	7	6
d/e	-0.57	-10.54	-7.46	-7.19	-0.37	0.10	0.98	-0.31	0.74	0.73	0.73	0.76	0.71	6	6	6
b/m	-9.69	0.45	1.41**	-0.53	-8.04	0.07	1.17	-1.72	0.74	0.46	0.75	0.79	0.66	1	4	4
<i>Panel B: Interest Rates and Spreads</i>																
tbl	0.14	0.31	-0.43	0.33	0.92	1.25	1.88	0.91	0.74	0.76	0.77	0.80	0.76	5	4	3
lty	1.32	1.27	0.60	1.21	1.17	1.49	2.10	1.05	0.74	0.78	0.79	0.81	0.78	4	4	3
ltr	0.60	1.21	1.06	1.00	-1.54	-0.78	0.97	-1.70	0.75	0.66	0.70	0.78	0.65	31	30	23
tms	-3.09	-1.33	-2.04	-1.47	-0.22	0.04	1.80	-0.35	0.74	0.73	0.73	0.80	0.71	7	6	5
dfy	-1.43	-0.93	-0.27	-0.39	-0.30	-0.17	1.75	-0.06	0.74	0.74	0.72	0.80	0.74	9	7	5
dfr	-4.37	-0.87	-1.48	-1.22	-1.76	-0.59	0.94	-1.17	0.75	0.65	0.70	0.77	0.68	13	13	10
crdstd	-4.82	-4.09	-3.39	-4.33	-2.42	-1.44	-1.53	-1.64	0.70	0.55	0.60	0.60	0.59	15	14	15
<i>Panel C: Macroeconomic Variables</i>																
infl	0.89	0.54	0.30	0.48	0.19	0.73	2.23	0.31	0.74	0.75	0.77	0.83	0.76	18	5	5
ntis	-12.41	-2.66	-5.74	-3.06	3.37	2.12	3.37	1.78	0.74	0.90	0.84	0.87	0.83	8	5	4
svar	0.36	-0.43	0.23	0.35	0.27	0.39	2.17	0.44	0.74	0.76	0.75	0.82	0.76	13	7	5
i/k	5.02**	5.41**	5.43**	5.32**	1.03	2.01	2.25	1.80	0.72	0.74	0.78	0.79	0.77	6	5	5
govik	-2.45	-0.54	-0.20	-1.16	-2.70	-0.87	-0.58	-1.58	0.72	0.57	0.68	0.69	0.64	5	5	5
ndrbl	2.06	2.94*	2.77*	2.85*	0.28	1.65	1.61	1.30	0.67	0.67	0.74	0.73	0.72	17	18	17
wtexas	-2.76	-1.24	-1.69	-1.10	-3.72	-1.06	0.22	-1.31	0.75	0.53	0.67	0.73	0.66	37	17	12
<i>Panel D: Variance and Risk Measures</i>																
vp	2.44	7.25*	6.80*	5.67	-2.91	0.01	1.89	-2.90	0.80	0.64	0.77	0.86	0.63	40	26	17
impvar	-6.32	-3.13	-1.89	-5.44	-0.01	1.28	1.49	0.03	0.63	0.63	0.69	0.70	0.63	20	22	21
vrp	-13.97	-4.82	-4.69	-7.79	-6.96	-2.38	-1.24	-5.15	0.67	0.22	0.53	0.59	0.36	53	50	47
tail	-0.28	0.15	-0.65	-0.03	0.59	1.06	2.49	0.51	0.74	0.77	0.79	0.83	0.76	6	5	4
skvw	-0.15	-2.15	-1.55	-1.55	-3.19	0.41	1.45	-0.07	0.74	0.56	0.74	0.79	0.72	32	16	11
<i>Panel E: Cross-Section and Technical</i>																
lzrt	-16.96	-3.15	-3.02	-4.56	-7.46	-2.85	-1.48	-4.03	0.75	0.31	0.58	0.65	0.51	41	22	20
avgcor	3.63	3.95	2.37	4.22	0.67	2.11	2.92	1.22	0.75	0.77	0.83	0.85	0.80	21	12	8
dtoy	0.69	-3.95	-2.00	-2.14	0.07	-1.08	1.00	-0.66	0.75	0.76	0.67	0.77	0.70	8	14	8
dtoat	0.26	0.51	0.73	0.25	-1.43	0.59	2.05	-0.30	0.75	0.69	0.77	0.83	0.74	9	4	4
ygap	-4.90	-4.45	-3.89	-5.97	-1.33	0.19	1.44	-0.83	0.70	0.62	0.70	0.75	0.65	6	6	6
rdsp	0.18	-1.48	-1.17	-1.01	-1.06	0.54	1.93	0.15	0.74	0.68	0.77	0.82	0.75	10	11	7
disag	-4.62	-3.60	-2.39	-4.00	-4.43	-2.48	-1.67	-3.04	0.73	0.47	0.59	0.63	0.56	25	18	15
Average	-3.26	-0.87	-0.65	-1.24	-1.99	-0.06	1.11	-0.82	0.73	0.62	0.72	0.76	0.68	15	12	10

Notes: Out-of-sample forecasting results for the quarterly U.S. equity premium, 2004Q4 to 2024Q4. R^2_{OOS} is measured relative to the historical mean (HM) benchmark. ΔCER is the annualized certainty equivalent return gain (percentage points) relative to HM. Portfolio weights derive from mean-variance optimization with risk aversion $\gamma = 3$, transaction costs of 1.00%, and a 5-years rolling window for variance; weights are constrained to $[0.0, 1.5]$. Sharpe ratios are annualized; turnover is reported as a quarterly average. BRT: Bayesian regression with Student- t errors ($\nu = 5$). GB: generalized Bayesian CQR with flat prior and calibrated η . GB_f : same model with fixed $\eta = 1$. Bold entries denote positive R^2_{OOS} or ΔCER , the highest Sharpe ratio, or the lowest turnover among the reported methods. *, **, ***: 10%, 5%, 1% significance.

Table 26: Out-of-Sample Equity Premium Forecasting: $\gamma = 5$, TC = 0.00% (2004Q4–2024Q4)

	R^2_{OOS} (%)				ΔCER (%)				Sharpe Ratio				Turnover (%)			
	OLS	BRT	GB	GB_f	OLS	BRT	GB	GB_f	HM	OLS	BRT	GB	GB_f	OLS	BRT	GB
<i>Panel A: Valuation Ratios</i>																
d/p	-11.21	-0.06	0.98**	-1.82	-4.07	-0.25	-0.02	-0.96	0.65	0.45	0.63	0.65	0.57	1	5	5
d/y	-13.57	-0.28	0.81*	-2.37	-4.18	-0.41	-0.01	-1.31	0.65	0.26	0.62	0.65	0.54	0	5	6
e/p	-5.01	-1.35	0.21	-2.88	-1.88	-0.49	-0.15	-1.11	0.65	0.49	0.61	0.64	0.56	7	6	6
d/e	-0.57	-10.54	-7.46	-7.19	0.19	-0.33	0.45	-0.61	0.65	0.66	0.63	0.69	0.61	7	6	6
b/m	-9.69	0.45	1.41**	-0.53	-4.03	-0.04	0.24	-0.26	0.65	0.51	0.64	0.67	0.63	1	5	5
<i>Panel B: Interest Rates and Spreads</i>																
tbl	0.14	0.31	-0.43	0.33	0.32	0.57	1.17	0.22	0.65	0.68	0.69	0.74	0.67	5	5	4
lty	1.32	1.27	0.60	1.21	0.29	0.55	1.26	0.20	0.65	0.67	0.69	0.74	0.67	5	5	4
ltr	0.60	1.21	1.06	1.00	-0.73	-0.24	0.94	-0.79	0.65	0.59	0.63	0.72	0.59	29	29	23
tms	-3.09	-1.33	-2.04	-1.47	0.89	0.91	0.90	0.88	0.65	0.72	0.71	0.72	0.71	8	6	6
dfy	-1.43	-0.93	-0.27	-0.39	0.95	-0.60	0.76	-0.29	0.65	0.73	0.61	0.71	0.63	8	8	5
dfr	-4.37	-0.87	-1.48	-1.22	-0.83	-0.60	0.53	-0.63	0.65	0.59	0.61	0.69	0.60	11	12	10
crdstd	-4.82	-4.09	-3.39	-4.33	-1.99	-1.45	-1.51	-1.49	0.62	0.50	0.54	0.53	0.53	14	14	14
<i>Panel C: Macroeconomic Variables</i>																
infl	0.89	0.54	0.30	0.48	-0.08	-0.02	1.12	-0.04	0.65	0.64	0.65	0.73	0.65	17	6	6
ntis	-12.41	-2.66	-5.74	-3.06	1.99	0.90	2.05	0.70	0.65	0.79	0.71	0.79	0.70	8	5	5
svar	0.36	-0.43	0.23	0.35	0.88	-0.20	1.24	0.27	0.65	0.71	0.64	0.74	0.67	13	7	6
i/k	5.02**	5.41**	5.43**	5.32**	0.83	0.66	0.73	0.73	0.64	0.70	0.71	0.71	0.71	8	8	8
govik	-2.45	-0.54	-0.20	-1.16	-1.17	-0.58	-0.56	-0.78	0.64	0.54	0.60	0.60	0.58	6	5	5
ndrbl	2.06	2.94*	2.77*	2.85*	-0.81	-0.05	-0.11	-0.29	0.62	0.58	0.64	0.64	0.62	15	15	15
wtexas	-2.76	-1.24	-1.69	-1.10	-0.81	-0.40	0.05	-0.24	0.65	0.60	0.63	0.67	0.63	28	15	13
<i>Panel D: Variance and Risk Measures</i>																
vp	2.44	7.25*	6.80*	5.67	0.06	1.49	2.33	0.26	0.69	0.69	0.79	0.84	0.70	35	34	27
impvar	-6.32	-3.13	-1.89	-5.44	0.67	1.27	1.29	0.79	0.57	0.62	0.67	0.68	0.63	16	19	20
vrp	-13.97	-4.82	-4.69	-7.79	-2.76	-0.73	-0.16	-1.67	0.60	0.34	0.55	0.60	0.45	40	47	46
tail	-0.28	0.15	-0.65	-0.03	0.28	0.23	1.31	0.20	0.65	0.67	0.67	0.74	0.66	6	5	5
skvw	-0.15	-2.15	-1.55	-1.55	-0.66	0.17	0.82	-0.26	0.65	0.60	0.67	0.71	0.63	25	18	12
<i>Panel E: Cross-Section and Technical</i>																
lzrt	-16.96	-3.15	-3.02	-4.56	-3.71	-1.81	-1.19	-2.13	0.65	0.38	0.52	0.59	0.49	28	18	18
avgcor	3.63	3.95	2.37	4.22	0.45	1.41	2.44	0.63	0.65	0.68	0.75	0.81	0.69	25	14	11
dtoy	0.69	-3.95	-2.00	-2.14	0.62	-0.91	0.54	-0.81	0.65	0.70	0.60	0.70	0.60	7	13	8
dtoat	0.26	0.51	0.73	0.25	-0.01	0.08	0.93	0.17	0.65	0.66	0.66	0.72	0.66	7	5	5
ygap	-4.90	-4.45	-3.89	-5.97	-0.48	-0.13	0.05	-0.32	0.63	0.59	0.63	0.65	0.61	6	6	6
rdsp	0.18	-1.48	-1.17	-1.01	-0.68	0.27	0.98	0.27	0.65	0.59	0.67	0.72	0.67	9	10	7
disag	-4.62	-3.60	-2.39	-4.00	-3.26	-1.97	-1.44	-2.36	0.63	0.38	0.50	0.55	0.46	20	17	14
Average	-3.26	-0.87	-0.65	-1.24	-0.77	-0.09	0.55	-0.36	0.64	0.59	0.64	0.69	0.62	13	12	11

Notes: Out-of-sample forecasting results for the quarterly U.S. equity premium, 2004Q4 to 2024Q4. R^2_{OOS} is measured relative to the historical mean (HM) benchmark. ΔCER is the annualized certainty equivalent return gain (percentage points) relative to HM. Portfolio weights derive from mean-variance optimization with risk aversion $\gamma = 5$, transaction costs of 0.00%, and a 5-years rolling window for variance; weights are constrained to $[0.0, 1.5]$. Sharpe ratios are annualized; turnover is reported as a quarterly average. BRT: Bayesian regression with Student- t errors ($\nu = 5$). GB: generalized Bayesian CQR with flat prior and calibrated η . GB_f: same model with fixed $\eta = 1$. Bold entries denote positive R^2_{OOS} or ΔCER , the highest Sharpe ratio, or the lowest turnover among the reported methods. *, **, ***: 10%, 5%, 1% significance.

Table 27: Out-of-Sample Equity Premium Forecasting: $\gamma = 5$, $TC = 0.50\%$ (2004Q4–2024Q4)

	R^2_{OOS} (%)				ΔCER (%)				Sharpe Ratio				Turnover (%)			
	OLS	BRT	GB	GB_f	OLS	BRT	GB	GB_f	HM	OLS	BRT	GB	GB_f	OLS	BRT	GB
Panel A: Valuation Ratios																
d/p	-11.21	-0.06	0.98**	-1.82	-3.98	-0.24	-0.02	-0.96	0.64	0.41	0.62	0.64	0.56	1	5	5
d/y	-13.57	-0.28	0.81*	-2.37	-4.07	-0.40	-0.02	-1.33	0.64	0.26	0.61	0.64	0.53	0	5	6
e/p	-5.01	-1.35	0.21	-2.88	-1.93	-0.51	-0.19	-1.15	0.64	0.47	0.60	0.63	0.54	7	6	6
d/e	-0.57	-10.54	-7.46	-7.19	0.17	-0.37	0.44	-0.65	0.64	0.65	0.62	0.68	0.60	7	6	6
b/m	-9.69	0.45	1.41**	-0.53	-3.93	-0.04	0.23	-0.26	0.64	0.48	0.64	0.66	0.62	1	5	5
Panel B: Interest Rates and Spreads																
tbl	0.14	0.31	-0.43	0.33	0.30	0.56	1.18	0.20	0.64	0.67	0.69	0.73	0.66	5	5	4
lty	1.32	1.27	0.60	1.21	0.28	0.54	1.28	0.19	0.64	0.67	0.69	0.74	0.66	5	5	4
ltr	0.60	1.21	1.06	1.00	-1.23	-0.73	0.58	-1.32	0.64	0.55	0.59	0.69	0.54	29	29	23
tms	-3.09	-1.33	-2.04	-1.47	0.87	0.89	0.88	0.87	0.64	0.71	0.70	0.71	0.70	8	6	6
dfy	-1.43	-0.93	-0.27	-0.39	0.89	-0.67	0.74	-0.32	0.64	0.71	0.60	0.70	0.62	8	8	5
dfr	-4.37	-0.87	-1.48	-1.22	-0.95	-0.75	0.43	-0.77	0.64	0.57	0.59	0.68	0.59	11	12	10
crdstd	-4.82	-4.09	-3.39	-4.33	-2.20	-1.65	-1.71	-1.68	0.61	0.48	0.52	0.51	0.51	14	14	14
Panel C: Macroeconomic Variables																
infl	0.89	0.54	0.30	0.48	-0.33	-0.04	1.11	-0.07	0.64	0.62	0.64	0.72	0.64	17	6	6
ntis	-12.41	-2.66	-5.74	-3.06	1.95	0.90	2.07	0.68	0.64	0.78	0.71	0.79	0.69	8	5	5
svar	0.36	-0.43	0.23	0.35	0.74	-0.24	1.22	0.23	0.64	0.69	0.63	0.73	0.66	13	7	6
i/k	5.02**	5.41**	5.43**	5.32**	0.76	0.58	0.66	0.65	0.63	0.69	0.70	0.70	0.70	8	8	8
govik	-2.45	-0.54	-0.20	-1.16	-1.18	-0.59	-0.57	-0.78	0.63	0.53	0.59	0.59	0.57	6	5	5
ndrbl	2.06	2.94*	2.77*	2.85*	-1.02	-0.26	-0.32	-0.50	0.61	0.56	0.62	0.62	0.60	15	15	15
wtexas	-2.76	-1.24	-1.69	-1.10	-1.19	-0.56	-0.07	-0.39	0.64	0.56	0.61	0.65	0.61	28	15	13
Panel D: Variance and Risk Measures																
vp	2.44	7.25*	6.80*	5.67	-0.48	0.95	1.92	-0.37	0.68	0.64	0.75	0.81	0.65	35	34	27
impvar	-6.32	-3.13	-1.89	-5.44	0.50	1.03	1.02	0.51	0.56	0.60	0.65	0.66	0.60	16	19	20
vrp	-13.97	-4.82	-4.69	-7.79	-3.43	-1.52	-0.95	-2.45	0.59	0.27	0.49	0.54	0.38	40	47	46
tail	-0.28	0.15	-0.65	-0.03	0.27	0.23	1.33	0.19	0.64	0.66	0.66	0.74	0.65	6	5	5
skvw	-0.15	-2.15	-1.55	-1.55	-1.06	-0.06	0.70	-0.45	0.64	0.55	0.65	0.70	0.61	25	18	12
Panel E: Cross-Section and Technical																
lzrt	-16.96	-3.15	-3.02	-4.56	-4.06	-2.00	-1.38	-2.34	0.64	0.34	0.50	0.57	0.46	28	18	18
avgcor	3.63	3.95	2.37	4.22	0.03	1.24	2.33	0.31	0.64	0.64	0.73	0.80	0.66	25	14	11
dtoy	0.69	-3.95	-2.00	-2.14	0.58	-1.04	0.50	-0.89	0.64	0.68	0.58	0.69	0.58	7	13	8
dtoat	0.26	0.51	0.73	0.25	-0.07	0.08	0.92	0.17	0.64	0.64	0.65	0.71	0.65	7	5	5
ygap	-4.90	-4.45	-3.89	-5.97	-0.51	-0.17	0.00	-0.35	0.63	0.58	0.61	0.64	0.60	6	6	6
rdsp	0.18	-1.48	-1.17	-1.01	-0.76	0.16	0.95	0.21	0.64	0.58	0.65	0.71	0.65	9	10	7
disag	-4.62	-3.60	-2.39	-4.00	-3.58	-2.21	-1.60	-2.60	0.63	0.35	0.47	0.53	0.43	20	17	14
Average	-3.26	-0.87	-0.65	-1.24	-0.92	-0.22	0.44	-0.50	0.63	0.57	0.62	0.67	0.60	13	12	11

Notes: Out-of-sample forecasting results for the quarterly U.S. equity premium, 2004Q4 to 2024Q4. R^2_{OOS} is measured relative to the historical mean (HM) benchmark. ΔCER is the annualized certainty equivalent return gain (percentage points) relative to HM. Portfolio weights derive from mean-variance optimization with risk aversion $\gamma = 5$, transaction costs of 0.50%, and a 5-years rolling window for variance; weights are constrained to $[0.0, 1.5]$. Sharpe ratios are annualized; turnover is reported as a quarterly average. BRT: Bayesian regression with Student- t errors ($\nu = 5$). GB: generalized Bayesian CQR with flat prior and calibrated η . GB_f : same model with fixed $\eta = 1$. Bold entries denote positive R^2_{OOS} or ΔCER , the highest Sharpe ratio, or the lowest turnover among the reported methods. *, **, ***: 10%, 5%, 1% significance.

Table 28: Out-of-Sample Equity Premium Forecasting: $\gamma = 5$, TC = 1.00% (2004Q4–2024Q4)

	R^2_{OOS} (%)				ΔCER (%)				Sharpe Ratio				Turnover (%)			
	OLS	BRT	GB	GB_f	OLS	BRT	GB	GB_f	HM	OLS	BRT	GB	GB_f	OLS	BRT	GB
<i>Panel A: Valuation Ratios</i>																
d/p	-11.21	-0.06	0.98**	-1.82	-3.88	-0.23	-0.03	-0.96	0.63	0.36	0.61	0.63	0.55	1	5	5
d/y	-13.57	-0.28	0.81*	-2.37	-3.95	-0.40	-0.03	-1.35	0.63	0.26	0.60	0.63	0.51	0	5	6
e/p	-5.01	-1.35	0.21	-2.88	-1.97	-0.54	-0.23	-1.18	0.63	0.45	0.58	0.62	0.53	7	6	6
d/e	-0.57	-10.54	-7.46	-7.19	0.14	-0.41	0.42	-0.68	0.63	0.64	0.61	0.68	0.59	7	6	6
b/m	-9.69	0.45	1.41**	-0.53	-3.83	-0.05	0.22	-0.26	0.63	0.46	0.63	0.65	0.61	1	5	5
<i>Panel B: Interest Rates and Spreads</i>																
tbl	0.14	0.31	-0.43	0.33	0.29	0.56	1.19	0.19	0.63	0.66	0.68	0.72	0.65	5	5	4
lty	1.32	1.27	0.60	1.21	0.28	0.54	1.30	0.18	0.63	0.66	0.68	0.73	0.65	5	5	4
ltr	0.60	1.21	1.06	1.00	-1.73	-1.24	0.23	-1.85	0.63	0.50	0.55	0.66	0.50	29	29	23
tms	-3.09	-1.33	-2.04	-1.47	0.85	0.88	0.85	0.86	0.63	0.69	0.69	0.70	0.69	8	6	6
dfy	-1.43	-0.93	-0.27	-0.39	0.84	-0.75	0.73	-0.34	0.63	0.70	0.59	0.69	0.61	8	8	5
dfr	-4.37	-0.87	-1.48	-1.22	-1.07	-0.90	0.32	-0.90	0.63	0.55	0.58	0.67	0.57	11	12	10
crdst	-4.82	-4.09	-3.39	-4.33	-2.41	-1.85	-1.91	-1.87	0.60	0.46	0.50	0.49	0.49	14	14	14
<i>Panel C: Macroeconomic Variables</i>																
infl	0.89	0.54	0.30	0.48	-0.57	-0.07	1.09	-0.10	0.63	0.59	0.63	0.71	0.62	17	6	6
ntis	-12.41	-2.66	-5.74	-3.06	1.91	0.89	2.09	0.67	0.63	0.77	0.70	0.78	0.68	8	5	5
svar	0.36	-0.43	0.23	0.35	0.59	-0.29	1.20	0.20	0.63	0.67	0.62	0.72	0.65	13	7	6
i/k	5.02**	5.41**	5.43**	5.32**	0.70	0.51	0.59	0.58	0.62	0.68	0.69	0.69	0.69	8	8	8
govik	-2.45	-0.54	-0.20	-1.16	-1.19	-0.59	-0.58	-0.78	0.62	0.52	0.58	0.58	0.56	6	5	5
ndrbl	2.06	2.94*	2.77*	2.85*	-1.23	-0.48	-0.53	-0.72	0.60	0.54	0.60	0.60	0.58	15	15	15
wtexas	-2.76	-1.24	-1.69	-1.10	-1.56	-0.72	-0.19	-0.53	0.63	0.53	0.59	0.64	0.60	28	15	13
<i>Panel D: Variance and Risk Measures</i>																
vp	2.44	7.25*	6.80*	5.67	-1.03	0.40	1.49	-1.01	0.67	0.59	0.71	0.78	0.60	35	34	27
impvar	-6.32	-3.13	-1.89	-5.44	0.33	0.79	0.75	0.22	0.55	0.57	0.63	0.64	0.57	16	19	20
vrp	-13.97	-4.82	-4.69	-7.79	-4.11	-2.31	-1.74	-3.23	0.58	0.20	0.42	0.48	0.31	40	47	46
tail	-0.28	0.15	-0.65	-0.03	0.26	0.22	1.35	0.18	0.63	0.65	0.65	0.73	0.65	6	5	5
skvw	-0.15	-2.15	-1.55	-1.55	-1.45	-0.30	0.58	-0.64	0.63	0.50	0.62	0.68	0.59	25	18	12
<i>Panel E: Cross-Section and Technical</i>																
lzrt	-16.96	-3.15	-3.02	-4.56	-4.41	-2.20	-1.57	-2.56	0.63	0.30	0.48	0.55	0.43	28	18	18
avgcor	3.63	3.95	2.37	4.22	-0.40	1.06	2.22	-0.01	0.63	0.61	0.71	0.79	0.63	25	14	11
dtoy	0.69	-3.95	-2.00	-2.14	0.53	-1.17	0.47	-0.97	0.63	0.67	0.56	0.68	0.57	7	13	8
dtoat	0.26	0.51	0.73	0.25	-0.12	0.07	0.92	0.18	0.63	0.63	0.64	0.70	0.64	7	5	5
ygap	-4.90	-4.45	-3.89	-5.97	-0.53	-0.21	-0.05	-0.39	0.62	0.57	0.60	0.63	0.59	6	6	6
rdsp	0.18	-1.48	-1.17	-1.01	-0.84	0.06	0.91	0.15	0.63	0.56	0.64	0.70	0.64	9	10	7
disag	-4.62	-3.60	-2.39	-4.00	-3.90	-2.45	-1.76	-2.84	0.62	0.31	0.44	0.51	0.41	20	17	14
Average	-3.26	-0.87	-0.65	-1.24	-1.08	-0.36	0.33	-0.64	0.62	0.54	0.61	0.66	0.58	13	12	11

Notes: Out-of-sample forecasting results for the quarterly U.S. equity premium, 2004Q4 to 2024Q4. R^2_{OOS} is measured relative to the historical mean (HM) benchmark. ΔCER is the annualized certainty equivalent return gain (percentage points) relative to HM. Portfolio weights derive from mean-variance optimization with risk aversion $\gamma = 5$, transaction costs of 1.00%, and a 5-years rolling window for variance; weights are constrained to $[0.0, 1.5]$. Sharpe ratios are annualized; turnover is reported as a quarterly average. BRT: Bayesian regression with Student- t errors ($\nu = 5$). GB: generalized Bayesian CQR with flat prior and calibrated η . GB_f : same model with fixed $\eta = 1$. Bold entries denote positive R^2_{OOS} or ΔCER , the highest Sharpe ratio, or the lowest turnover among the reported methods. *, **, ***: 10%, 5%, 1% significance.

8 Empirical Illustration: Comparison with Frequentist CQR

Table 29 compares the calibrated GBCQR (GB), fixed- η GBCQR (GB_f), and frequentist CQR in the equity premium forecasting application. Each cell reports the arithmetic average across all 31 predictors. Panel A evaluates over 1994Q4–2024Q4; Panel B over 2004Q4–2024Q4.

GB and CQR deliver very similar economic performance: their Sharpe ratios and turnover are nearly identical across all 18 configurations, and their ΔCER values track each other closely. GB does hold a small but consistent edge — it achieves the least negative R^2_{OOS} in both windows (-0.61 vs. -1.28 over 30 years; -0.65 vs. -1.16 over 20 years), and its ΔCER improves in the shorter window (e.g., 1.22 vs. 1.02 at $\gamma = 3$, $\text{TC} = 0.5\%$).

Table 29: Summary of Average Out-of-Sample Results: GB, GB_f , and CQR

γ	TC (%)	ΔCER (%)			Sharpe Ratio			Turnover (%)		
		GB	GB_f	CQR	GB	GB_f	CQR	GB	GB_f	CQR
<i>Panel A: 1994Q4–2024Q4.</i>										
3	0.00	0.83	0.42	0.95	0.40	0.35	0.40	11	14	11
3	0.50	0.72	0.26	0.83	0.39	0.33	0.39	11	14	11
3	1.00	0.61	0.09	0.71	0.38	0.32	0.38	11	14	11
5	0.00	0.27	0.77	0.44	0.32	0.28	0.33	12	13	12
5	0.50	0.18	0.65	0.36	0.31	0.26	0.31	12	13	12
5	1.00	0.09	0.54	0.27	0.29	0.24	0.30	12	13	12
Avg. R^2_{OOS} (%): GB = -0.61 , GB_f = -0.77 , CQR = -1.28										
<i>Panel B: 2004Q4–2024Q4.</i>										
3	0.00	1.33	-0.48	1.14	0.79	0.71	0.78	10	13	10
3	0.50	1.22	-0.65	1.02	0.78	0.70	0.77	10	13	10
3	1.00	1.11	-0.82	0.90	0.76	0.68	0.76	10	13	10
5	0.00	0.55	-0.36	0.51	0.69	0.62	0.69	11	12	11
5	0.50	0.44	-0.50	0.40	0.67	0.60	0.67	11	12	11
5	1.00	0.33	-0.64	0.29	0.66	0.58	0.66	11	12	11
Avg. R^2_{OOS} (%): GB = -0.65 , GB_f = -1.24 , CQR = -1.16										

Notes: Each entry reports the arithmetic mean across all 31 predictors. R^2_{OOS} (reported at the bottom of each panel) is measured relative to the historical mean benchmark and is invariant to (γ, TC) . ΔCER is the annualized certainty equivalent return gain (percentage points) relative to the historical mean. Portfolio weights derive from mean-variance optimization with a 5-years rolling window for variance; weights are constrained to $[0.0, 1.5]$. Sharpe ratios are annualized; turnover is a quarterly average (%). GB: generalized Bayesian CQR with flat prior and calibrated η . GB_f : same model with fixed $\eta = 1$. CQR: frequentist composite quantile regression. Bold entries denote positive ΔCER , the highest Sharpe ratio, or the lowest turnover among the three methods.